Propagation of high-energy particles: Cosmic-ray Protons and nuclei

(Lecture 8)

Satyendra Thoudam
Khalifa University

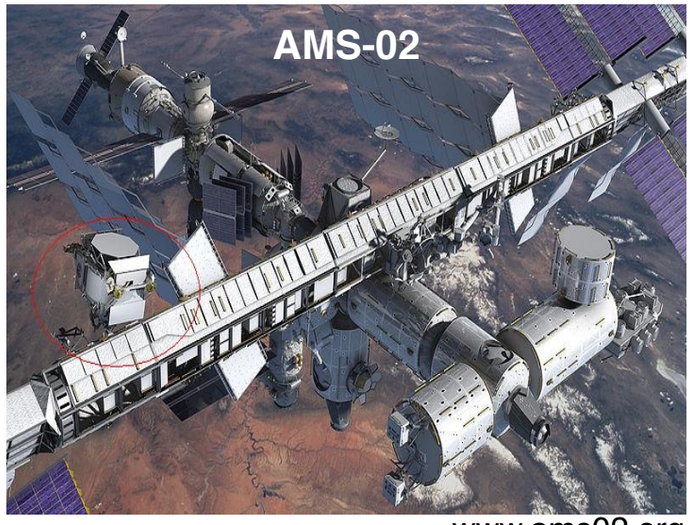
Outline of this chapter

- Introduction on the properties of cosmic rays:
  - Composition
  - Spectrum
  - Anisotropy
  - Cosmic-ray distribution
  - Cosmic-ray sources
  - Maximum energy of cosmic rays
- Galactic propagation of cosmic-ray nuclei

Introduction: General

- Cosmic rays were discovered by Victor Hess in 1912.
- Over the last 100 years, several space- and ground-based experiments have been built.
- Several properties of cosmic rays have been measured.

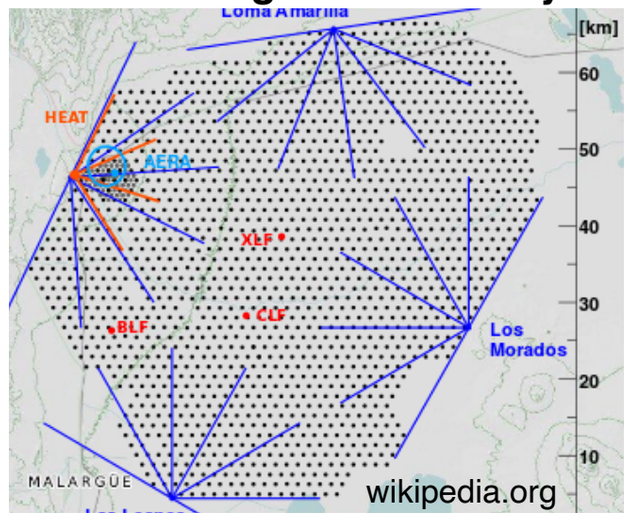
cosmicray.umd.edu



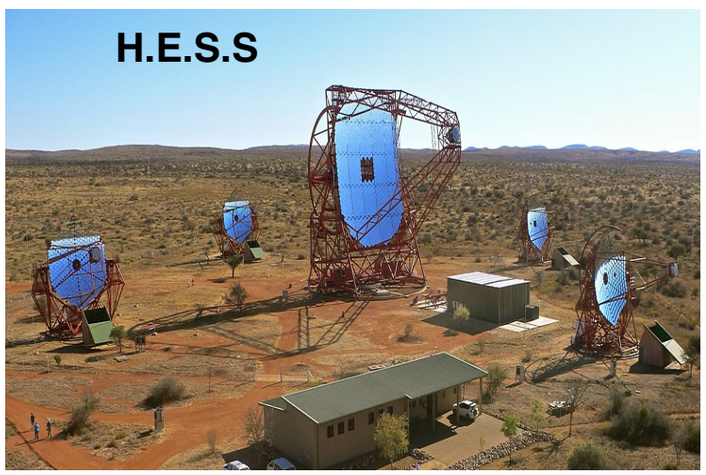
www.ams02.org

www.ikp.kit.edu

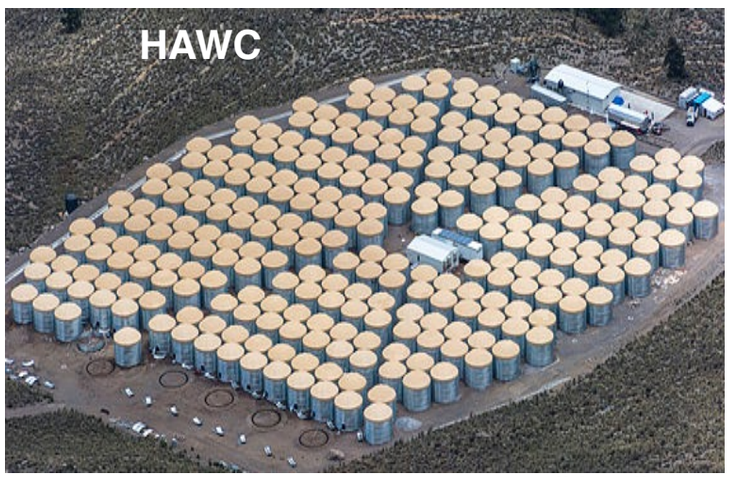
Pierre-Auger Observatory



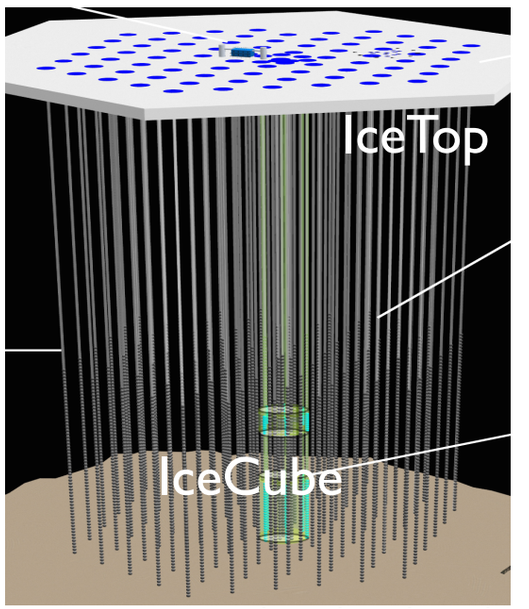
fermi.gsfc.nasa.gov



www.mpi-hd.mpg.de



wikipedia.org



icecube.wisc.edu

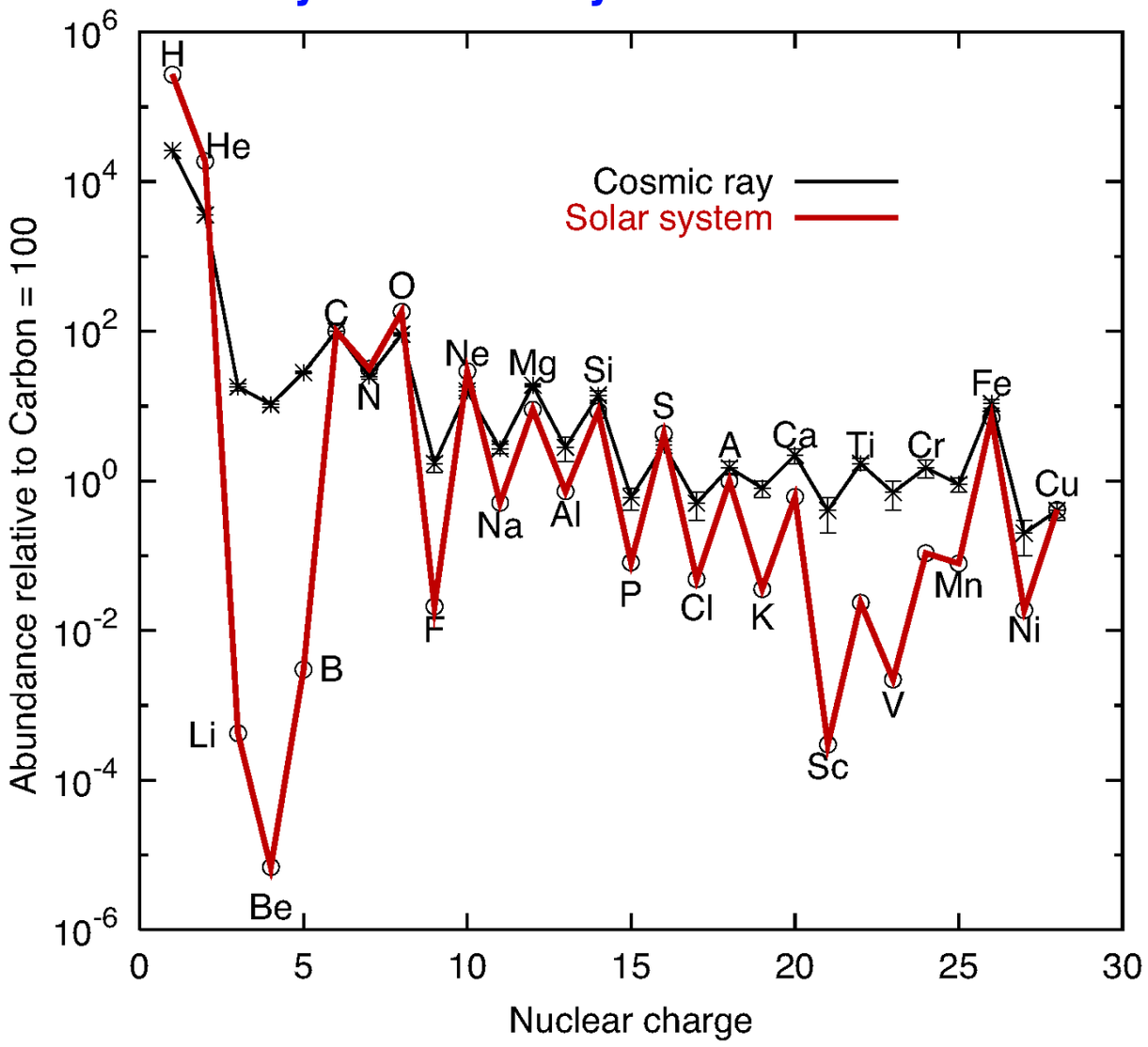
Introduction: Cosmic-ray composition

Cosmic ray protons, nuclei and electrons are the only particles which have been detected from sources outside the Solar System. They consists mostly of charged particles:

- ✦Protons ~ 88%
- ✦Helium nuclei ~ 10%
- ✦Heavier nuclei ~ 1%
- ✦Electrons ~ 1%

Introduction: Cosmic-ray composition

Cosmic-ray and Solar system element abundance

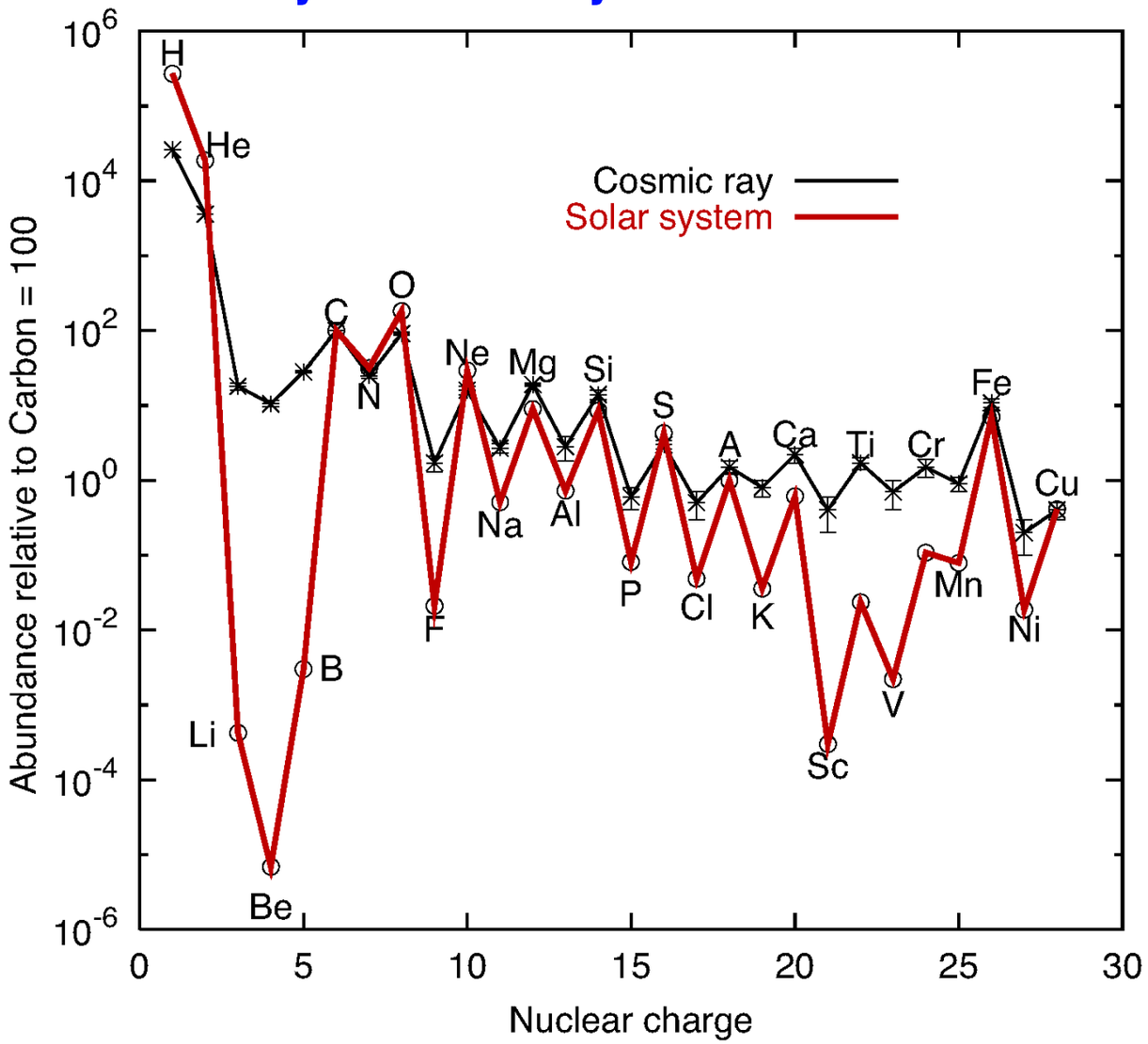


The following features are apparent:

* The abundance peaks at carbon, oxygen, Neon etc. and at the iron group are present both in the cosmic ray and Solar System abundances. These nuclei are referred to as "Primary cosmic rays". They are directly produced by diffusive shock acceleration inside the sources.

Introduction: Cosmic-ray composition

Cosmic-ray and Solar system element abundance



The following features are apparent:

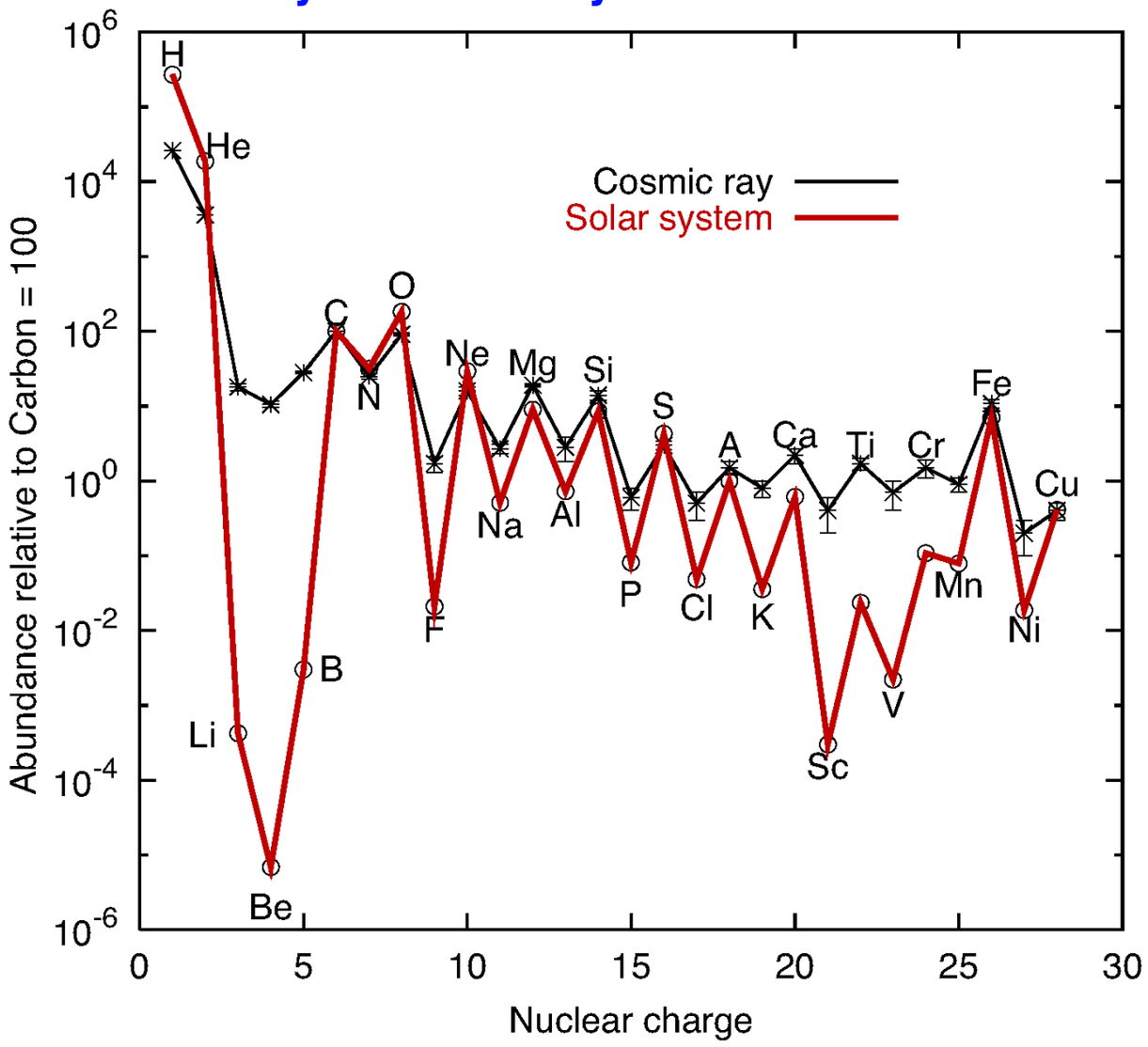
* The abundance peaks at carbon, oxygen, Neon etc. and at the iron group are present both in the cosmic ray and Solar System abundances. These nuclei are referred to as "Primary cosmic rays". They are directly produced by diffusive shock acceleration inside the sources.

* Elements such as lithium, beryllium, boron and sub-Fe (between Calcium and iron) are overabundant in the cosmic rays relative to the Solar System. These nuclei are referred to as "Secondary cosmic rays". They are produced from the spallation (fragmentation) of heavier primary cosmic-ray nuclei in the interstellar medium during their propagation from their sources to the Earth.

[If not mentioned otherwise, "cosmic rays" means "primary cosmic-ray nuclei"]

Introduction: Cosmic-ray composition

Cosmic-ray and Solar system element abundance

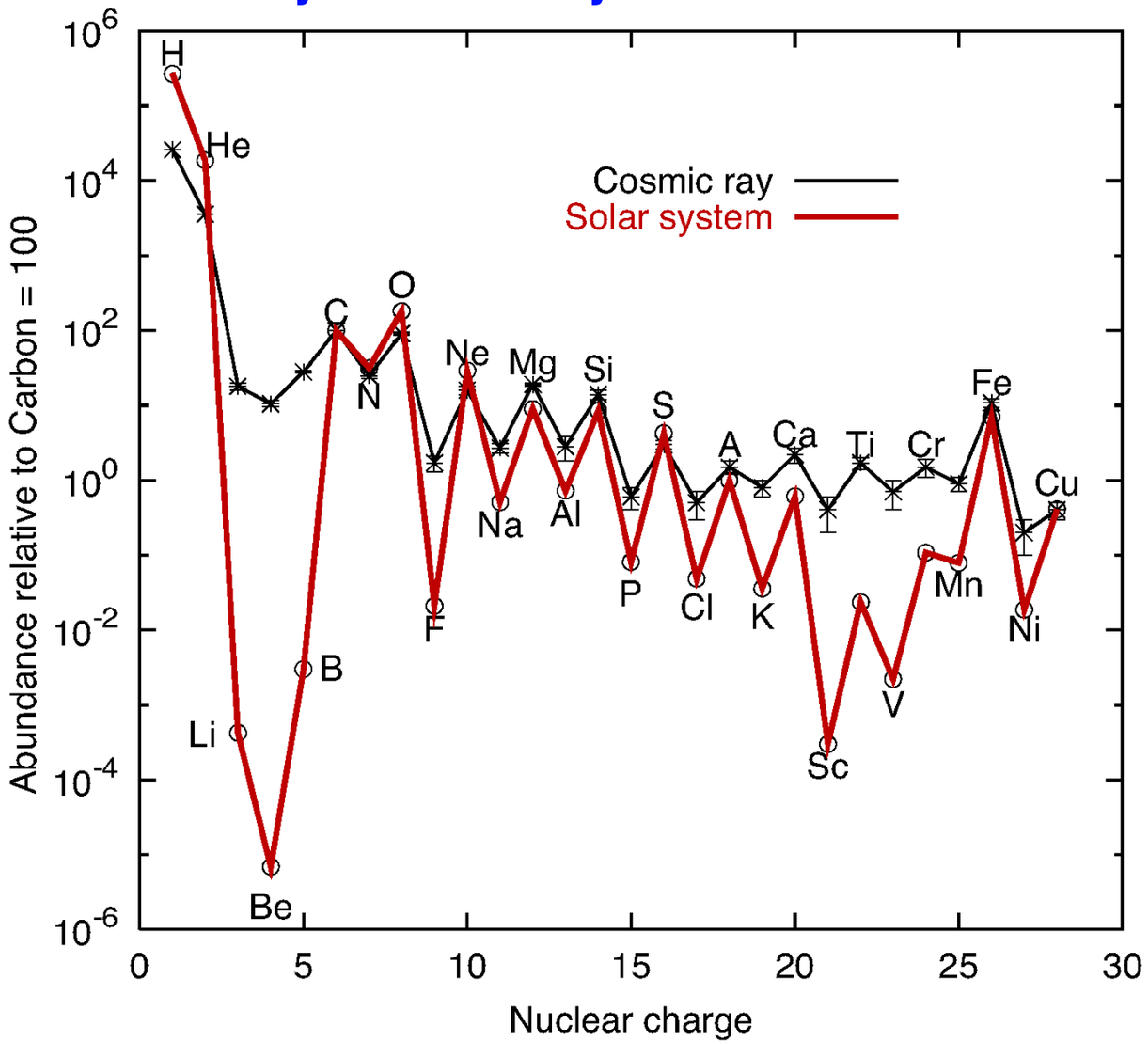


The following features are apparent:

* Odd–even effect in the relative stabilities of the nuclei according to atomic number are present both in the Solar System and in the cosmic rays. Even nuclei are more stable than odd nuclei, and therefore expected to be more numerous.

Introduction: Cosmic-ray composition

Cosmic-ray and Solar system element abundance



The following features are apparent:

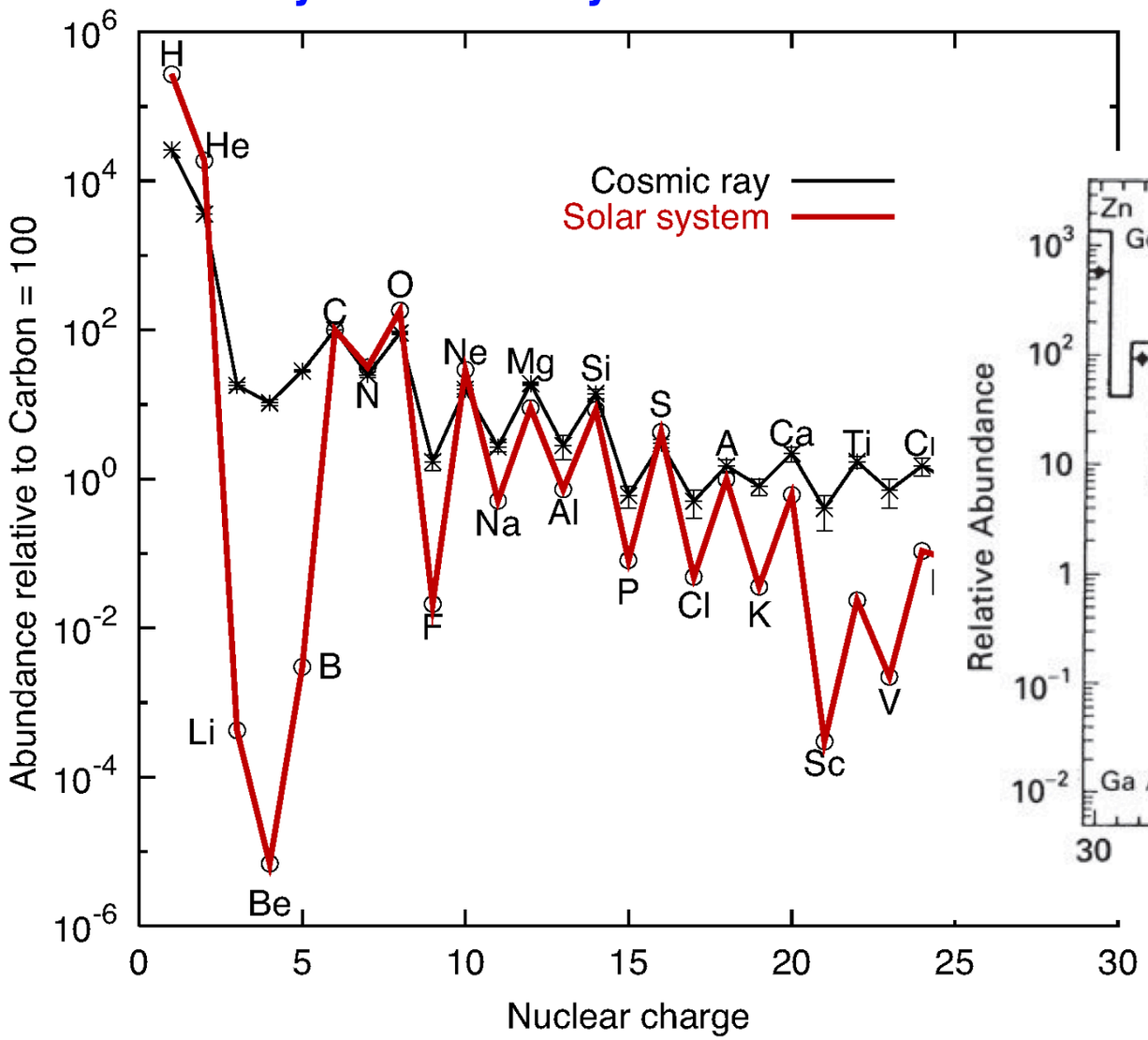
* Odd–even effect in the relative stabilities of the nuclei according to atomic number are present both in the Solar System and in the cosmic rays. Even nuclei are more stable than odd nuclei, and therefore expected to be more numerous.

*This similarity is a strong indication that the seed particles of cosmic rays are indeed the thermal particles already present in the interstellar medium.

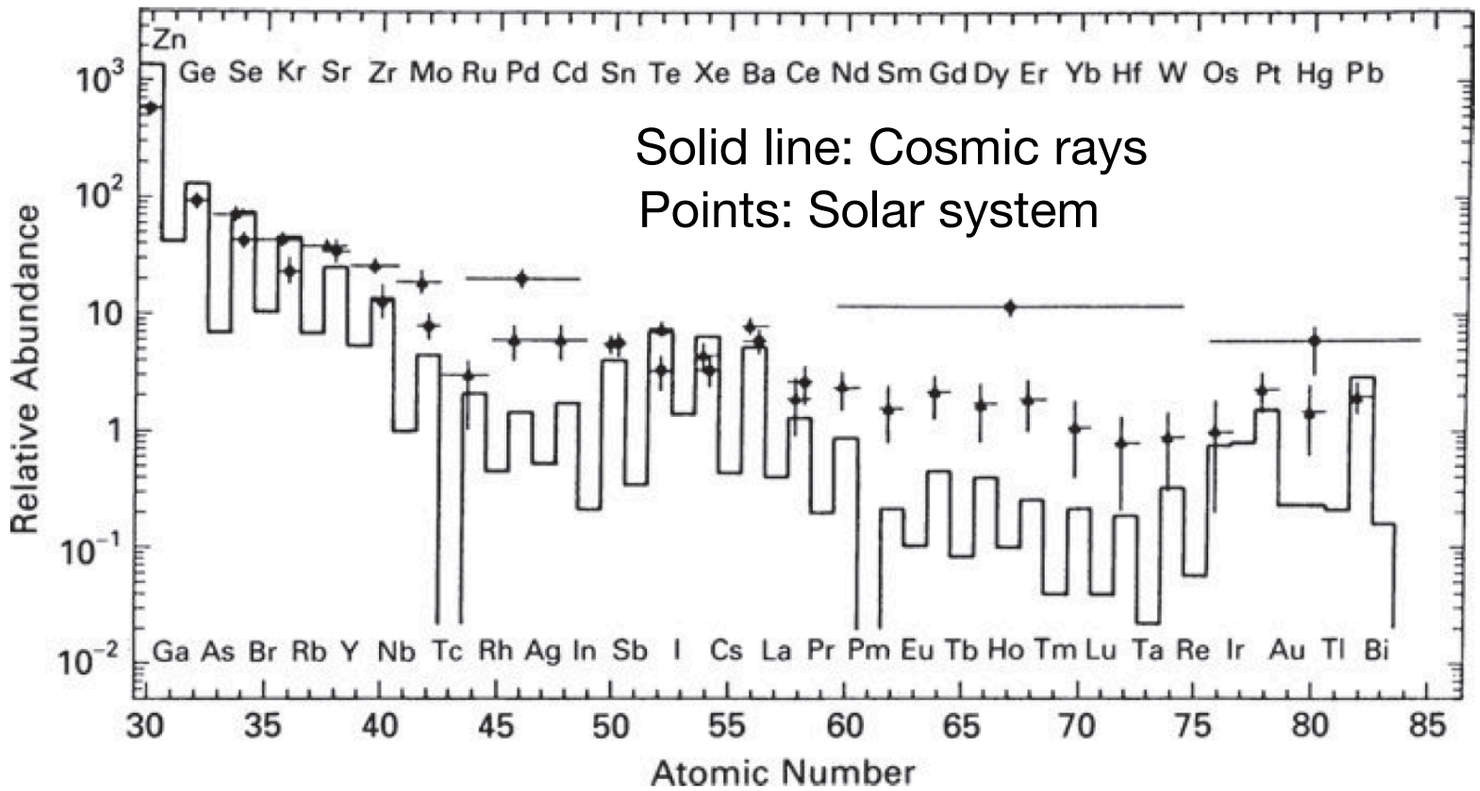
* Supernova shocks run through the interstellar medium and pick up thermal particles which gets accelerated through the diffusive shock acceleration process.

Introduction: Cosmic-ray composition

Cosmic-ray and Solar system element abundance



Elements heavier than iron



Overall, the distribution of element abundances in the cosmic rays is not so different from those in the Solar System / interstellar medium.

Introduction: Cosmic-ray spectrum

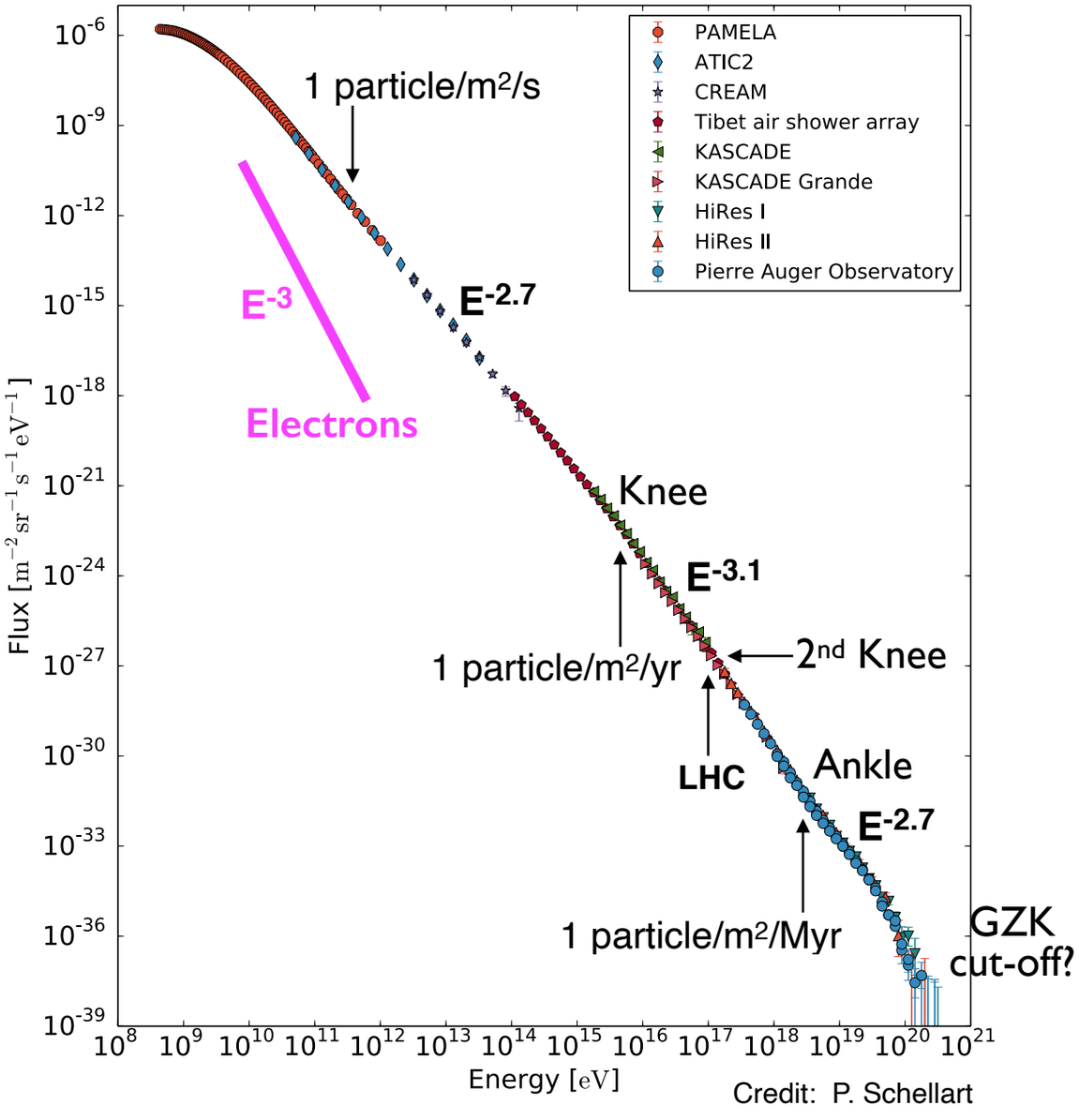
An overview of the spectrum:

*A very wide range of energies is observed and the spectrum can be described by power-law distributions over many decades in energy.

* Mostly nuclei with changing composition as energy increases.

* Important features in the spectrum:
  'knee' at ~ 10^{15} eV
  'ankle' at ~10^{18} eV
  'GZK cut-off' at ~ 10^{20} eV.

All-particle spectrum (mostly nuclei)



Introduction: Cosmic-ray spectrum

An overview of the spectrum:

*A very wide range of energies is observed and the spectrum can be described by power-law distributions over many decades in energy.

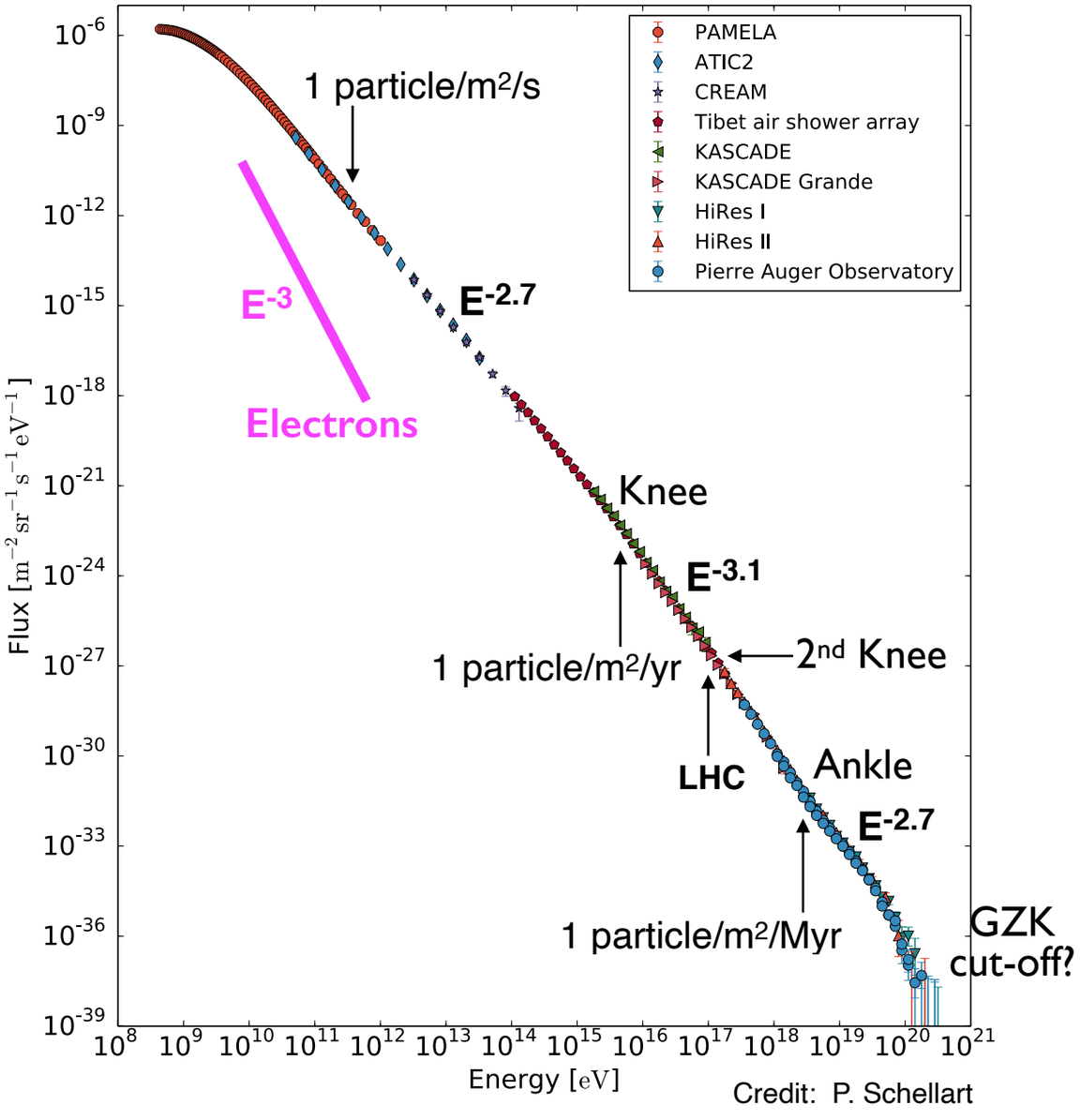
* Mostly nuclei with changing composition as energy increases.

* Important features in the spectrum:
  'knee' at ~ 10^{15} eV
  'ankle' at ~10^{18} eV
  'GZK cut-off' at ~ 10^{20} eV.

* Flux suppression below ~ 10^{10} eV (10 GeV) due to Solar modulation (will discuss in detail later).

* Below ~ 10^{14} eV, mostly measured by space-based experiments. Above, mostly using ground-based experiments.

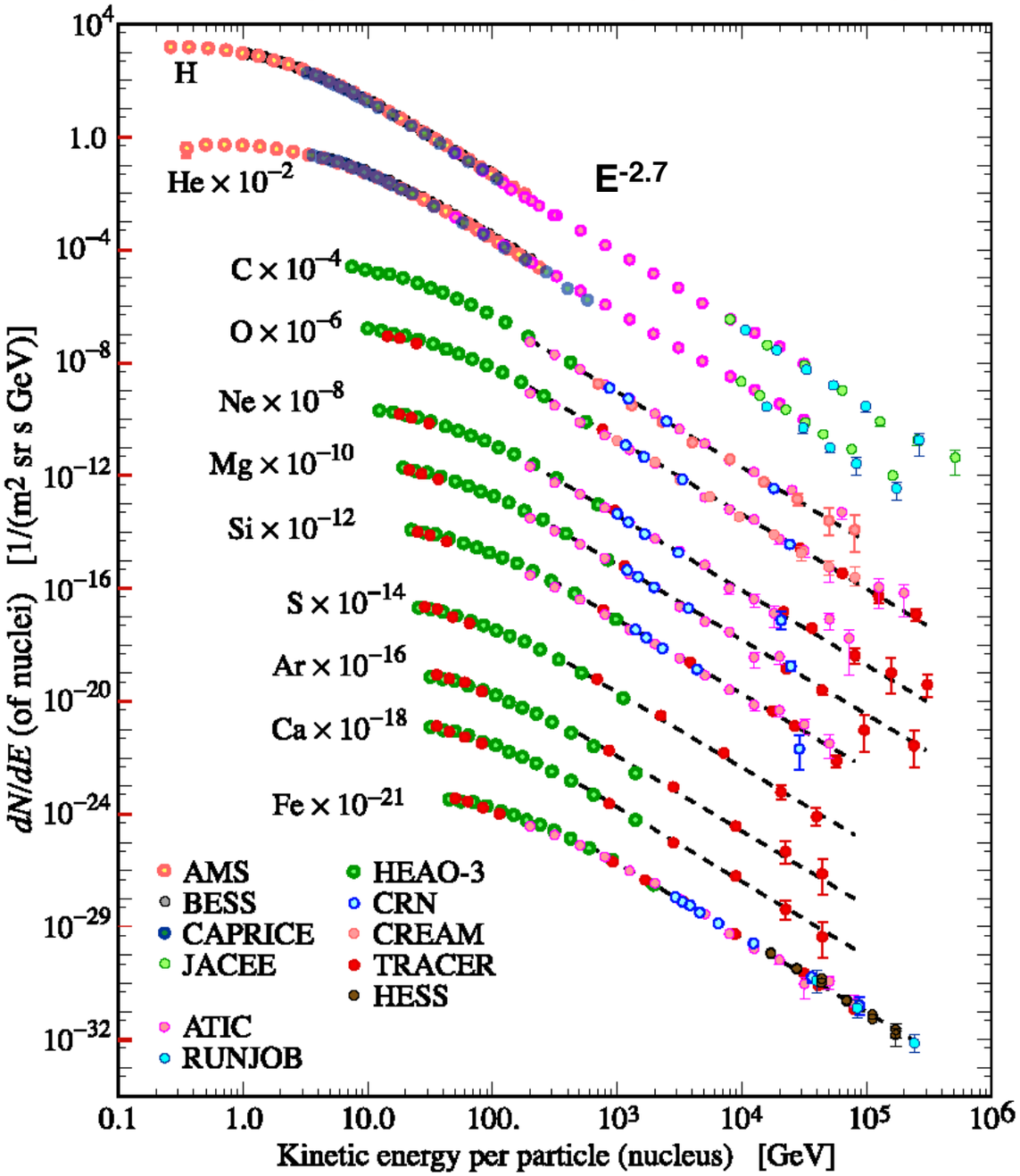
All-particle spectrum (mostly nuclei)



Introduction: Cosmic-ray spectrum

An overview of the spectrum:

Elemental spectra (Primary cosmic-ray nuclei)



Credit: Boyle & Muller, Particle Data Group

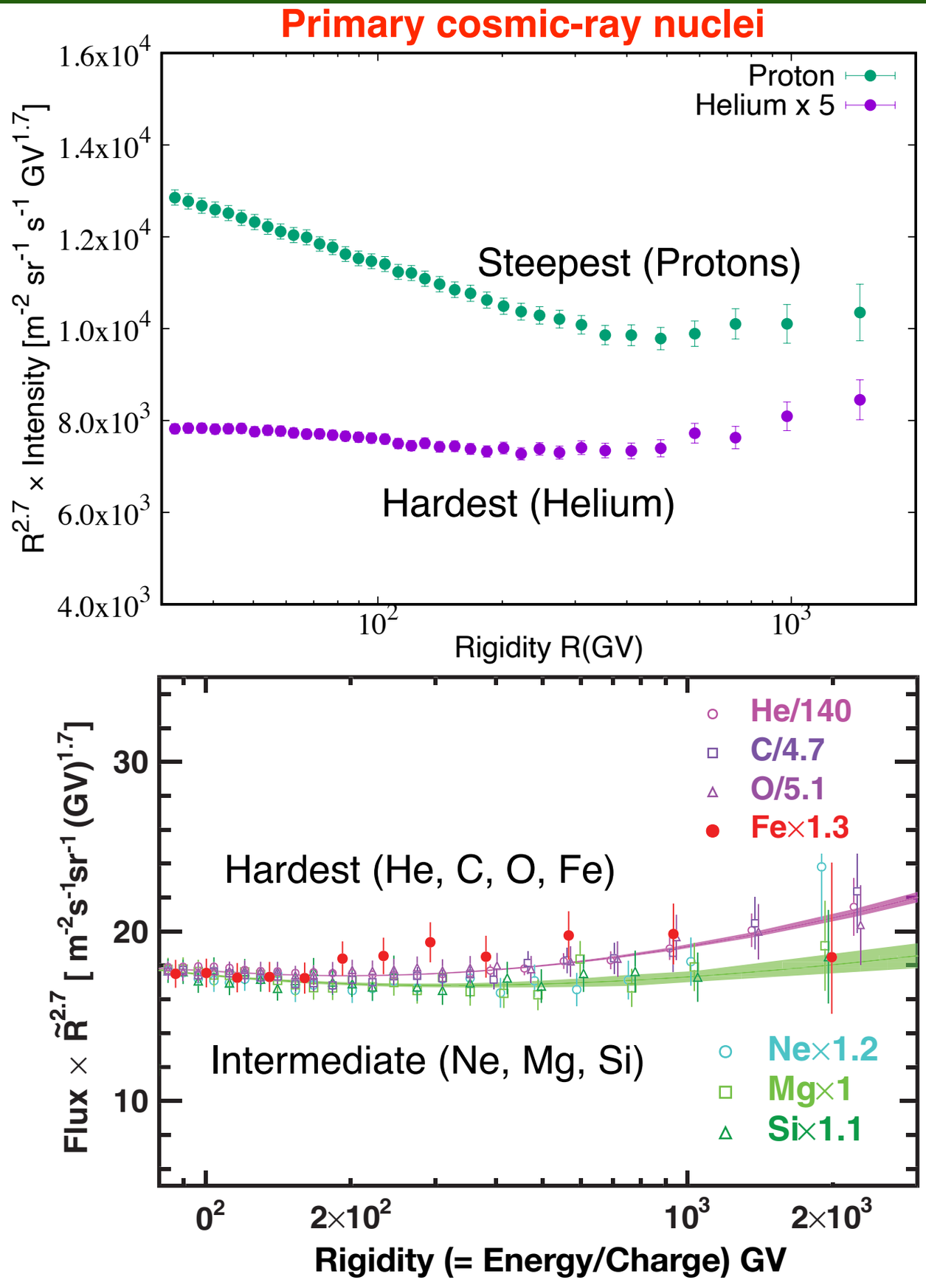
- All types of nuclei seem to follow a perfect power law: $E^{-\Gamma}$

Introduction: Cosmic-ray spectrum

An overview of the spectrum:

However, recent measurements have discovered something different:

- AMS-02, CALET, CREAM, DAMPE experiments have found "Different cosmic-ray primary species showing different spectral shapes".
  - Protons: Steepest spectrum.
  - Ne, Mg, Si: Intermediate.
  - He, C, O, Fe: Hardest.



AMS-02 Collaboration, 2015, PRL, 114,171103
AMS-02 Collaboration, 2015, PRL, 115, 211101

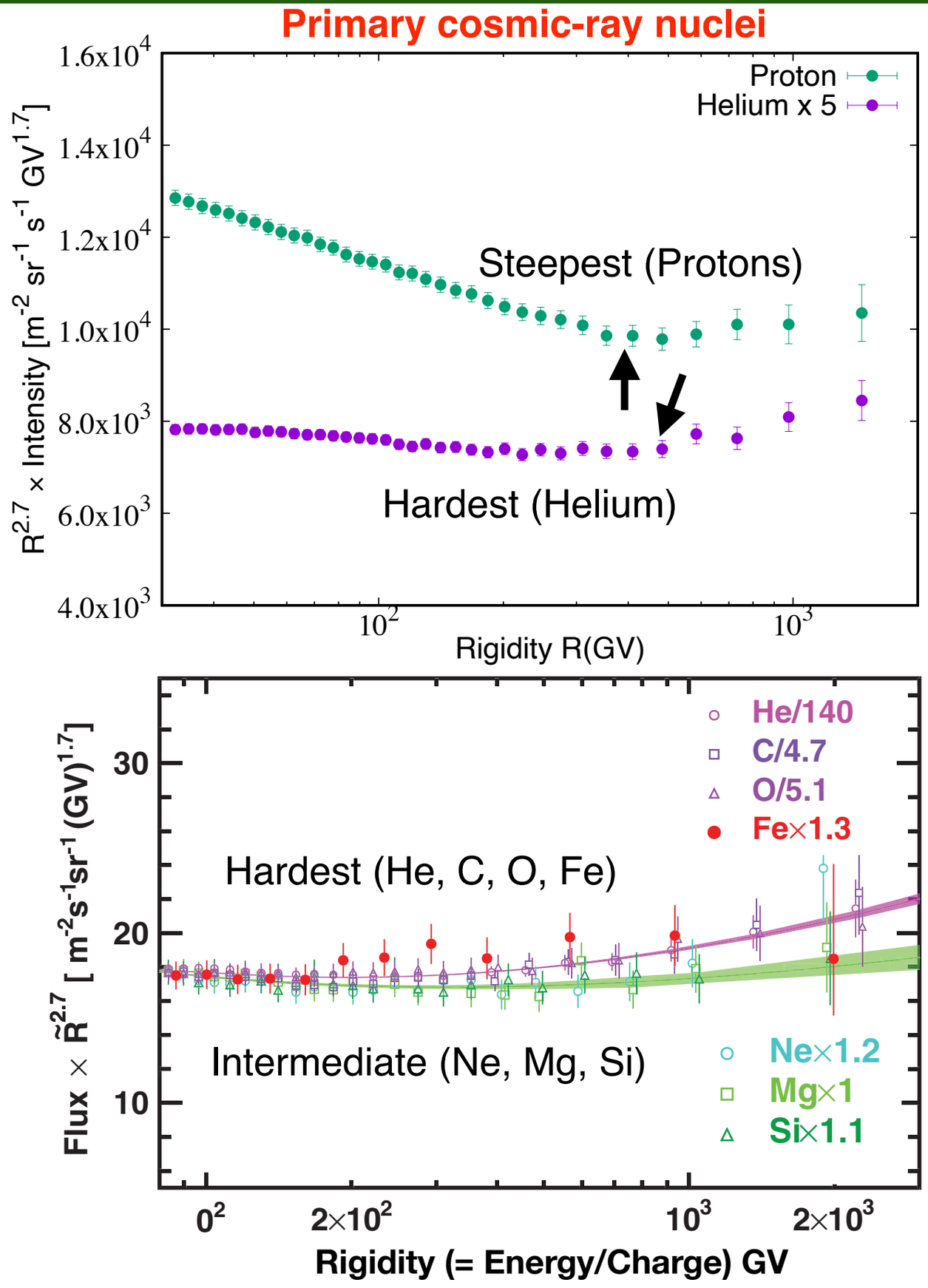
Introduction: Cosmic-ray spectrum

An overview of the spectrum:

However, recent measurements have discovered something different:

- AMS-02, CALET, CREAM, DAMPE experiments have found "Different cosmic-ray primary species showing different spectral shapes".
  - Protons: Steepest spectrum.
  - Ne, Mg, Si: Intermediate.
  - He, C, O, Fe: Hardest.
- Also discovered a break (hardening) in the spectrum at ~ 200-300 GeV/nucleon.

  Diffusive shock acceleration theory and propagation: Predicts a Spectral index independent of the type of nuclei!



AMS-02 Collaboration, 2015, PRL, 114,171103
AMS-02 Collaboration, 2015, PRL, 115, 211101

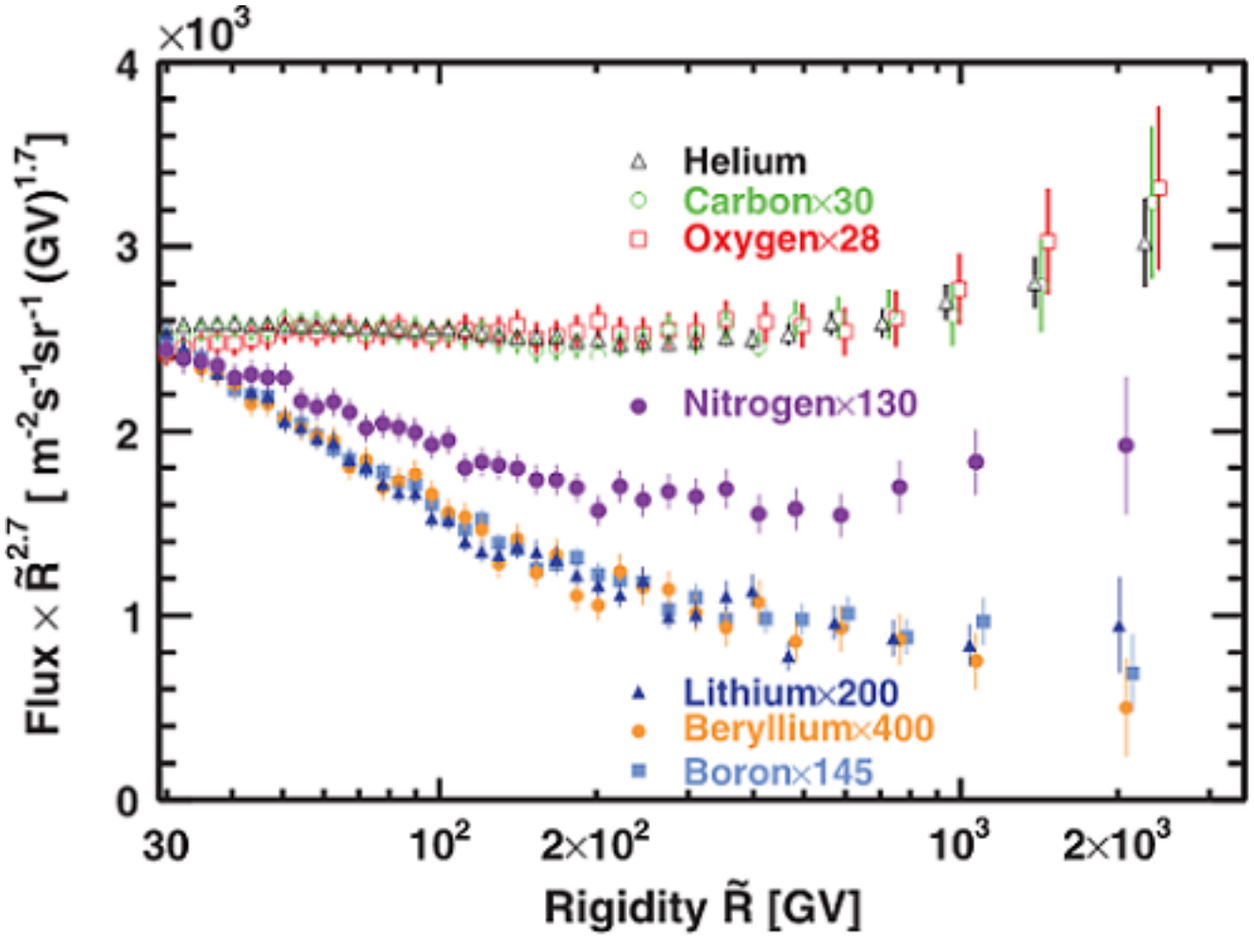
Introduction: Cosmic-ray spectrum

An overview of the spectrum:

* Secondaries cosmic rays (Li, Be, B) have spectra steeper than the primaries (He, C, O) as shown in the figure.

* This is well known. Mainly because cosmic-ray secondaries are produced from the spallation of primary cosmic rays during propagation through the Galaxy (details will be discussed later).

Primary and secondary cosmic-ray spectra



https://www.ssdc.asi.it/ams/

Introduction: Cosmic-ray spectrum

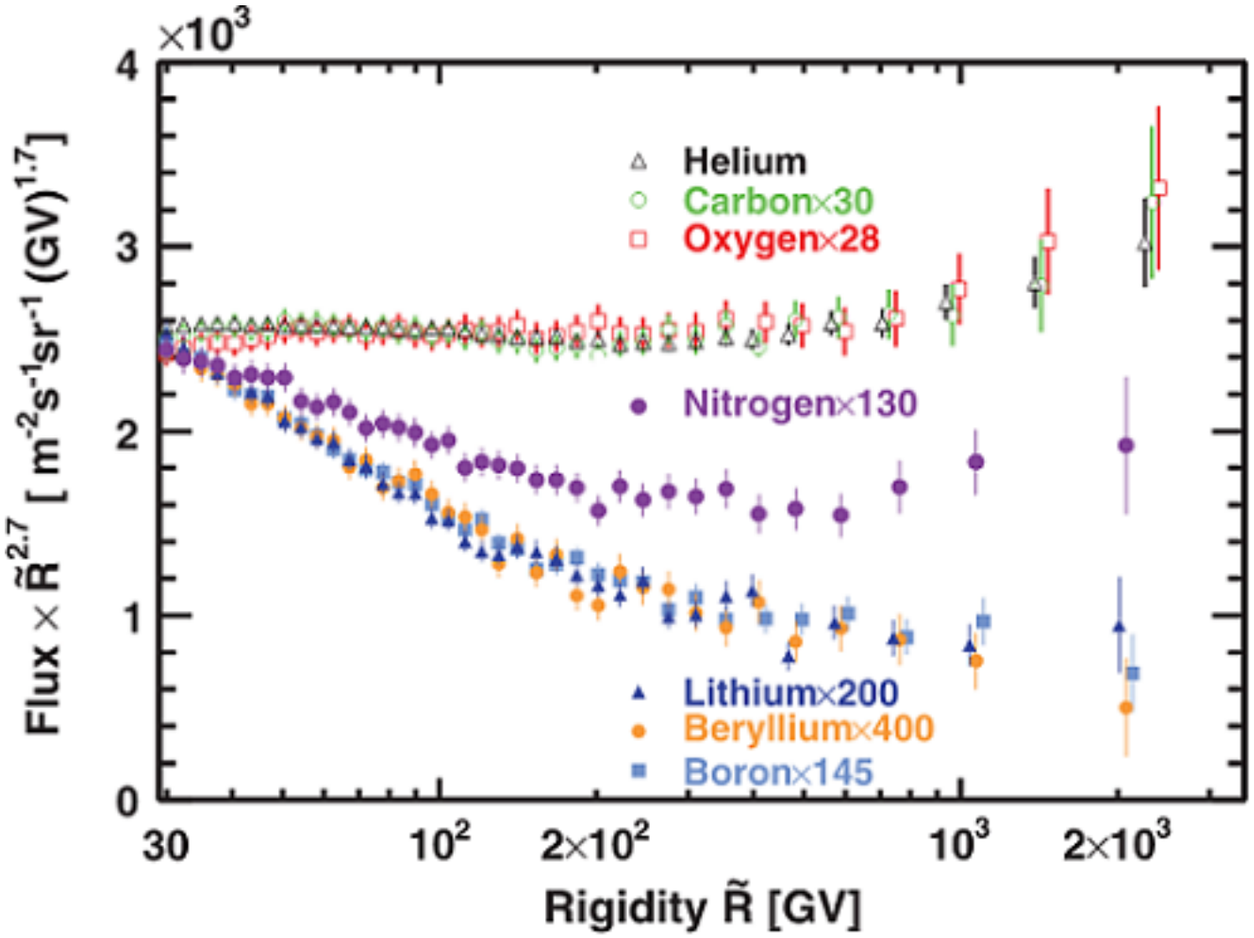
An overview of the spectrum:

* Secondaries cosmic rays (Li, Be, B) have spectra steeper than the primaries (He, C, O) as shown in the figure.

* This is well known. Mainly because cosmic-ray secondaries are produced from the spallation of primary cosmic rays during propagation through the Galaxy (details will be discussed later).

* Nitrogen spectrum has a shape in between the primary and secondary spectra. This might indicate nitrogen has both primary and secondary components.

Primary and secondary cosmic-ray spectra

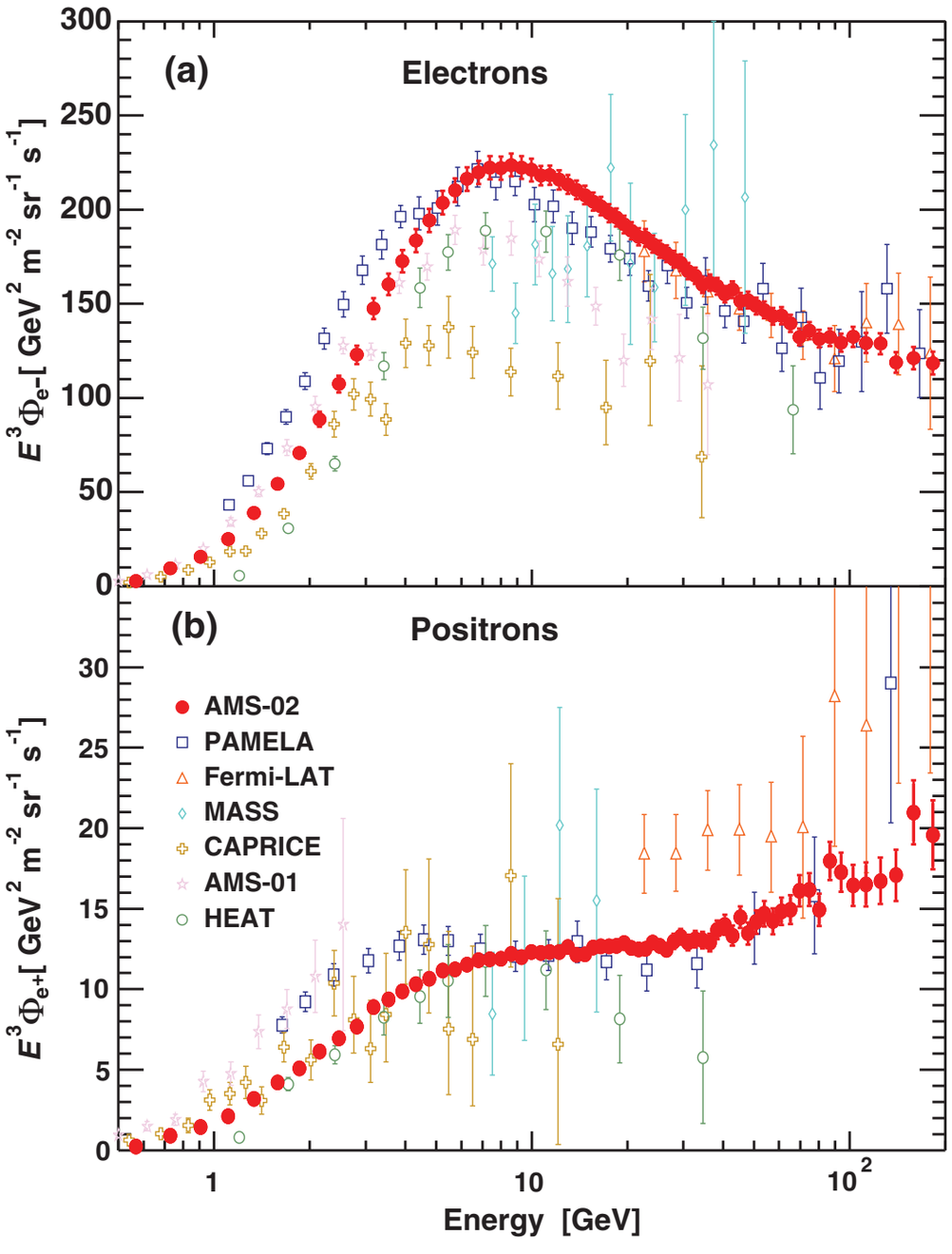


https://www.ssdc.asi.it/ams/

Introduction: Cosmic-ray spectrum

An overview of the spectrum:

Electron and positron spectra

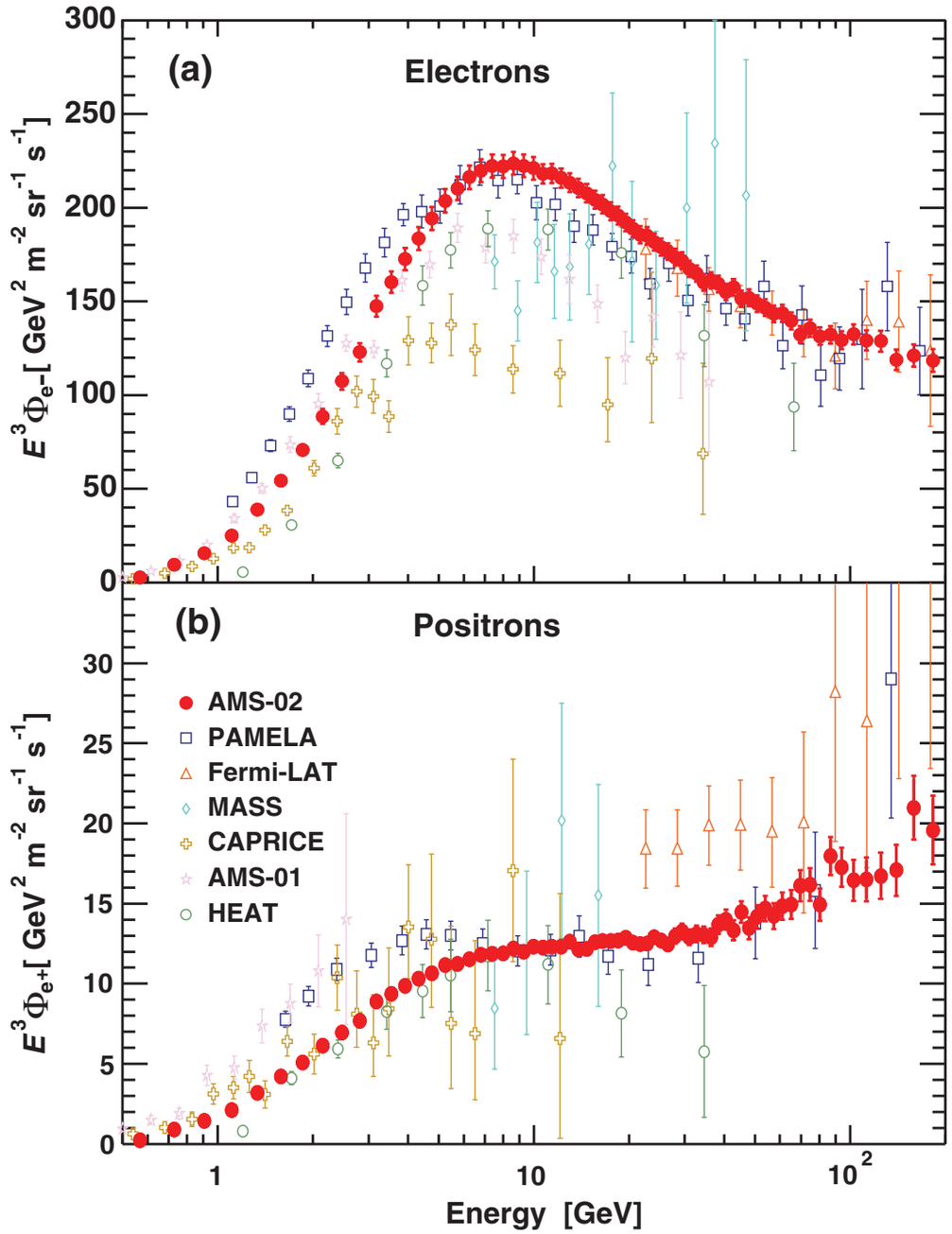


AMS-02 Collaboration (2014, PRL, 113 121102)

Introduction: Cosmic-ray spectrum

An overview of the spectrum:

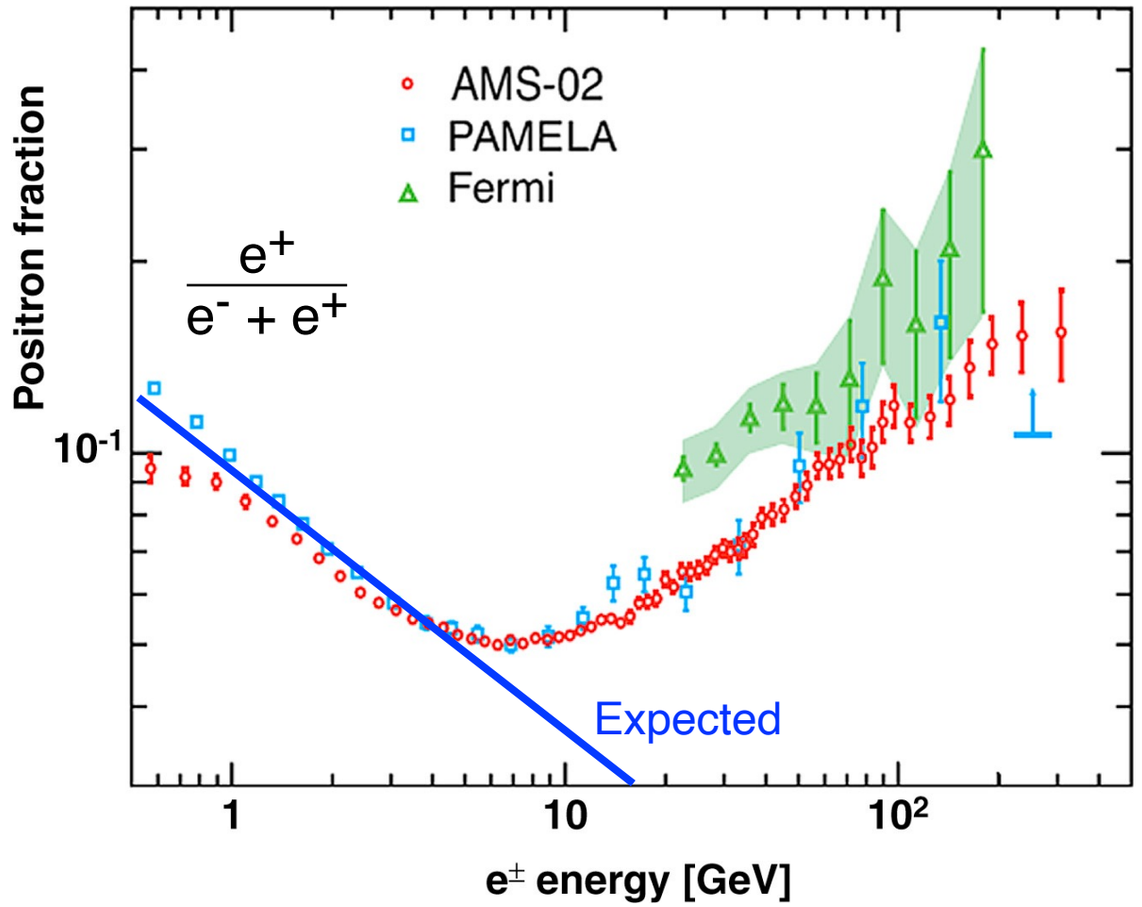
Electron and positron spectra



AMS-02 Collaboration (2014, PRL, 113 121102)

Positron fraction

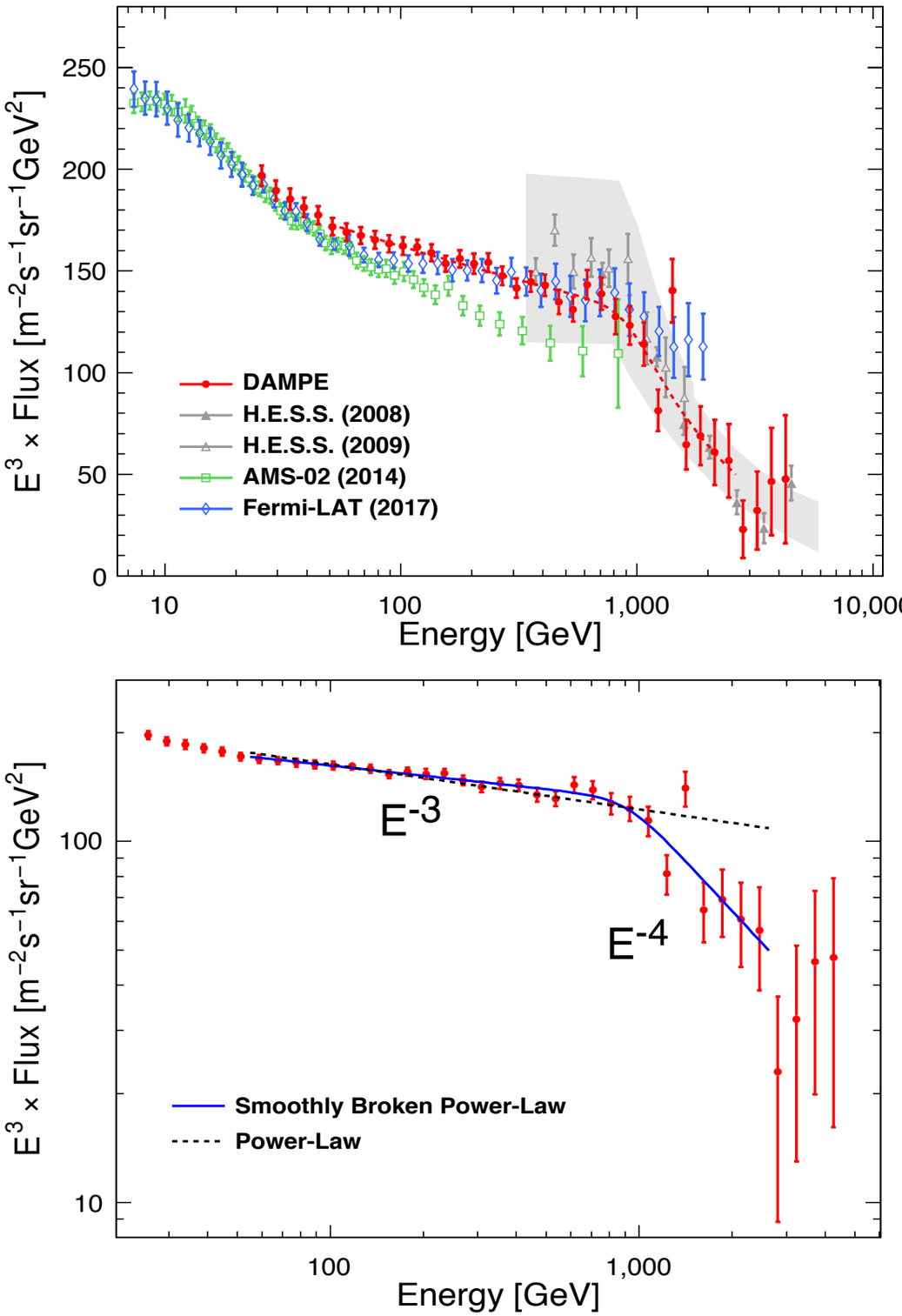
AMS-02 Collaboration, 2013, PRL, 110, 141102



- ✦ Observed by PAMELA, AMS-02, Fermi, ..
- ✦ Continuous rise above ~ 10 GeV
- ✦ Theoretically expected to decrease as ~ $E^{-0.33}$
- ✦ Rise due to increase in the e+ flux.
- ✦ Could be due to pulsars, e+ production inside sources, dark matter …..

Introduction: Cosmic-ray spectrum

All-electron spectrum (up to higher energies)

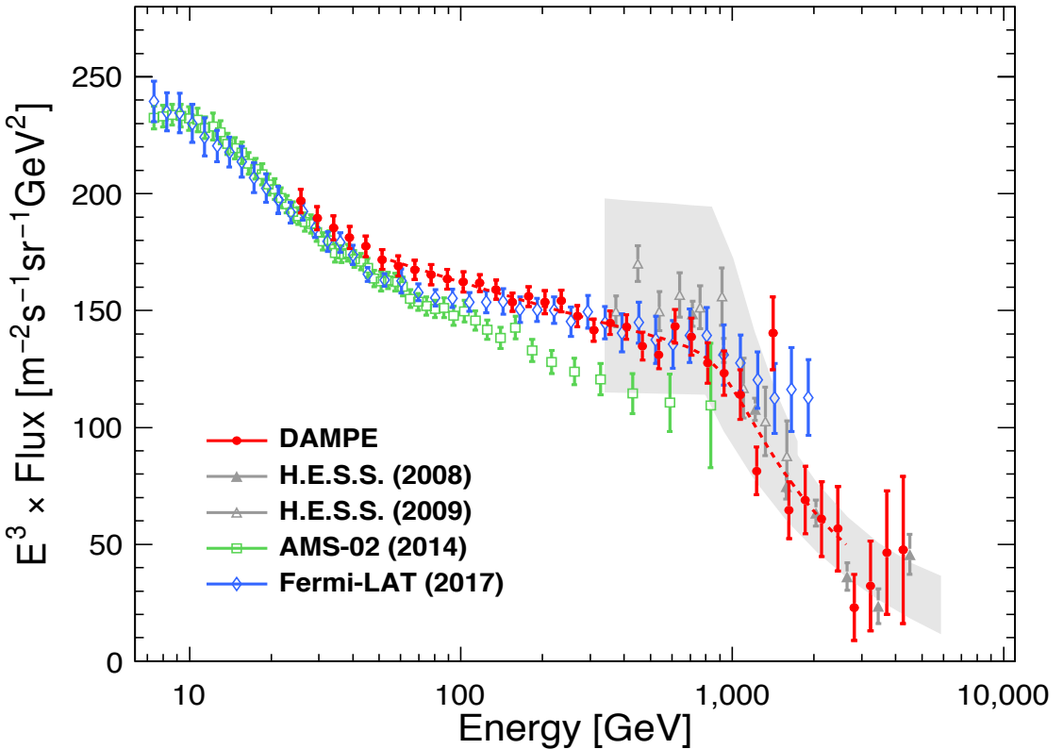


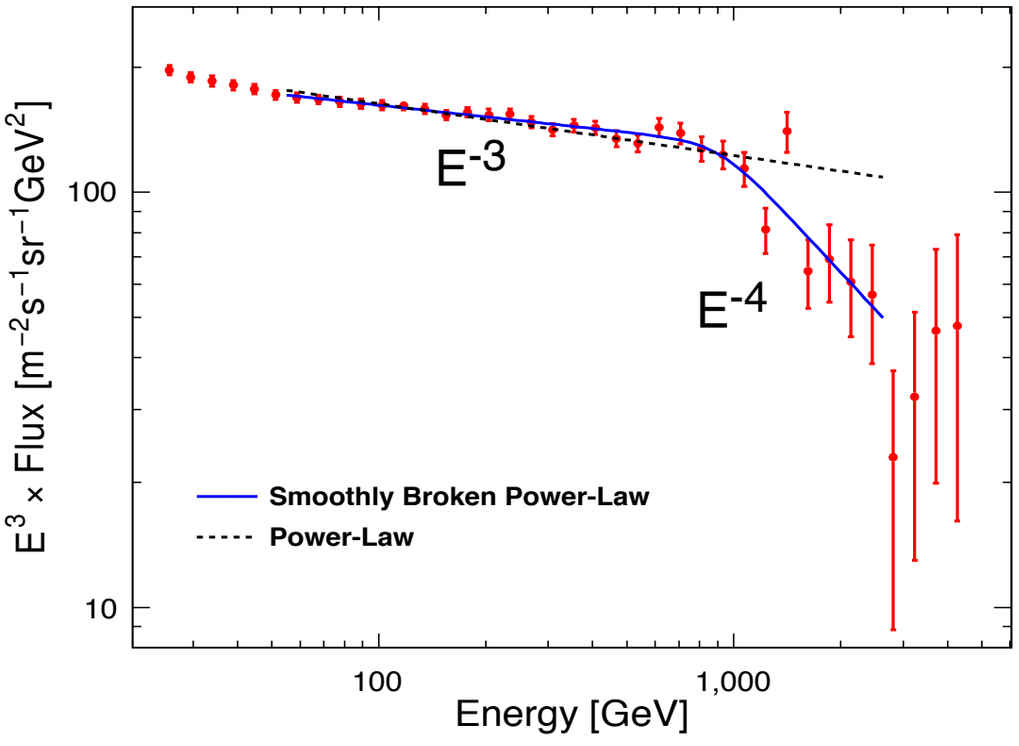
- ✦ Observed by H.E.S.S., CALET and DAMPE.
- ✦ Spectral index:
  - $\Gamma \approx -3$, below break.
  - $\Gamma \approx -4$, above break.
- ✦ Unlikely to be propagation effect in the Galaxy.
- ✦ Unlikely due to cut-off in the positron flux.

DAMPE Collaboration, 2017, Nature, 552, 63

Introduction: Cosmic-ray spectrum

All-electron spectrum (up to higher energies)



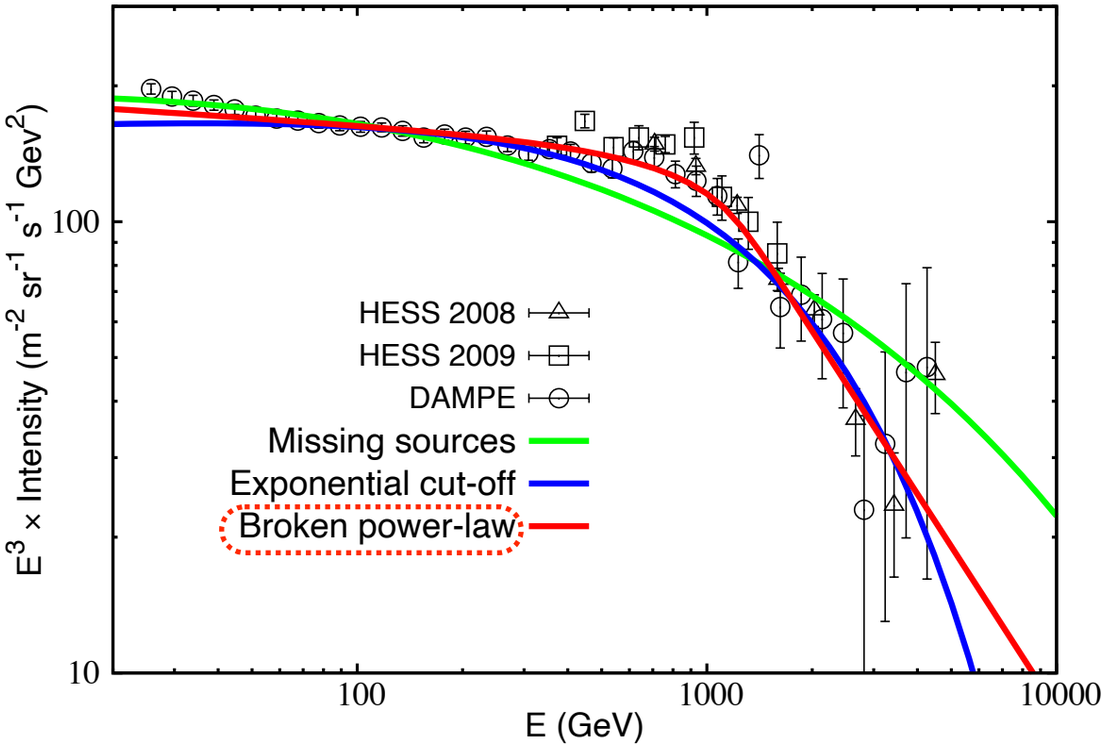


DAMPE Collaboration, 2017, Nature, 552, 63

- ✦ Observed by H.E.S.S., CALET and DAMPE.
- ✦ Spectral index:
  - $\Gamma \approx -3$, below break.
  - $\Gamma \approx -4$, above break.
- ✦ Unlikely to be propagation effect in the Galaxy.
- ✦ Unlikely due to cut-off in the positron flux.

Possible astrophysical origin

- ✦ Missing nearby sources
- ✦ Exponential cut-off in the source spectrum
- ✦ **Broken power-law source spectrum**



Thoudam & Hörandel, Mon. Not. R. Astron. Soc. 414, 1432–1438 (2011)

Introduction: Cosmic-ray arrival direction (anisotropy)

* Arrival direction of cosmic rays at the Earth is expected to be highly isotropic because of the diffusive nature of their propagation through the Galaxy.

* Below ~ 10 GeV, cosmic rays are highly Solar modulated. So, they are not expected to preserve their direction while entering the Solar system. Only high energy cosmic rays can penetrate to the vicinity of the Earth undeflected by the magnetic field in the interplanetary medium.

Introduction: Cosmic-ray arrival direction (anisotropy)

* Arrival direction of cosmic rays at the Earth is expected to be highly isotropic because of the diffusive nature of their propagation through the Galaxy.

* Below ~ 10 GeV, cosmic rays are highly Solar modulated. So, they are not expected to preserve their direction while entering the Solar system. Only high energy cosmic rays can penetrate to the vicinity of the Earth undeflected by the magnetic field in the interplanetary medium.

For a relativistic proton, the gyroradius is

$$r_g = 3 \times 10^9 \gamma (B/10^{-9}\,\mathrm{T})\,\mathrm{m}$$

where the magnetic flux density B is measured in tesla and $\gamma = (1 - v^2/c^2)^{-1/2}$ is the Lorentz factor (Sect. 7.1). Adopting the local value of the magnetic flux density in the interplanetary medium $B = 10^{-9}$ T, relativistic protons with $\gamma = 10^3$, that is, energies of 10^{12} eV, have gyroradii which are 3×10^{12} = 20 AU, that is, 20 times the distance from the Sun to the Earth, corresponding roughly to the radius of the orbit of the planet Uranus about the Sun. Thus, particles with these energies and greater are likely to preserve information about their arrival directions at the Solar System when they arrive at the top of the Earth's atmosphere.

High Energy Astrophysics, Longair: Page 493-520

Introduction: Cosmic-ray arrival direction (anisotropy)

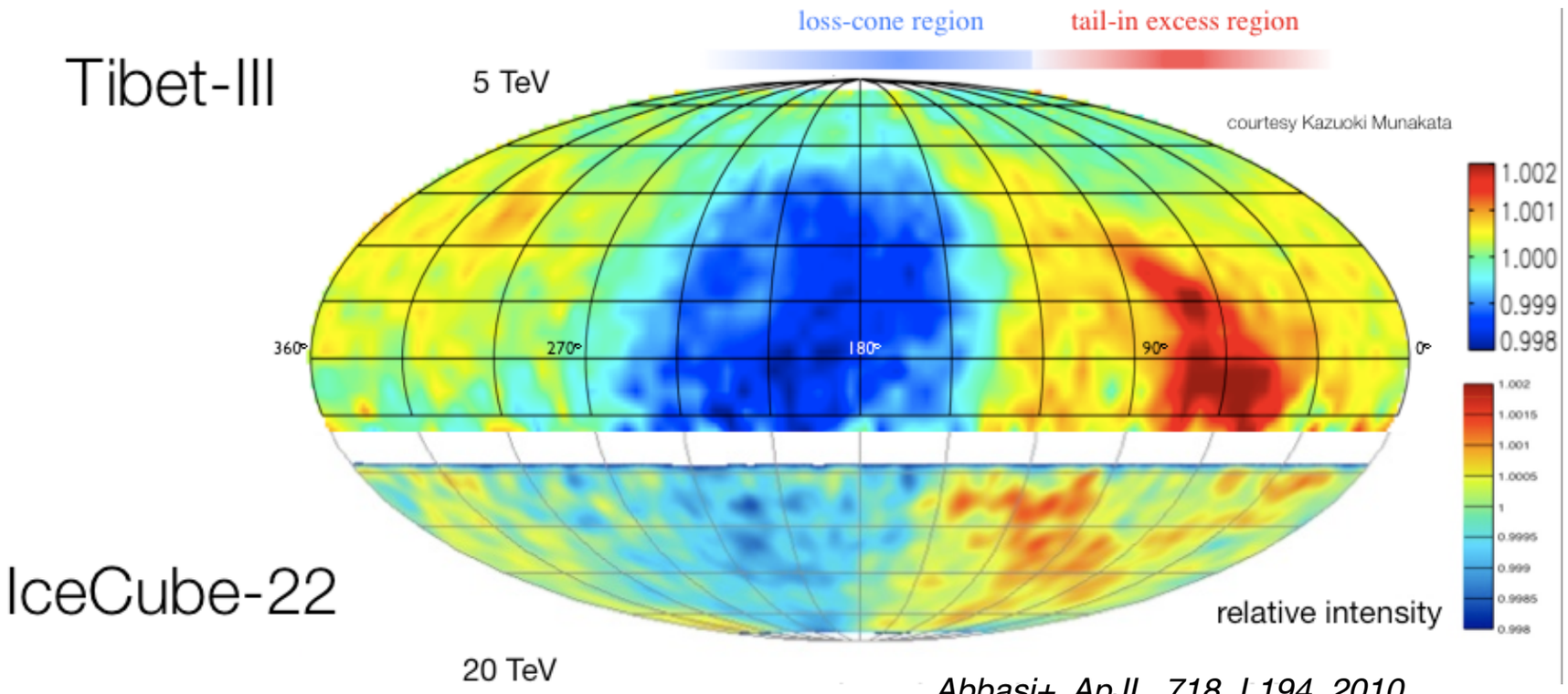
* Observations shows some level of anisotropy in the arrival direction of cosmic rays.
* Anisotropy "A" is usually quantified in terms of dipole anisotropy as follows:

$$I(\theta) = I_0 + I_0 A \cos\theta$$

where I_0 is averaged intensity,
$\theta = 0$ corresponds to the direction in which the intensity is maximum.

* Measurements have found anisotropy in the range of $A \sim 10^{-4}$ to 10^{-3}

Cosmic-ray angular distribution (equatorial coordinates)

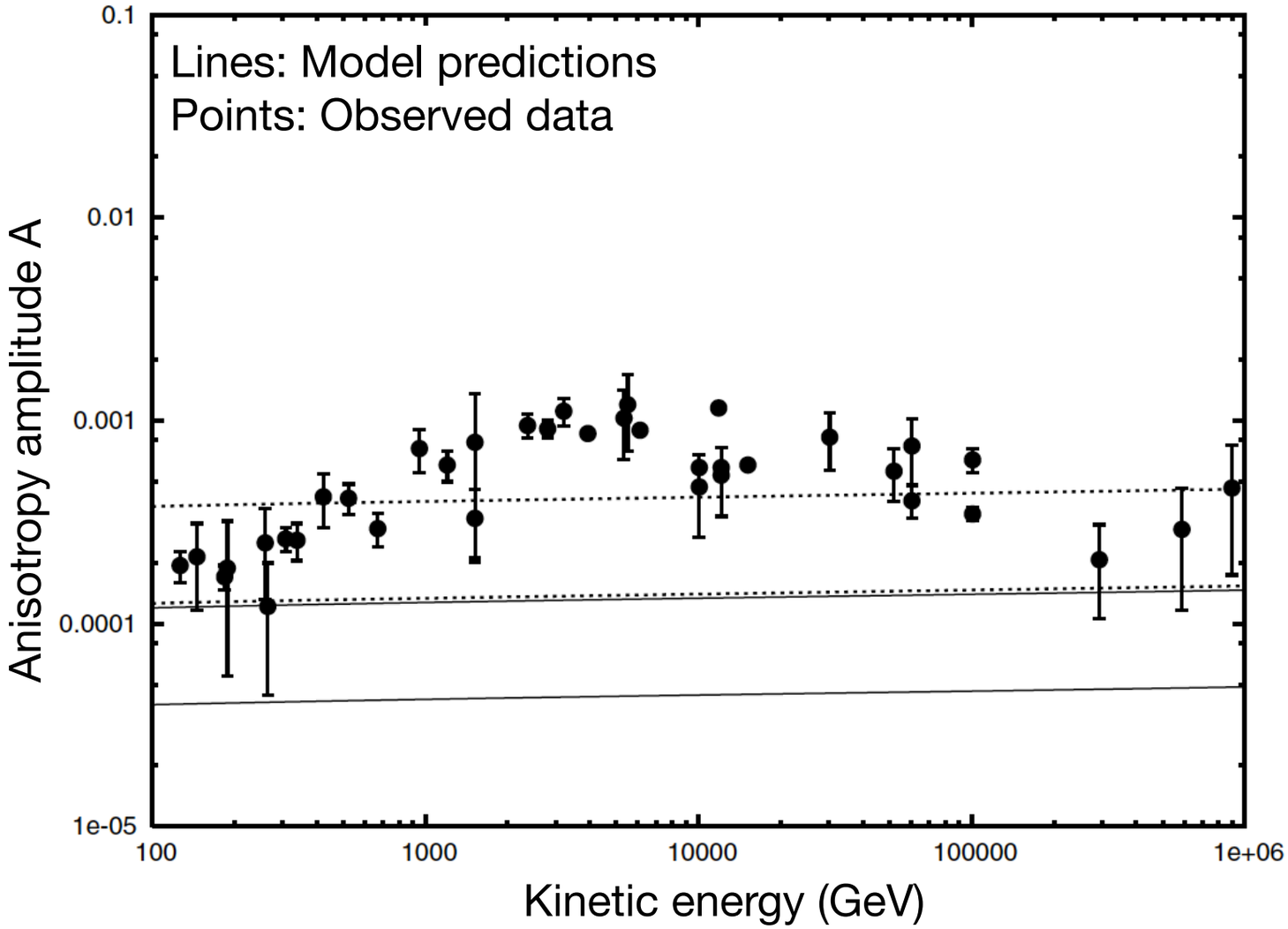


Abbasi+, ApJL, 718, L194, 2010

High Energy Astrophysics, Longair: Page 493-520

Introduction: Cosmic-ray arrival direction (anisotropy)

Observed anisotropy as a function of cosmic-ray energy

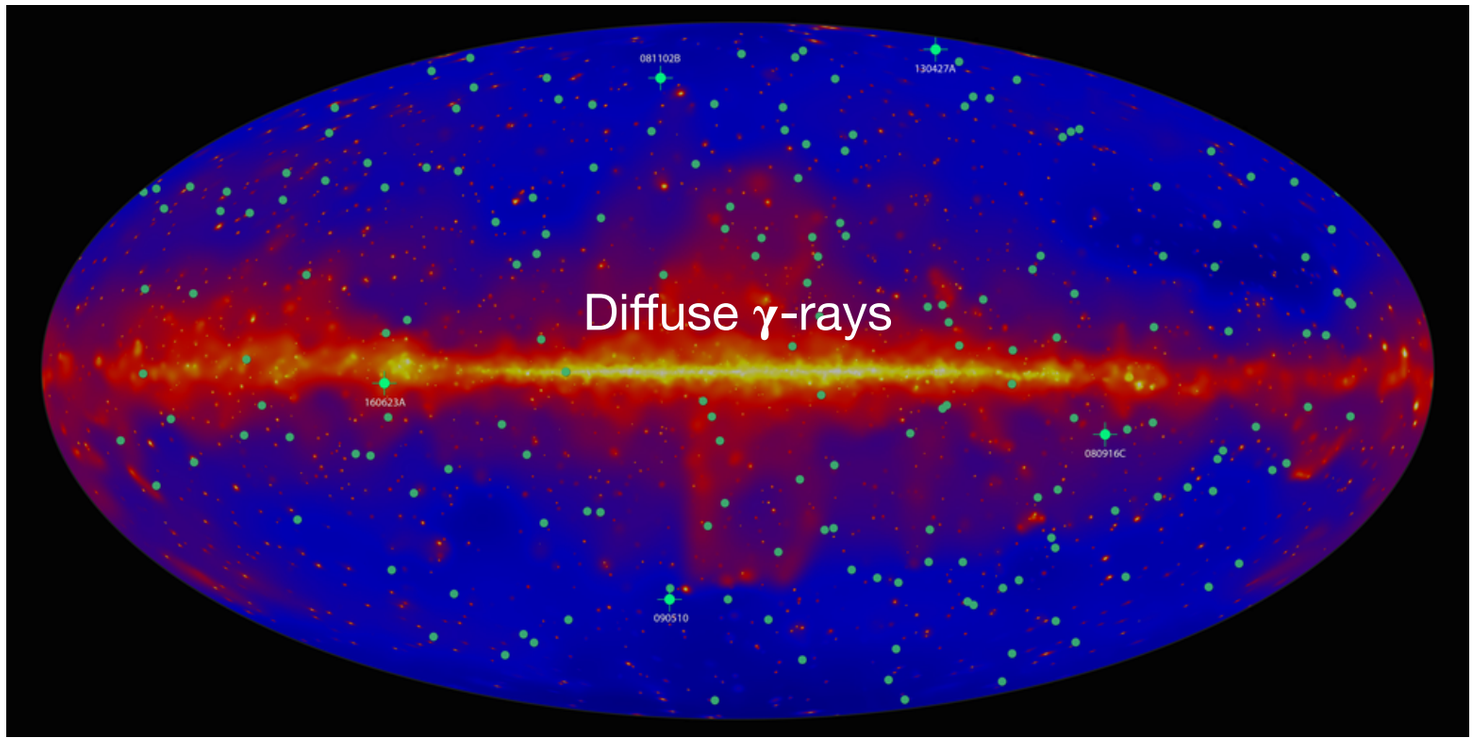


- ✦ **Cause of Isotropy:** Due to diffusive propagation.
- ✦ **Cause of Anisotropy:** Nearby sources, non-uniform source distribution, magnetic field structure, ….

Thoudam, S., Mon. Not. R. Astron. Soc., 2007, 378, 48

Introduction: Cosmic-ray distribution

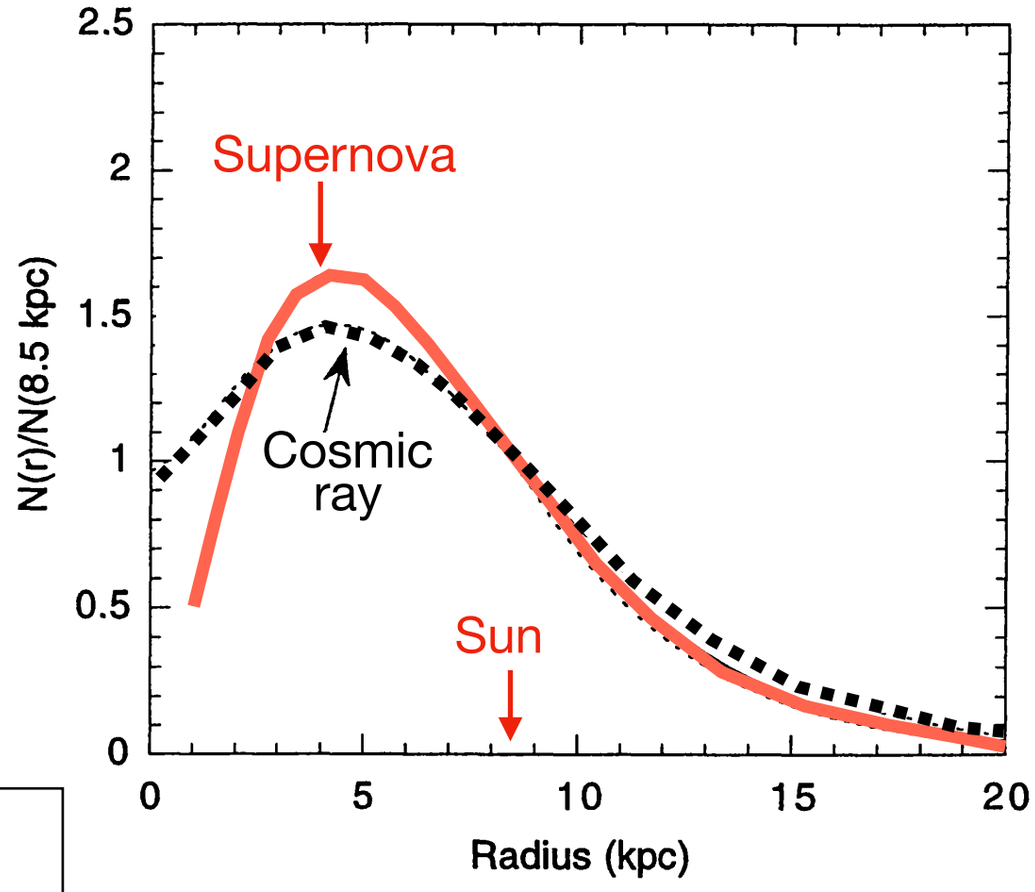
Fermi γ-ray sky map (>10 GeV)



Credit: NASA/Fermi Collaboration/ESA

- Gamma-ray observations indicate presence of cosmic rays through out the Galactic disk.
- Tension between the radial distributions of cosmic rays and that of the supernova remnants in the Galaxy.
- Note that such studies assume gamma rays to be produced mainly from proton-proton interaction (π^0-decay gamma rays).

Radial distribution of supernova remnants and cosmic rays in the Galactic disk



Case & Bhattacharya, 1996, A&A, 120, 437

Introduction: Galactic cosmic-ray sources

Theoretical argument in favor of cosmic-ray acceleration inside supernova remnants

✣ **Minimum power requirement for a source in the Galaxy:**

✦ **Galactic CR luminosity =** $\frac{V_{gal} W_{CR}}{T_{CR}}$ **= 10^{41} erg/s**

V_{gal}: Galactic volume ~ 1200 kpc^3

W_{CR}: CR energy density=1 eV/cm^3

T_{CR}: CR confinement time ~10 Million Years

✣ **Supernova remnants are the most plausible sources of cosmic rays in the Galaxy:**

✦ **Luminosity of supernova =** E_{SN} R_{SN} **= 10^{42} erg/s**

E_{SN}: Kinetic energy from a supernova explosion

~ 10^{51} ergs

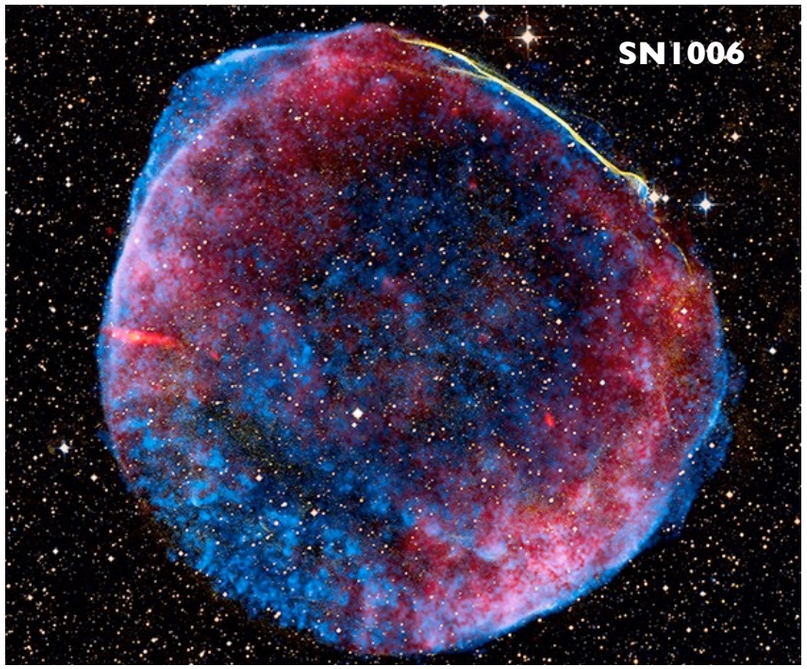
R_{SN}: Rate of supernova ~ 1/30 yr

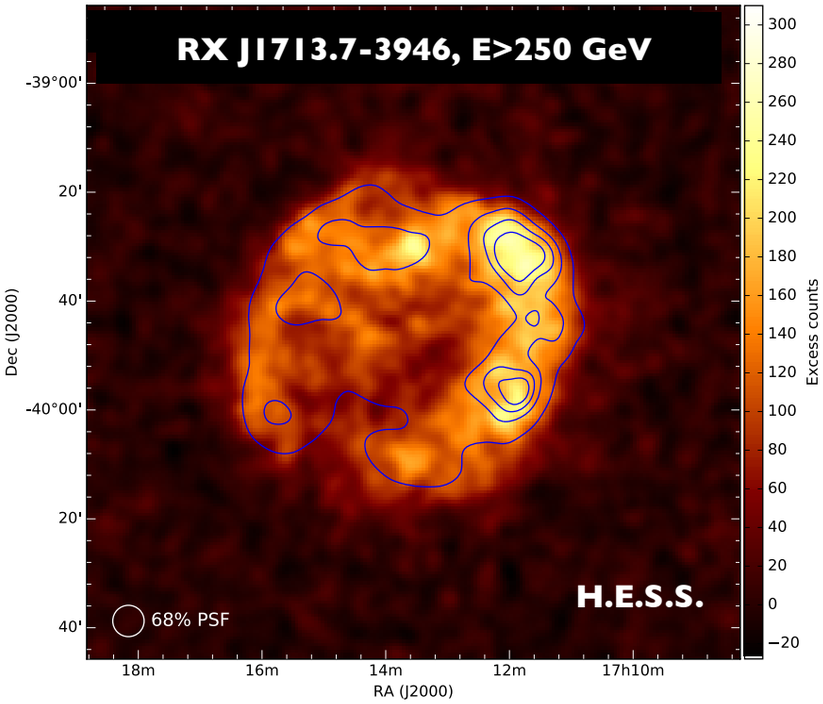
✦ **10% of SN kinetic energy —> Cosmic rays**

Introduction: Galactic cosmic-ray sources

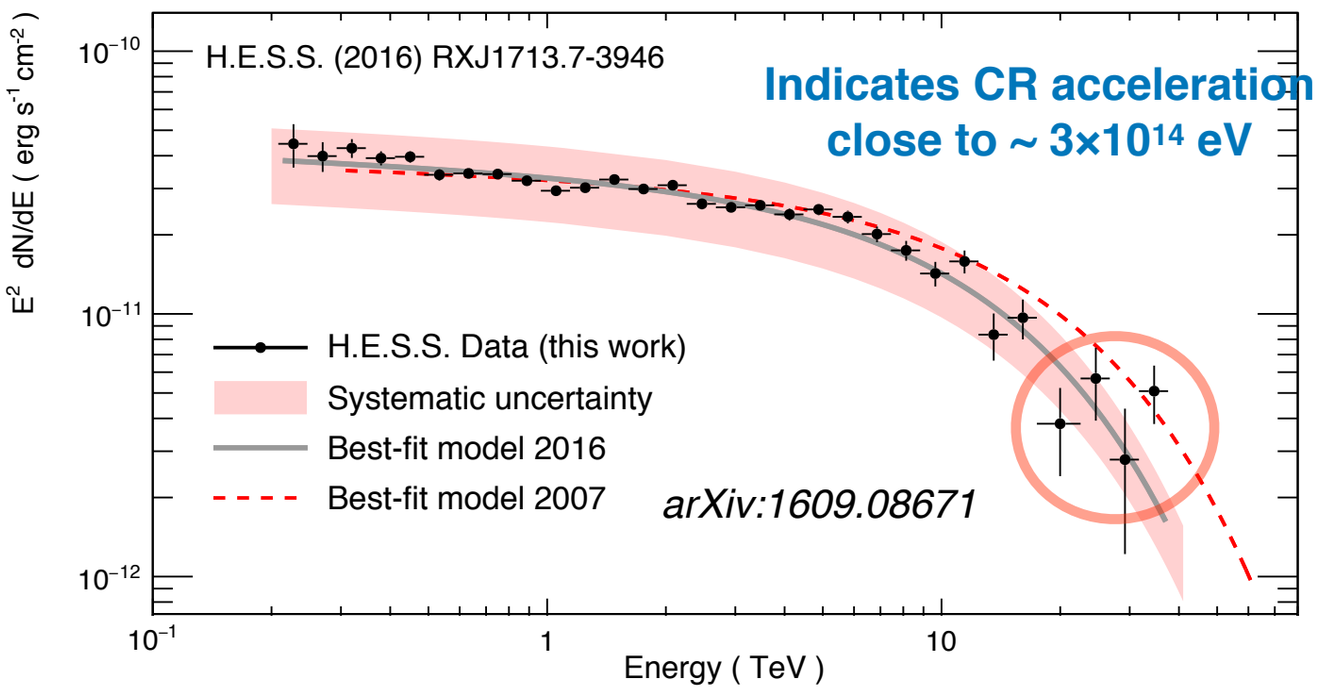
Observational evidence of cosmic-ray acceleration inside supernova remnants

Non-thermal radiation from supernova remnants





- Detection of non-thermal emissions such as X-rays, GeV-TeV gamma rays from supernova remnants.
- This indicates the presence of high-energy particles.
- But, they do not confirm hadronic acceleration!



Introduction: Galactic cosmic-ray sources

Observational evidence of cosmic-ray acceleration inside supernova remnants

First detection of π^0-decay bump?

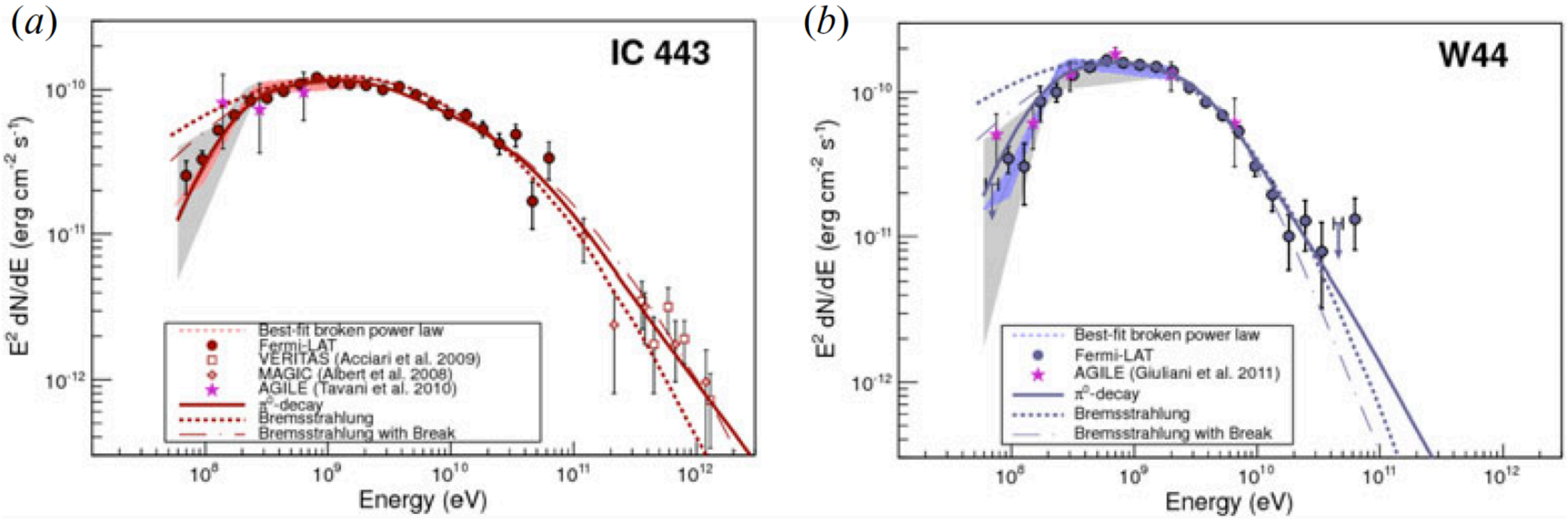


FIGURE 3. Gamma-ray spectra of IC443 (*a*) and W44 (*b*) as measured with the Fermi LAT (Abdo *et al.* 2010*a*), in blue, and AGILE (Giuliani *et al.* 2011), in magenta. The pion bump feature at about 1 GeV is evident in the spectra of both SNRs. For IC433 the data points of MAGIC (Albert *et al.* 2007*b*) and VERITAS (Acciari *et al.* 2009) are also shown. Colour-shaded areas denote the best-fit broadband smooth broken power law (60 MeV to 2 GeV); grey-shaded bands show Fermi LAT systematic errors below 2 GeV. The data points at the highest energy (4×10^{10}–1×10^{11} eV) suggest a hardening of the spectrum, which might be produced by runaway CRs illuminating a molecular cloud (MC) located in front of the shells.

Amato & Casanova, J. Plasma Phys. (2021), vol. 87, 845870101

Introduction: Maximum energy of cosmic rays from a source

Magnetic confinement within a source region

For *confinement* in a source with linear dimension L (I avoid the symbol r_g as it is often used for the gyro-radius) the gyration radius r_B should be less than the source dimension:

$$r_B \simeq \frac{E}{ZeB} < L \iff E < ZeB\,L\,.$$

From private communication with A. Achterberg, 2016

Introduction: Maximum energy of cosmic rays from a source

Magnetic confinement within a source region

For *confinement* in a source with linear dimension L (I avoid the symbol r_g as it is often used for the gyro-radius) the gyration radius r_B should be less than the source dimension:

$$r_B \simeq \frac{E}{ZeB} < L \iff E < ZeB\,L\,.$$

For *acceleration*, on the other hand, you get a different limit:

$$E < ZeB\,\beta\,L\,,$$

with $\beta = V/c$ and V the typical velocity associated with the accelerating agent. In your application $\beta \ll 1$!

This is called Hillas limit for acceleration. Derivation on the next slide.

From private communication with A. Achterberg, 2016

Introduction: Maximum energy of cosmic rays from a source

Let me derive this second limit for shock acceleration. If particles with energy E diffuse with diffusion coefficient $D(E)$, and the flow speed into the shock is V (= the shock speed if it expands into a stationary medium), the accelerated particles form a precursor with a scale equal to the diffusion length,

$$L \simeq \frac{D(E)}{V} = \frac{D(E)}{V_s} ,$$

with V_s the shock speed. If the precursor scale becomes comparable to the shock radius R_s (assuming a spherical shock) particles will not always encounter the shock again: they can miss it while diffusing in the flow. The maximum energy therefore follows from:

$$D(E) < V_s R_s .$$

If the particles diffuse at the Bohm rate (scattering mean-free-path ~ gyration radius) we have $D(E) = D_B \sim c r_B/3 = cE/3ZeB$. Putting this into the above condition yields:

$$\boxed{E < 3ZeB \left(\frac{V_s}{c}\right) R_s .}$$

From private communication with A. Achterberg, 2016

Introduction: Maximum energy of cosmic rays from a source

Analysis of the maximum energy based on Hillas condition

✣ **Hillas limit**

- ✦ $E_{max} \sim \beta ZBL$

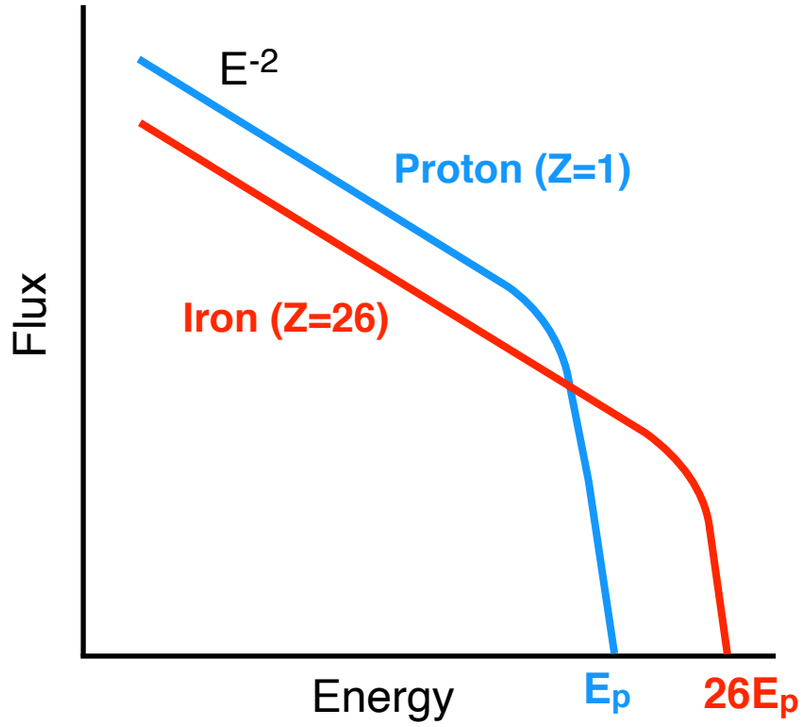
  β=u/c

  u: Shock velocity

  Z: Charge

  B: Magnetic field

  L: Size of the acceleration region



Hillas, 1984, ARAA, 22, 425

Introduction: Maximum energy of cosmic rays from a source

Analysis of the maximum energy based on Hillas condition

✣ **Hillas limit**

- ✦ **E_{max}~ βZBL**

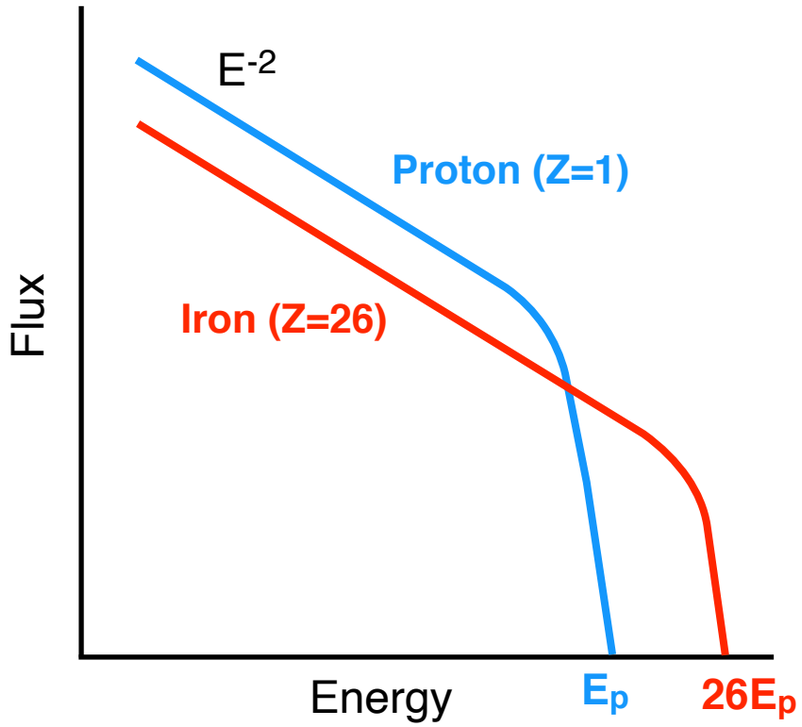
  β=u/c

  u: Shock velocity

  Z: Charge

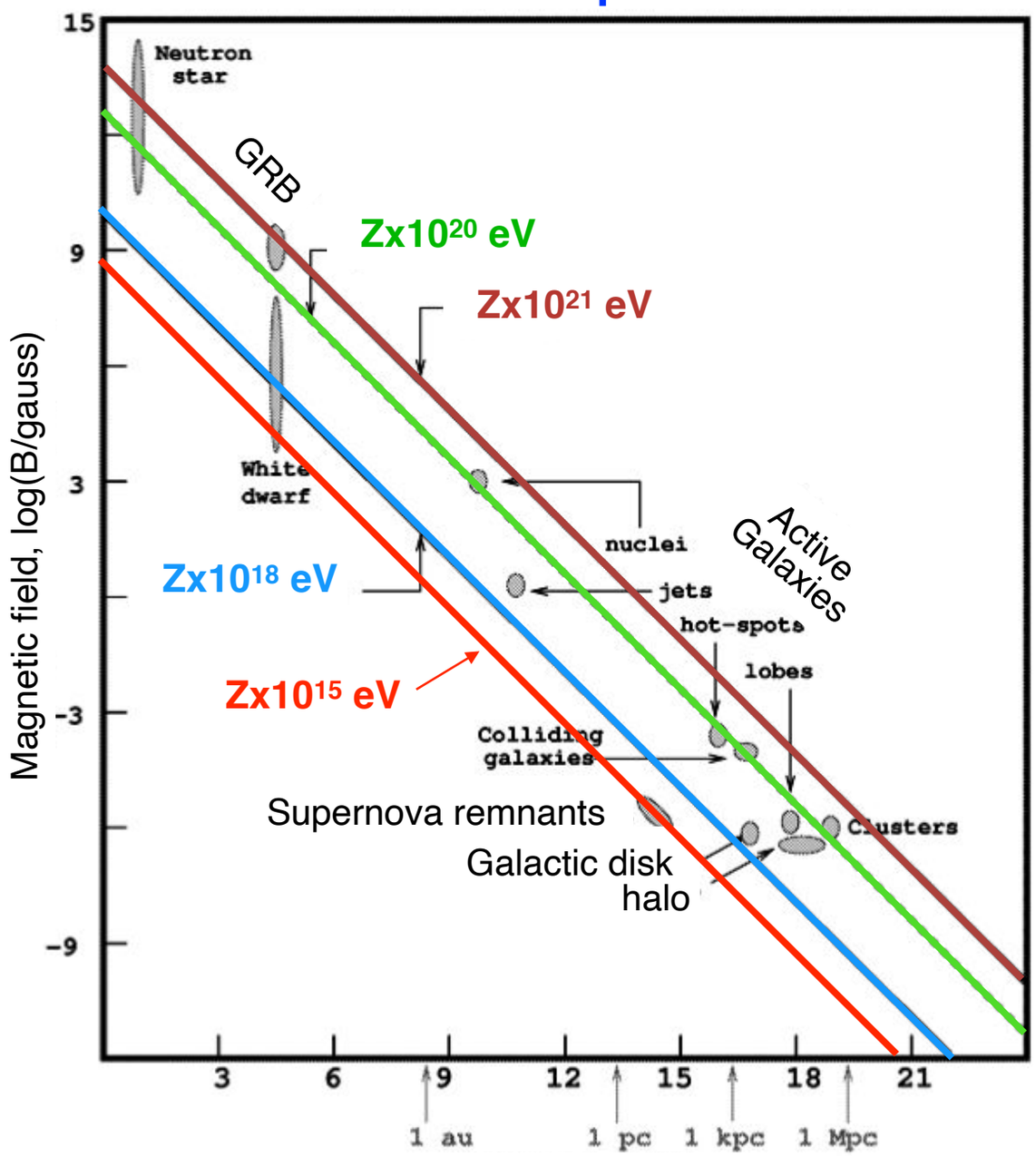
  B: Magnetic field

  L: Size of the acceleration region

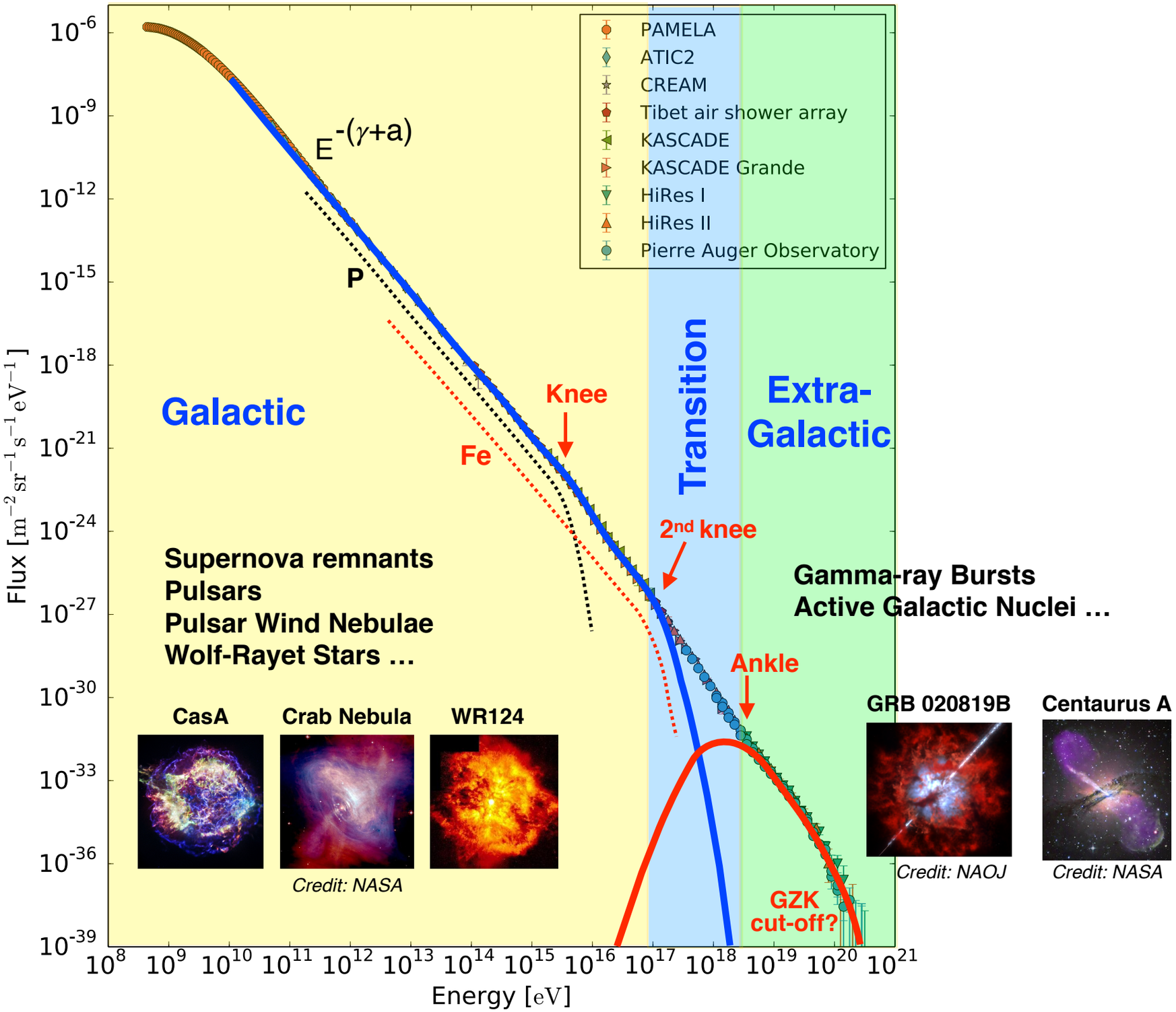


Hillas, 1984, ARAA, 22, 425

Hillas plot



Origin of Cosmic rays: The general picture



Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

For cosmic-ray primaries:

The simplest model of cosmic-ray propagation is the leaky box model (see [70] for a review). Under this model, our Galaxy is considered as a closed-box with semi-reflecting walls which confine cosmic rays for some time before they escape into intergalactic space. The spatial distribution of cosmic rays and their sources are all assumed to be uniform in this model. Under stationary condition, the equilibrium density of cosmic rays in this model is given by the following balance equation,

$$\frac{N_p}{T_e} + \frac{N_p}{T_p} = Q \qquad (1)$$

where N_p represents the cosmic-ray density, T_e is the escape (also confinement) time from the Galaxy, T_p is the spallation timescale and Q is the source injection rate.

Thoudam, PhD thesis (2012), https://repository.ubn.ru.nl/bitstream/handle/2066/93568/93568.pdf, Pages 4-6

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

For cosmic-ray secondaries:

Source term can be written as $Q = nc\sigma N_p$, where n is the average matter density in the Galaxy, c is the velocity of light and N_p is given by Eq. (1). The equilibrium density for the secondaries N_s is given by,

$$\frac{N_s}{T_e} + \frac{N_s}{T_s} = nc\sigma N_p \tag{2}$$

where T_s is the spallation timescale. From Eq. (2), we can write,

$$\boxed{\frac{N_s}{N_p} \approx nc\sigma T_e} \tag{3}$$

Eq. (3) is referred to as the secondary-to-primary ratio and it is a quantity that can be measured at the Earth. For example: Boron/carbon ratio.

Thoudam, PhD thesis (2012), https://repository.ubn.ru.nl/bitstream/handle/2066/93568/93568.pdf, Pages 4-6

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

Inference from secondary/primary ratio:

Total amount of matter traversed by cosmic-ray nuclei in the Galaxy

$$\frac{N_s}{N_p} \approx nc\sigma T_e \tag{3}$$

N_s/N_p is given by measurements,

σ is known from particle physics measurements,

c is the velocity of light,

=> ncT_e can be obtained

Then, it is possible to determine the total amount of matter traversed by cosmic rays in the Galaxy which is given by $X = nmcT_e$, where m is the mass of a proton. X is measured to be ~ 5 g/cm^2 at ~ 1 GeV/n.

Thoudam, PhD thesis (2012), https://repository.ubn.ru.nl/bitstream/handle/2066/93568/93568.pdf, Pages 4-6

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

Inference from secondary/primary ratio:

Average number density of matter traversed by cosmic-ray nuclei in the Galaxy

$$\frac{N_s}{N_p} \approx nc\sigma T_e \qquad (3)$$

N_s/N_p is given by measurements,

σ is known from particle physics measurements,

c is the velocity of light,

Also, if the cosmic-ray escape (confinement) time T_e is known, the average number density of matter n traversed by cosmic rays in the Galaxy can be determined.

Thoudam, PhD thesis (2012), https://repository.ubn.ru.nl/bitstream/handle/2066/93568/93568.pdf, Pages 4-6

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

Determination of cosmic-ray confinement time in the Galaxy:

Measurements of abundance ratios of radioactive secondaries to the stable secondaries give an estimate of the confinement time of cosmic rays in the Galaxy

* Spallation of C and O in the interstellar medium can produce Beryllium isotopes: ^{10}Be and ^{9}Be.
* ^{10}Be is radioactive (half-life about two million years), but ^{9}Be is stable.

High Energy Astrophysics, Longair: Page 516-517

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

Direct measurement of cosmic-ray confinement time in the Galaxy:

* The steady state abundance for stable secondary (^{9}Be) is given by

$$-\frac{N_i}{\tau_e(i)} + C_i - \frac{N_i}{\tau_{spal}(i)} = 0; \qquad N_i = \frac{C_i}{\frac{1}{\tau_e(i)} + \frac{1}{\tau_{spal}(i)}},$$

$$C_i = \sum_p n v_p \sigma_{pi} N_p$$

is the production rate of the secondary

where $\tau_{spal}(i)$ is the time-scale over which the isotope i is destroyed by inelastic collisions.

High Energy Astrophysics, Longair: Page 516-517

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

Direct measurement of cosmic-ray confinement time in the Galaxy:

* The steady state abundance for stable secondary (^{9}Be) is given by

$$-\frac{N_i}{\tau_{\rm e}(i)} + C_i - \frac{N_i}{\tau_{\rm spal}(i)} = 0; \qquad N_i = \frac{C_i}{\frac{1}{\tau_{\rm e}(i)} + \frac{1}{\tau_{\rm spal}(i)}} ,$$

$$C_i = \sum_p n v_p \sigma_{pi}\, N_p$$

is the production rate of the secondary

where $\tau_{\rm spal}(i)$ is the time-scale over which the isotope i is destroyed by inelastic collisions.

* For radioactive secondary (^{10}Be) of decay time $\tau_{\rm r}$:

$$-\frac{N_j}{\tau_{\rm e}(j)} + C_j - \frac{N_j}{\tau_{\rm spal}(j)} - \frac{N_j}{\tau_{\rm r}(j)} = 0; \qquad N_j = \frac{C_j}{\frac{1}{\tau_{\rm e}(j)} + \frac{1}{\tau_{\rm r}(j)} + \frac{1}{\tau_{\rm spal}(j)}} .$$

High Energy Astrophysics, Longair: Page 516-517

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

Direct measurement of cosmic-ray confinement time in the Galaxy:

The steady-state ratio of ^{10}Be and ^{9}Be isotopes is therefore

$$\frac{N(^{10}\mathrm{Be})}{N(^{9}\mathrm{Be})} = \frac{\dfrac{1}{\tau_{\mathrm{e}}(^{9}\mathrm{Be})} + \dfrac{1}{\tau_{\mathrm{spal}}(^{9}\mathrm{Be})}}{\dfrac{1}{\tau_{\mathrm{e}}(^{10}\mathrm{Be})} + \dfrac{1}{\tau_{\mathrm{r}}(^{10}\mathrm{Be})} + \dfrac{1}{\tau_{\mathrm{spal}}(^{10}\mathrm{Be})}} \frac{C(^{10}\mathrm{Be})}{C(^{9}\mathrm{Be})} .$$

High Energy Astrophysics, Longair: Page 516-517

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

Direct measurement of cosmic-ray confinement time in the Galaxy:

The steady-state ratio of ^{10}Be and ^{9}Be isotopes is therefore

$$\frac{N(^{10}\mathrm{Be})}{N(^{9}\mathrm{Be})} = \frac{\dfrac{1}{\tau_{\mathrm{e}}(^{9}\mathrm{Be})} + \dfrac{1}{\tau_{\mathrm{spal}}(^{9}\mathrm{Be})}}{\dfrac{1}{\tau_{\mathrm{e}}(^{10}\mathrm{Be})} + \dfrac{1}{\tau_{\mathrm{r}}(^{10}\mathrm{Be})} + \dfrac{1}{\tau_{\mathrm{spal}}(^{10}\mathrm{Be})}} \frac{C(^{10}\mathrm{Be})}{C(^{9}\mathrm{Be})} .$$

If the time-scale for the destruction of the beryllium isotopes by spallation is much greater than their escape times $\tau_{\mathrm{spal}} \gg \tau_{\mathrm{e}}$, we obtain a simpler expression,

$$\frac{N(^{10}\mathrm{Be})}{N(^{9}\mathrm{Be})} = \frac{\dfrac{1}{\tau_{\mathrm{e}}(^{9}\mathrm{Be})}}{\dfrac{1}{\tau_{\mathrm{e}}(^{10}\mathrm{Be})} + \dfrac{1}{\tau_{\mathrm{r}}(^{10}\mathrm{Be})}} \frac{C(^{10}\mathrm{Be})}{C(^{9}\mathrm{Be})} . \qquad (15.25)$$

High Energy Astrophysics, Longair: Page 516-517

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

Direct measurement of cosmic-ray confinement time in the Galaxy:

$$\frac{N(^{10}\mathrm{Be})}{N(^{9}\mathrm{Be})} = \frac{\dfrac{1}{\tau_e(^{9}\mathrm{Be})}}{\dfrac{1}{\tau_e(^{10}\mathrm{Be})} + \dfrac{1}{\tau_r(^{10}\mathrm{Be})}} \frac{C(^{10}\mathrm{Be})}{C(^{9}\mathrm{Be})} . \qquad (15.25)$$

- Measurements can give the ratio $N(^{10}\mathrm{Be})/N(^{9}\mathrm{Be})$.
- The production ratio $C(^{10}\mathrm{Be})/C(^{9}\mathrm{Be})$ is known from particle physics measurements.
- The ^{10}Be decay time $\tau_r(^{10}\mathrm{Be})$ = 2 Million years.
- Then, the escape time $\tau_e(^{10}\mathrm{Be})$ from the galaxy can be calculated.

High Energy Astrophysics, Longair: Page 516-517

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model

Direct measurement of cosmic-ray confinement time in the Galaxy:

$$\frac{N(^{10}\mathrm{Be})}{N(^{9}\mathrm{Be})} = \frac{\dfrac{1}{\tau_e(^{9}\mathrm{Be})}}{\dfrac{1}{\tau_e(^{10}\mathrm{Be})} + \dfrac{1}{\tau_r(^{10}\mathrm{Be})}} \frac{C(^{10}\mathrm{Be})}{C(^{9}\mathrm{Be})} . \qquad (15.25)$$

- Measurements can give the ratio $N(^{10}\mathrm{Be})/N(^{9}\mathrm{Be})$.
- The production ratio $C(^{10}\mathrm{Be})/C(^{9}\mathrm{Be})$ is known from particle physics measurements.
- The ^{10}Be decay time $\tau_r(^{10}\mathrm{Be})$ = 2 Million years.
- Then, the escape time $\tau_e(^{10}\mathrm{Be})$ from the galaxy can be calculated.

- Measurements give $N(^{10}\mathrm{Be})/N(^{9}\mathrm{Be})$ = 0.15 at energy about 0.5/nucleon.
- This gives an escape time from the galaxy of about 10^7 years.
- Note that it represents the escape time for all other nuclei at the same energy/nucleon (to be precise at the same rigidity/nucleon).

High Energy Astrophysics, Longair: Page 516-517

Introduction: Cosmic-ray propagation in the Galaxy

Once we know the escape time, the matter density traversed can be determined …

The amount of matter traversed by cosmic rays in the Galaxy is measured to be

$$X = nmcT_e \sim 5 \text{ g/cm}^2 \text{ at} \sim 1 \text{ GeV/n}$$

Measurements of abundance ratios of radioactive secondaries to the stable secondaries give an estimate of the confinement time of cosmic rays in the Galaxy to be $T_e \sim 1.7 \times 10^7$ yrs. This, along with the measured value of X, gives the averaged density of matter as $n \sim 0.2 \text{ cm}^{-3}$ [97]. This value of n is much less than the density of matter in the Galactic disk which is measured to be $\sim 1 \text{ cm}^{-3}$. This suggests that cosmic rays spend most of their lifetime in the Galactic halo where the matter density is as small as $\sim 0.01 \text{ cm}^{-3}$. In addition, if cosmic rays travel in straight paths with the speed of light, they would travel a distance of $\sim 5 \times 10^3$ kpc. To be compared, the vertical boundary of our Galaxy is only a few kpc say ~ 5 kpc and the radial boundary is $\sim (15 - 20)$ kpc. This suggests that cosmic rays do not travel in straight lines, they must undergo some sort of random motions in the Galaxy.

Thoudam, PhD thesis (2012), https://repository.ubn.ru.nl/bitstream/handle/2066/93568/93568.pdf, Pages 4-6

Introduction: Cosmic-ray propagation in the Galaxy

Leaky-Box model —> Diffusion model

It is now well understood that cosmic rays, being charged particles, can be deflected or scattered by the magnetic fields present in our Galaxy. Our Galactic magnetic field consists of both, a regular component and a random component. The random component can cause pitch angle scattering leading to diffusive motion of cosmic rays. The nature of the scattering is that of wave-particle resonant interactions [98]. A cosmic-ray particle interacts with magnetic inhomogeneities of wave number $k = 1/r_l$, where r_l is the Larmor radius of the particle.

You can read below if you want more information

The random magnetic field consists of a wide spectrum of inhomogeneities $W(k)$ with scale lengths $l = 2\pi/k$ appropriate to produce resonant scattering of cosmic rays with energies up to $\sim 10^{17}$ eV which corresponds to a scale length of ~ 100 pc. The spectrum peaks at large scale lengths and decreases at smaller lengths, which as a function of k can be represented as $W(k) \propto k^{-2+a}$ where a is a positive constant. The cosmic-ray diffusion coefficient D is related to the wave spectrum as $D \propto b(E/Z)^a$, where b is a constant which depends on the level of inhomogeneity. For a Kolmogorov-type of wave spectrum, $a = 1/3$ and for a Kraichnan-type wave spectrum, $a = 1/2$.

Thoudam, PhD thesis (2012), https://repository.ubn.ru.nl/bitstream/handle/2066/93568/93568.pdf, Pages 4-6

Introduction: Cosmic-ray propagation in the Galaxy

Diffusion model

In the diffusion model, Eq (1) from slide 35 can be written as:

$$\frac{N_p}{T_e} + \frac{N_p}{T_p} = Q \quad \Rightarrow \quad -\nabla \cdot (D \nabla N_p) + \frac{N_p}{T_p} = Q \qquad (4)$$

where the escape term in Eq. (1) is replaced by the diffusion term. $D(E)$ is diffusion coefficient.

Thoudam, PhD thesis (2012), https://repository.ubn.ru.nl/bitstream/handle/2066/93568/93568.pdf, Pages 4-6

Introduction: Cosmic-ray propagation in the Galaxy

Diffusion model

In the diffusion model, Eq (1) from slide 35 can be written as:

$$\frac{N_p}{T_e} + \frac{N_p}{T_p} = Q \quad \Rightarrow \quad -\nabla \cdot (D \nabla N_p) + \frac{N_p}{T_p} = Q \qquad (4)$$

where the escape term in Eq. (1) is replaced by the diffusion term. $D(E)$ is diffusion coefficient.

Similarly, Eq (2) for the secondaries can be written as:

$$\frac{N_s}{T_e} + \frac{N_s}{T_s} = nc\sigma N_p \quad \Rightarrow \quad -\nabla \cdot (D \nabla N_s) + \frac{N_s}{T_s} = nc\sigma N_p \qquad (5)$$

Thoudam, PhD thesis (2012), https://repository.ubn.ru.nl/bitstream/handle/2066/93568/93568.pdf, Pages 4-6

Introduction: Cosmic-ray propagation in the Galaxy

Diffusion model

In the diffusion model, Eq (1) from slide 35 can be written as:

$$\frac{N_p}{T_e} + \frac{N_p}{T_p} = Q \quad \Rightarrow \quad -\nabla \cdot (D \nabla N_p) + \frac{N_p}{T_p} = Q \qquad (4)$$

where the escape term in Eq. (1) is replaced by the diffusion term. $D(E)$ is diffusion coefficient.

Similarly, Eq (2) for the secondaries can be written as:

$$\frac{N_s}{T_e} + \frac{N_s}{T_s} = nc\sigma N_p \quad \Rightarrow \quad -\nabla \cdot (D \nabla N_s) + \frac{N_s}{T_s} = nc\sigma N_p \qquad (5)$$

From Eq. (5), it can be checked that the secondary-to-primary ratio under the diffusion model is given by

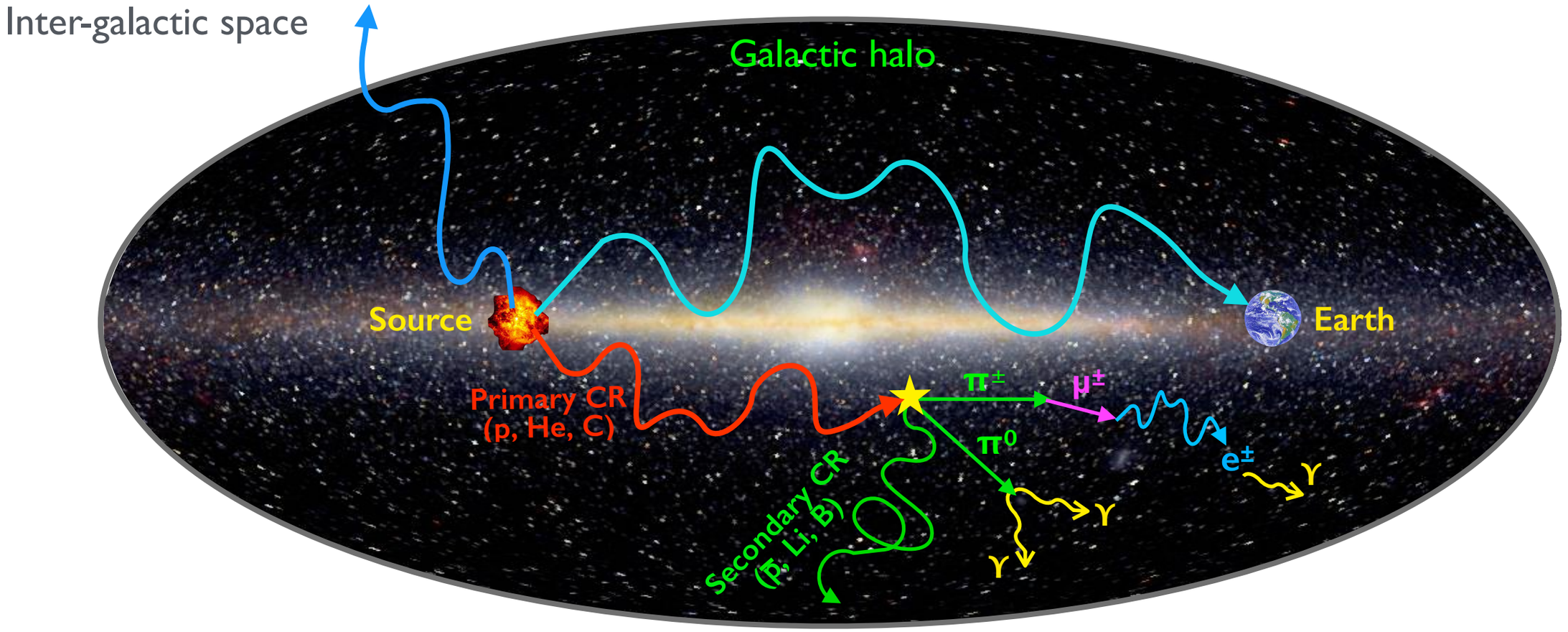
$$\frac{N_s}{N_p} \propto \frac{1}{D} \qquad (6)$$

Thus, measurements of secondary-to-primary ratios can give an estimate of the values of the diffusion coefficient. Compared to the Leaky box model, the diffusion model seems to be more realistic because it can describe a non-uniform distribution of cosmic rays in the Galaxy.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335
Thoudam, PhD thesis (2012), https://repository.ubn.ru.nl/bitstream/handle/2066/93568/93568.pdf, Pages 4-6

Diffusive propagation of cosmic rays in the Galaxy: In detail

A schematic of the propagation



- ✤ Particles scattered by magnetic field turbulence.
- ✤ Wave-particle interaction: Larmor radius, $R_L \sim \lambda$
- ✤ $\lambda \sim 10^{12}$-10^{20} cm <==> E~ 10^{9}-10^{17} eV
- ✤ Wave spectrum, $w(k) \propto k^{-d}$, $k \sim 1/\lambda$
- ✤ Diffusion coefficient, $D \propto E^{2-d}$
- ✤ For Kolmogorov-type spectrum: d=5/3, $D \propto E^{0.33}$
- ✤ For Kraichnan-type spectrum: d=3/2, $D \propto E^{0.5}$

- ✤ Hillas limit on confinement
  $E_{max} \sim ZBH$
- ✤ For Z=1, B=6 uG, H=2 kpc
  $E_{max} \sim Z \times 10^{19}$ eV

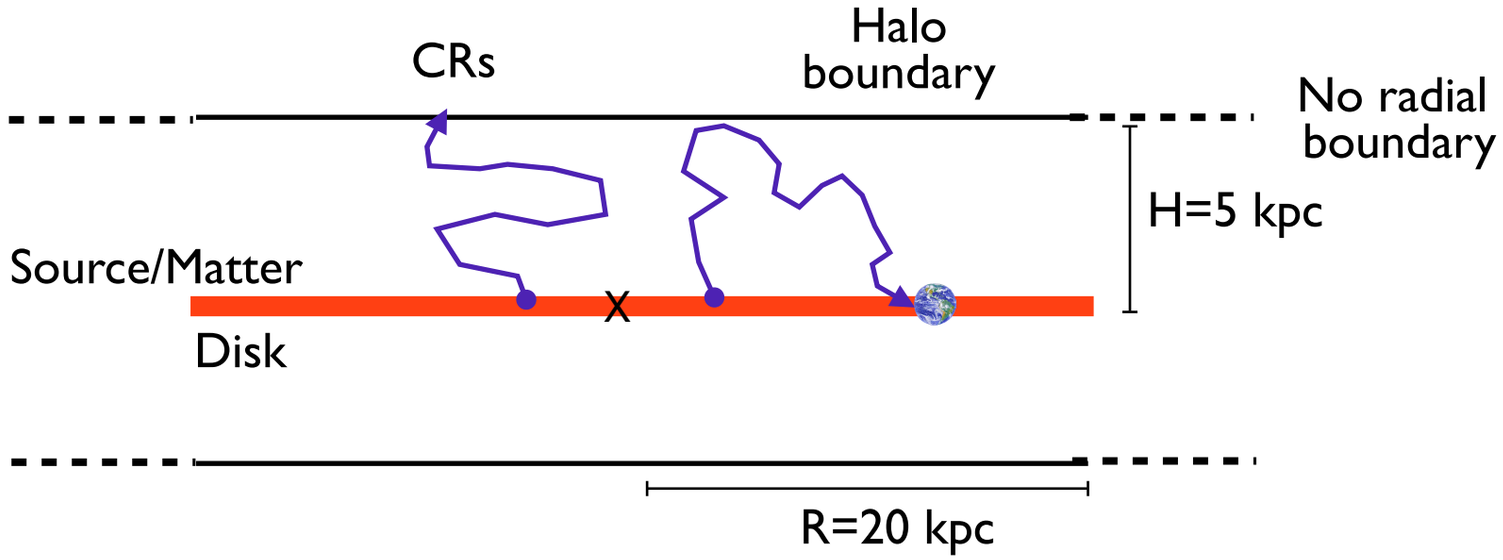
Power law

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR primary spectrum from a uniform distribution of sources

***We assume a cylindrical geometry for the propagation region with vertical boundaries and no radial boundary.**



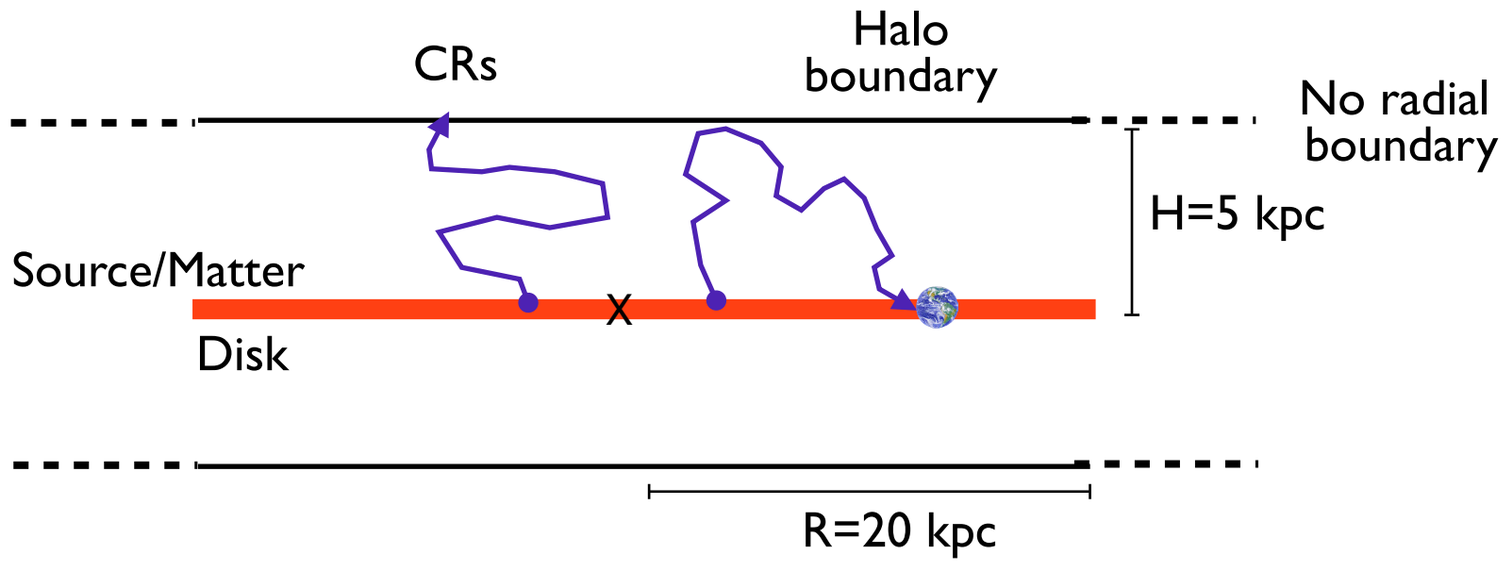
Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR primary spectrum from a uniform distribution of sources

***We assume a cylindrical geometry for the propagation region with vertical boundaries and no radial boundary.**



For cosmic rays from the bulk of sources in the Galaxy, the propagation of CR primaries can be described by a steady state transport equation:

$$\nabla \cdot (D_{\rm p} \nabla N_{\rm p}) - 2hn v_{\rm p} \sigma_{\rm p} \delta(z) N_{\rm p} = -S_{\rm p},$$

where $S_{\rm p}(\boldsymbol{r}, E) = \Re Q_{\rm p}(E)\delta(z)$ is the source term assumed to be extended up to the radial distance R.

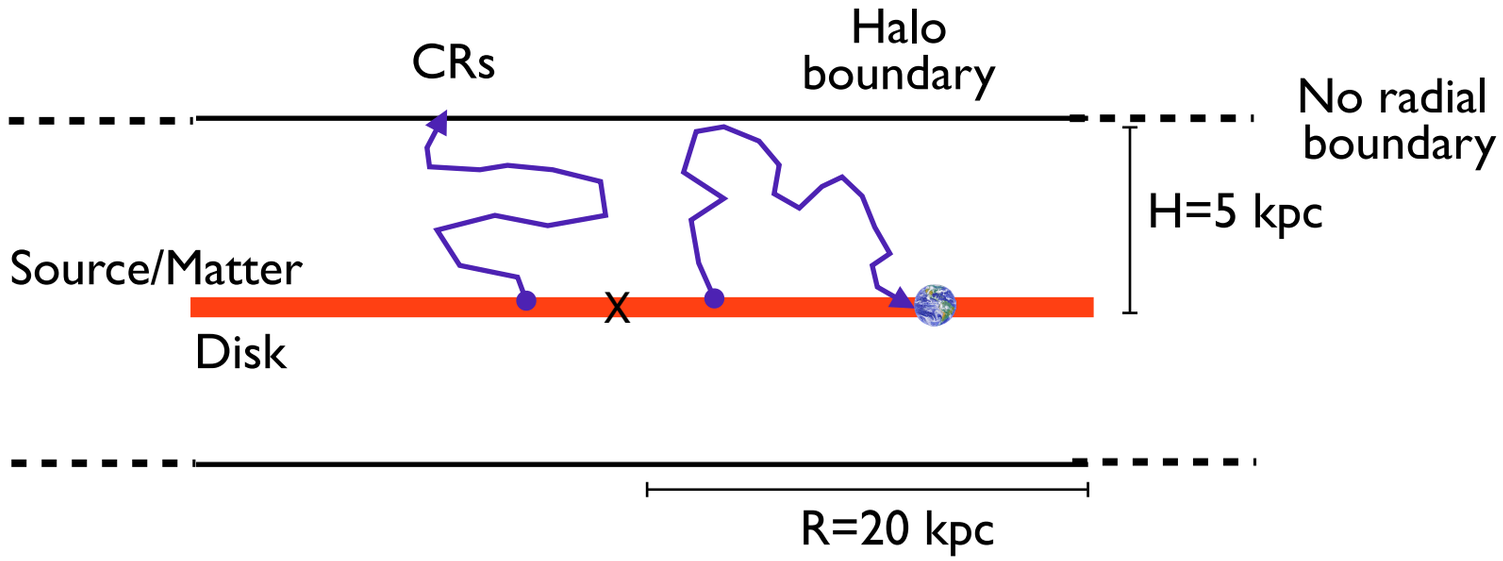
Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR primary spectrum from a uniform distribution of sources

***We assume a cylindrical geometry for the propagation region with vertical boundaries and no radial boundary.**



For cosmic rays from the bulk of sources in the Galaxy, the propagation of CR primaries can be described by a steady state transport equation:

$$\nabla \cdot (D_{\rm p} \nabla N_{\rm p}) - 2hn v_{\rm p} \sigma_{\rm p} \delta(z) N_{\rm p} = -S_{\rm p},$$

where $S_{\rm p}(\boldsymbol{r}, E) = \Re Q_{\rm p}(E)\delta(z)$ is the source term assumed to be extended up to the radial distance R.

For the presence of a Galactic vertical boundary at $z = \pm H$, the solution is given by

$$N_{\rm p}(z, E) = \frac{R \Re Q_{\rm p}(E)}{2D_{\rm p}} \int_0^\infty \frac{\sinh[K(H-z)]}{\sinh(KH)\left[K\coth(KH) + \frac{2hnc\sigma_{\rm p}}{2D_{\rm p}}\right]} J_1(KR)\,{\rm d}K,$$

Note the "only" z-dependence

You may check this as an exercise

where J_1 is the Bessel function of the order of 1

$\Re = 25\ {\rm Myr}^{-1}\ {\rm kpc}^{-2}$ denotes the supernova explosion rate in the Galaxy

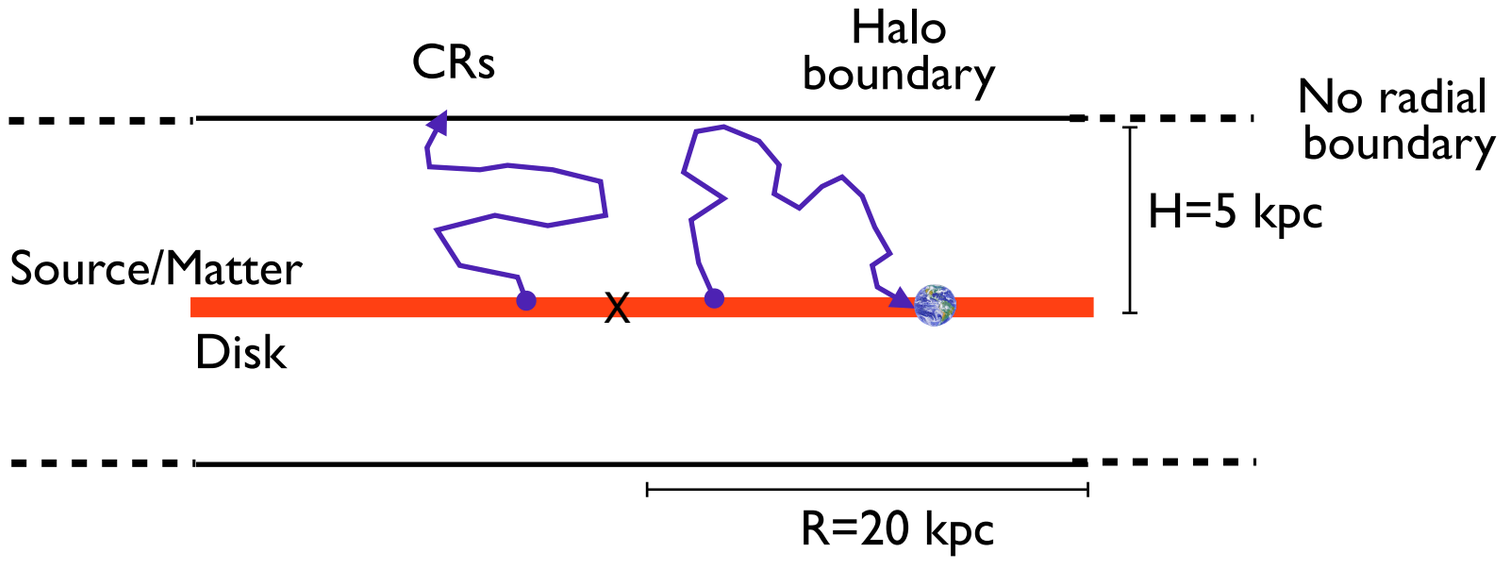
Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR primary spectrum from a uniform distribution of sources

***We assume a cylindrical geometry for the propagation region with vertical boundaries and no radial boundary.**



For cosmic rays from the bulk of sources in the Galaxy, the propagation of CR primaries can be described by a steady state transport equation:

$$\nabla \cdot (D_{\rm p} \nabla N_{\rm p}) - 2hn v_{\rm p} \sigma_{\rm p} \delta(z) N_{\rm p} = -S_{\rm p},$$

where $S_{\rm p}(\boldsymbol{r}, E) = \Re Q_{\rm p}(E)\delta(z)$ is the source term assumed to be extended up to the radial distance R.

For the presence of a Galactic vertical boundary at $z = \pm H$, the solution is given by

$$N_{\rm p}(z, E) = \frac{R \Re Q_{\rm p}(E)}{2 D_{\rm p}} \int_0^\infty \frac{\sinh[K(H - z)]}{\sinh(KH)\left[K \coth(KH) + \frac{2hnc\sigma_{\rm p}}{2D_{\rm p}}\right]} J_1(KR)\, {\rm d}K,$$

Note the "only" z-dependence

You may check this as an exercise

For $Q_p(E) \propto E^{-\gamma}$ and $D_p(E) \propto E^{a}$, $N_p(E) \propto E^{-(\gamma+a)}$

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR secondary spectrum from a uniform distribution of sources

For the CR secondaries produced from the interaction of CR primaries during the propagation in the Galaxy, the propagation can be described by a similar transport equation:

$$\nabla \cdot (D_s \nabla N_s) - 2hnv_s\sigma_s\delta(z)N_s = -S_s$$

where $S_s(\boldsymbol{r}, E) = 2hnc\sigma_{ps}\delta(z)N_p(\boldsymbol{r}, E)$ is the secondary source term.

σ_{ps} is the cross-section for the production of secondary nuclei 's' from primary 'p'.

σ_s is the total spallation (loss) cross-section for the secondaries.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR secondary spectrum from a uniform distribution of sources

For the CR secondaries produced from the interaction of CR primaries during the propagation in the Galaxy, the propagation can be described by a similar transport equation:

$$\nabla \cdot (D_s \nabla N_s) - 2hnv_s\sigma_s\delta(z)N_s = -S_s$$

where $S_s(\boldsymbol{r}, E) = 2hnc\sigma_{ps}\delta(z)N_p(\boldsymbol{r}, E)$ is the secondary source term.

σ_{ps} is the cross-section for the production of secondary nuclei 's' from primary 'p'.

σ_s is the total spallation (loss) cross-section for the secondaries.

The solution is also similar to that of the primaries:

$$N_s(z, E) = 2hnc\sigma_{ps}N_p(z, E)\frac{R}{2D_s}\int_0^\infty \frac{\sinh[K(H-z)]}{\sinh(KH)\,[K\coth(KH) + 2hnc\sigma_s/(2D_s)]} J_1(KR)\,\mathrm{d}K,$$

The CR density at the position of the Earth can be obtained by taking z=0.

For $N_p(E) \propto E^{-(\gamma+a)}$ and $D_s(E) \propto E^a$, $N_s(E) \propto E^{-(\gamma+2a)}$ steeper than the primaries by the diffusion index "a".

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR secondary spectrum from a uniform distribution of sources

For the CR secondaries produced from the interaction of CR primaries during the propagation in the Galaxy, the propagation can be described by a similar transport equation:

$$\nabla \cdot (D_s \nabla N_s) - 2hnv_s\sigma_s\delta(z)N_s = -S_s$$

where $S_s(\boldsymbol{r}, E) = 2hnc\sigma_{ps}\delta(z)N_p(\boldsymbol{r}, E)$ is the secondary source term.

σ_{ps} is the cross-section for the production of secondary nuclei 's' from primary 'p'.

σ_s is the total spallation (loss) cross-section for the secondaries.

The solution is also similar to that of the primaries:

$$N_s(z, E) = 2hnc\sigma_{ps}N_p(z, E)\frac{R}{2D_s}\int_0^\infty \frac{\sinh[K(H-z)]}{\sinh(KH)\,[K\coth(KH) + 2hnc\sigma_s/(2D_s)]} J_1(KR)\,\mathrm{d}K,$$

The CR density at the position of the Earth can be obtained by taking z=0.

At very high energies, it can be shown from the above solution that:

$$\frac{N_s}{N_p} \propto \frac{1}{D_s}.$$ as mentioned on Slide 50.

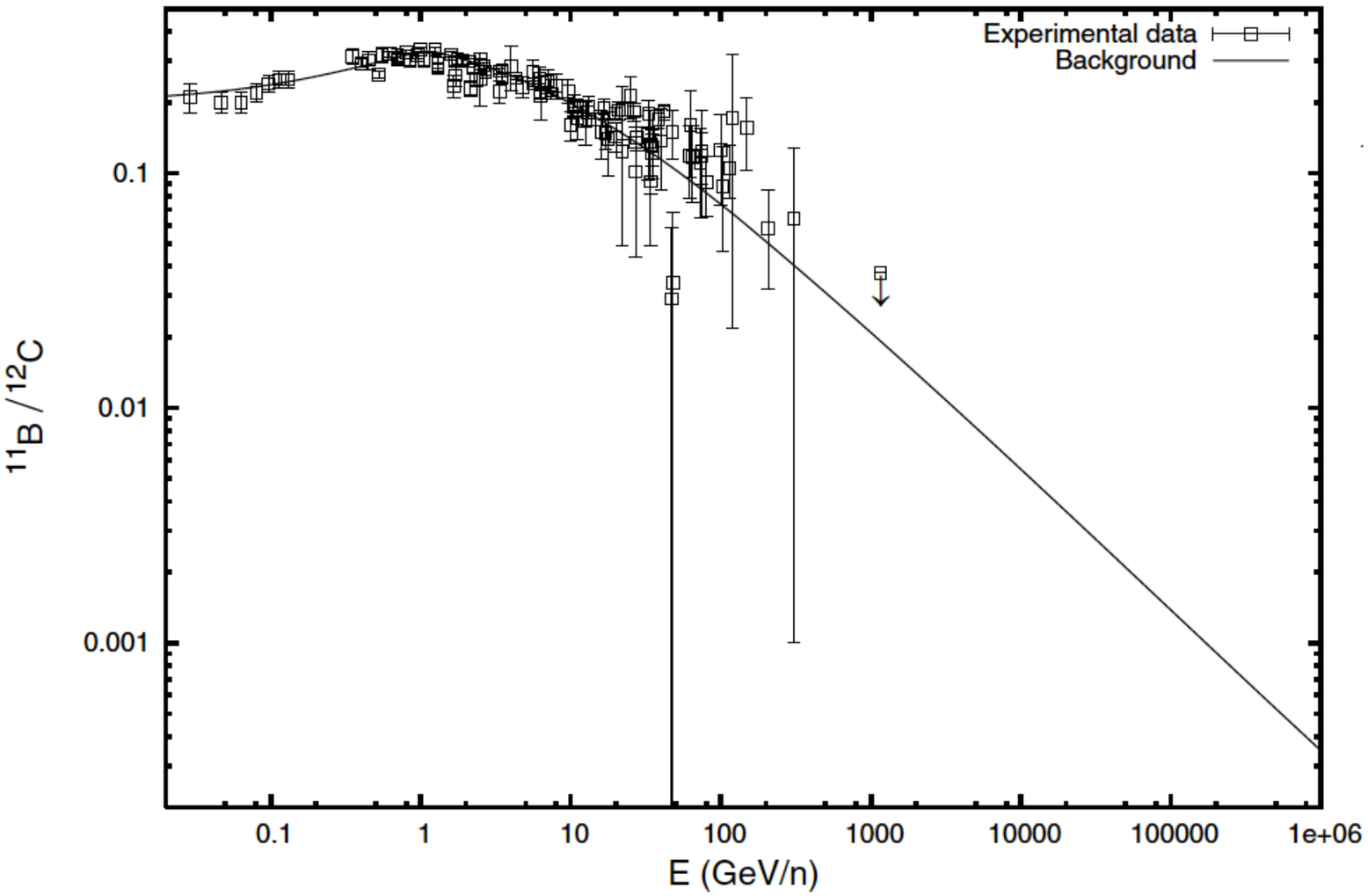
This ratio is independent of the source parameters!

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR secondary/primary ratio for a uniform distribution of sources: E.g., Boron/Carbon



$$D(E) = D_0 \left(\frac{E_0}{E}\right)^a \quad ; \quad (E < E_0)$$

$$= D_0 \left(\frac{E}{E_0}\right)^a \quad ; \quad (E > E_0),$$

The best-fit parameters are:

$D_0 = 2.9 \times 10^{28}\ \mathrm{cm^2\ s^{-1}}$
$E_0 = 3$ GeV/n
a = 0.6

- Note that this value of D(E) depends on the choice of the propagation model.

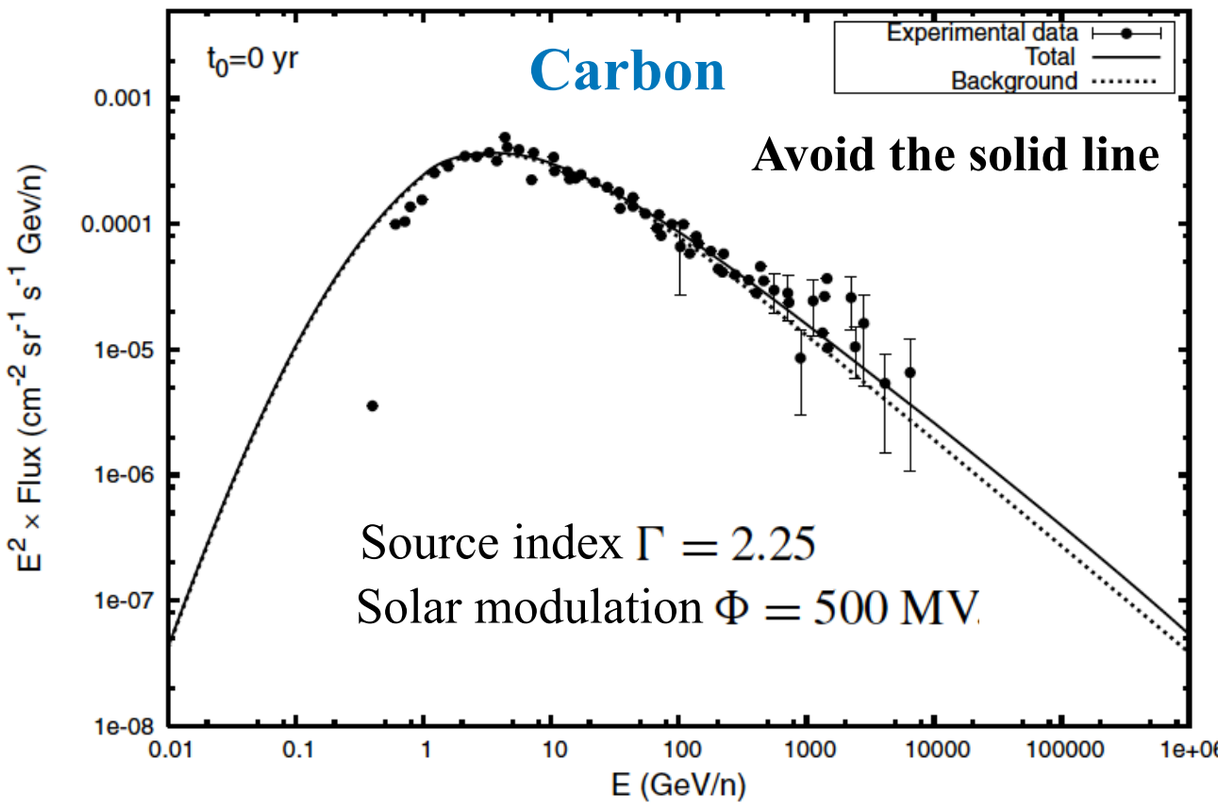
- We assume that the boron secondaries (^{11}B) are produced from the fragmentation (spallation) of the ^{12}C and ^{16}O progenitors with the ISM.

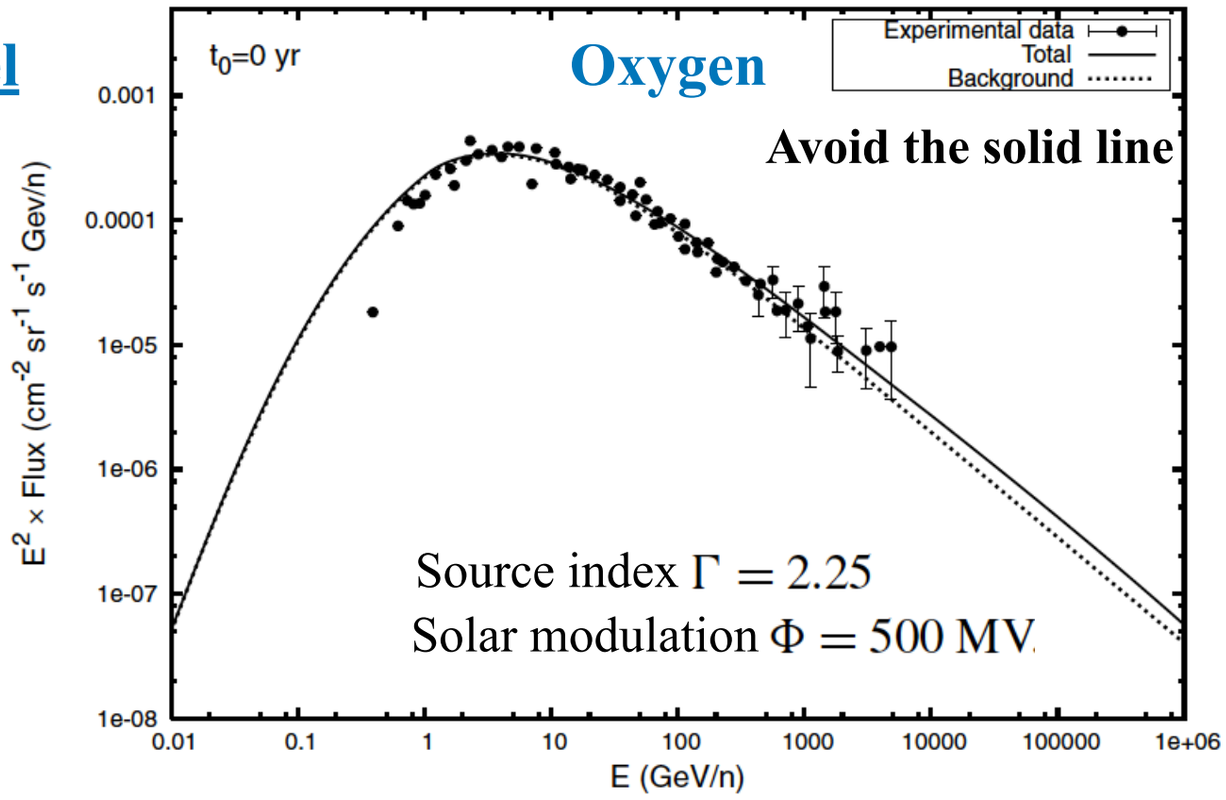
Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Particle spectra from the pure diffusion model





Fragmentation cross-sections

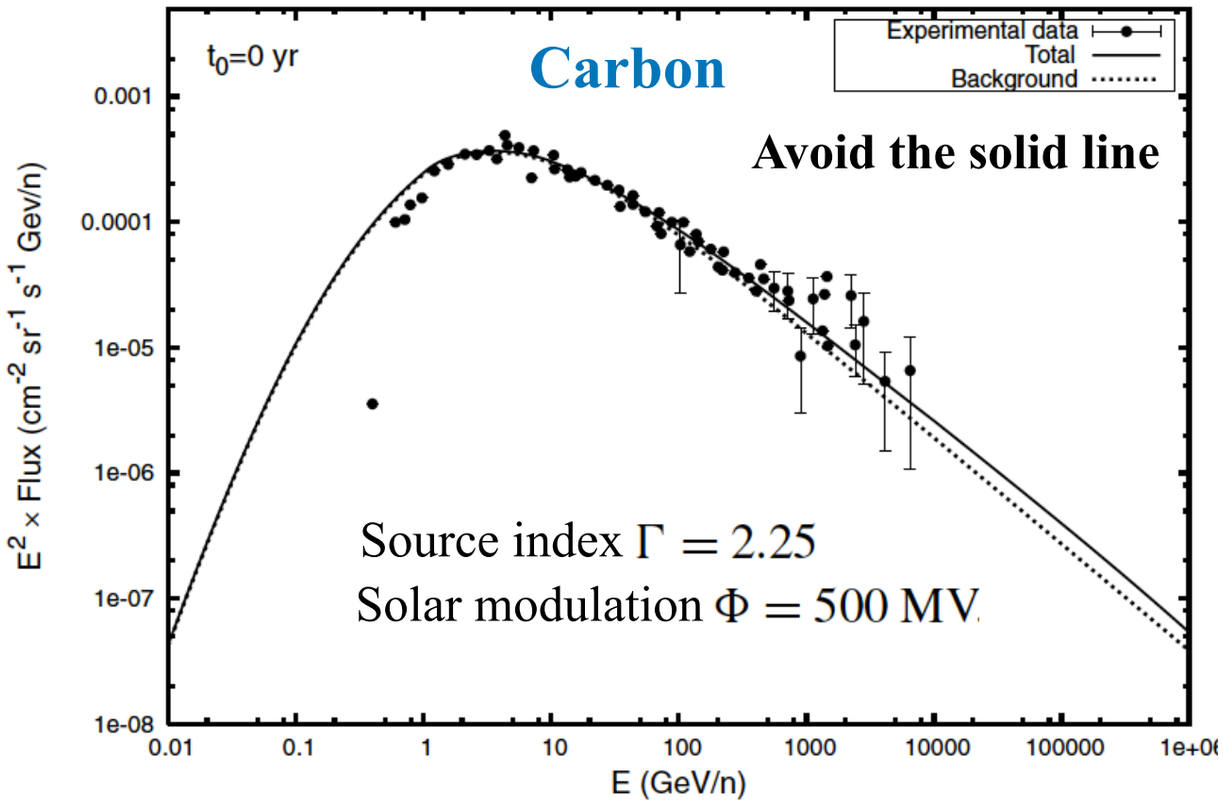
Cross-section	Value (mbarn)
σ_C	250
σ_O	308
σ_B	232
σ_{CB}	76.8
σ_{OB}	38.5

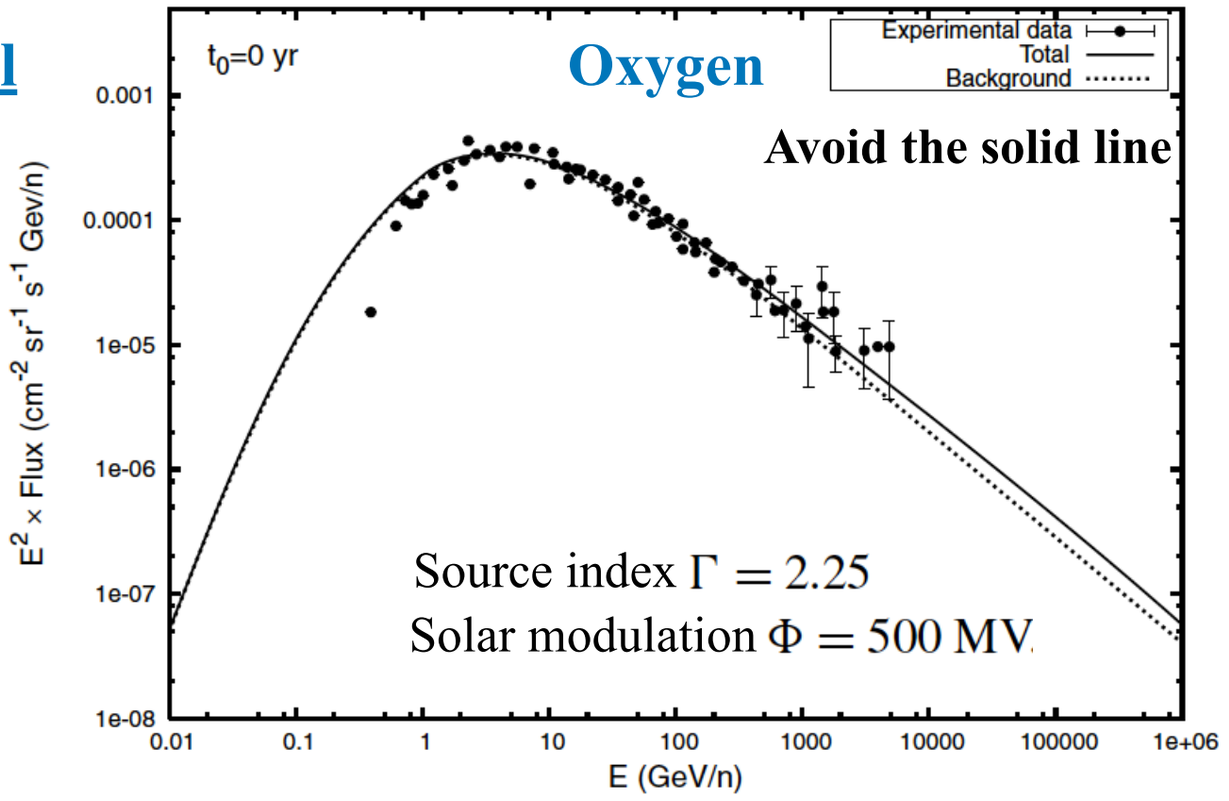
Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

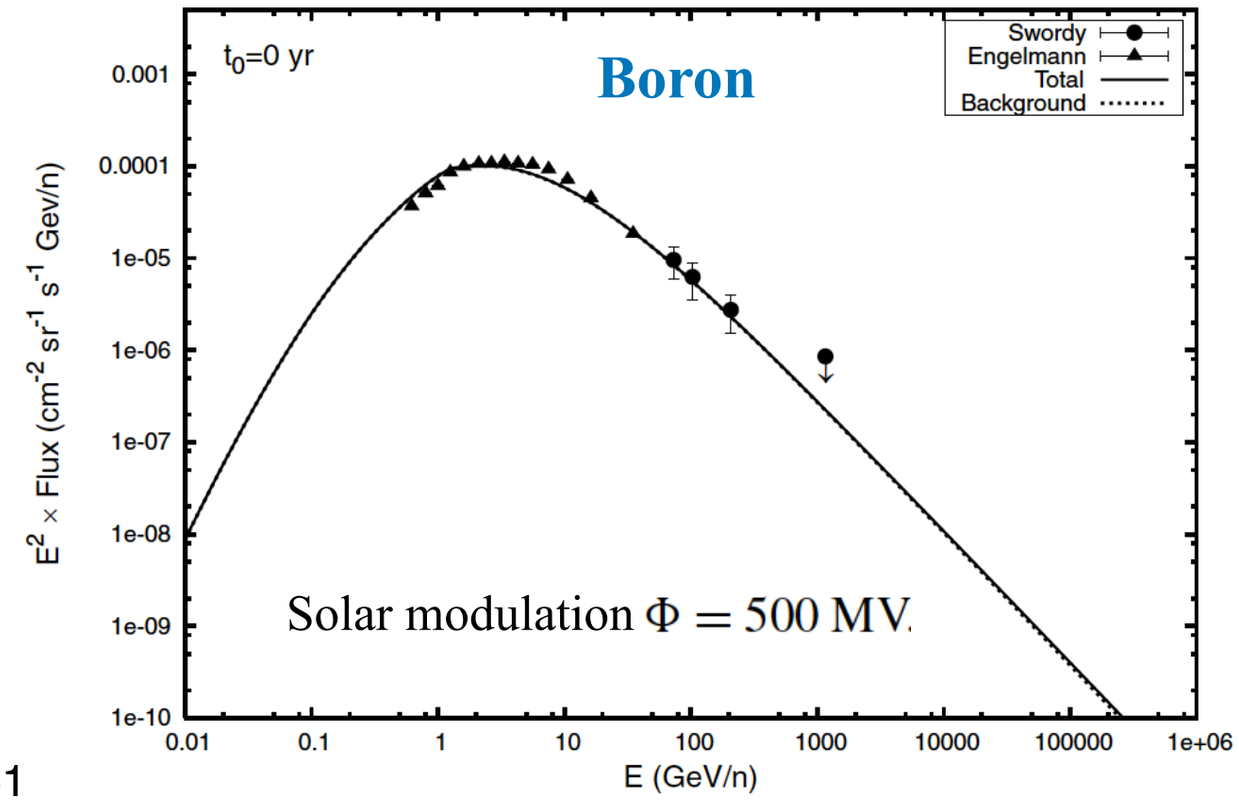
Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Particle spectra from the pure diffusion model







Fragmentation cross-sections

Cross-section	Value (mbarn)
σ_C	250
σ_O	308
σ_B	232
σ_{CB}	76.8
σ_{OB}	38.5

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Anisotropy under the pure diffusion mode with "uniform" source distribution

Anisotropy can be calculated as, $\delta = \frac{3D}{c}\frac{|\nabla N_i|}{N_i}$

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Anisotropy under the pure diffusion mode with "uniform" source distribution

Anisotropy can be calculated as, $\delta = \frac{3D}{c}\frac{|\nabla N_i|}{N_i}$

* Note that for the radially uniform source distribution, anisotropy along the radial direction will be "zero" as there is no cosmic-ray density gradient along "r".

* But, there is anisotropy along the vertical 'z' direction as you can see the z-dependence of the cosmic-ray density from the solutions presented in the previous slides.

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Anisotropy under the pure diffusion mode with "uniform" source distribution

Anisotropy can be calculated as, $\delta = \frac{3D}{c}\frac{|\nabla N_i|}{N_i}$

* Note that for the radially uniform source distribution, anisotropy along the radial direction will be "zero" as there is no cosmic-ray density gradient along "r".

* But, there is anisotropy along the vertical 'z' direction as you can see the z-dependence of the cosmic-ray density from the solutions presented in the previous slides.

* The energy dependence of anisotropy depends on the energy dependence of D and $\frac{|\nabla N_i|}{N_i}$

* In the simplest case, if $\nabla N_i = (N_i\text{-}0)/H$ where H is the scale height over which the density changes by N_i, $\frac{|\nabla N_i|}{N_i} = 1/H$

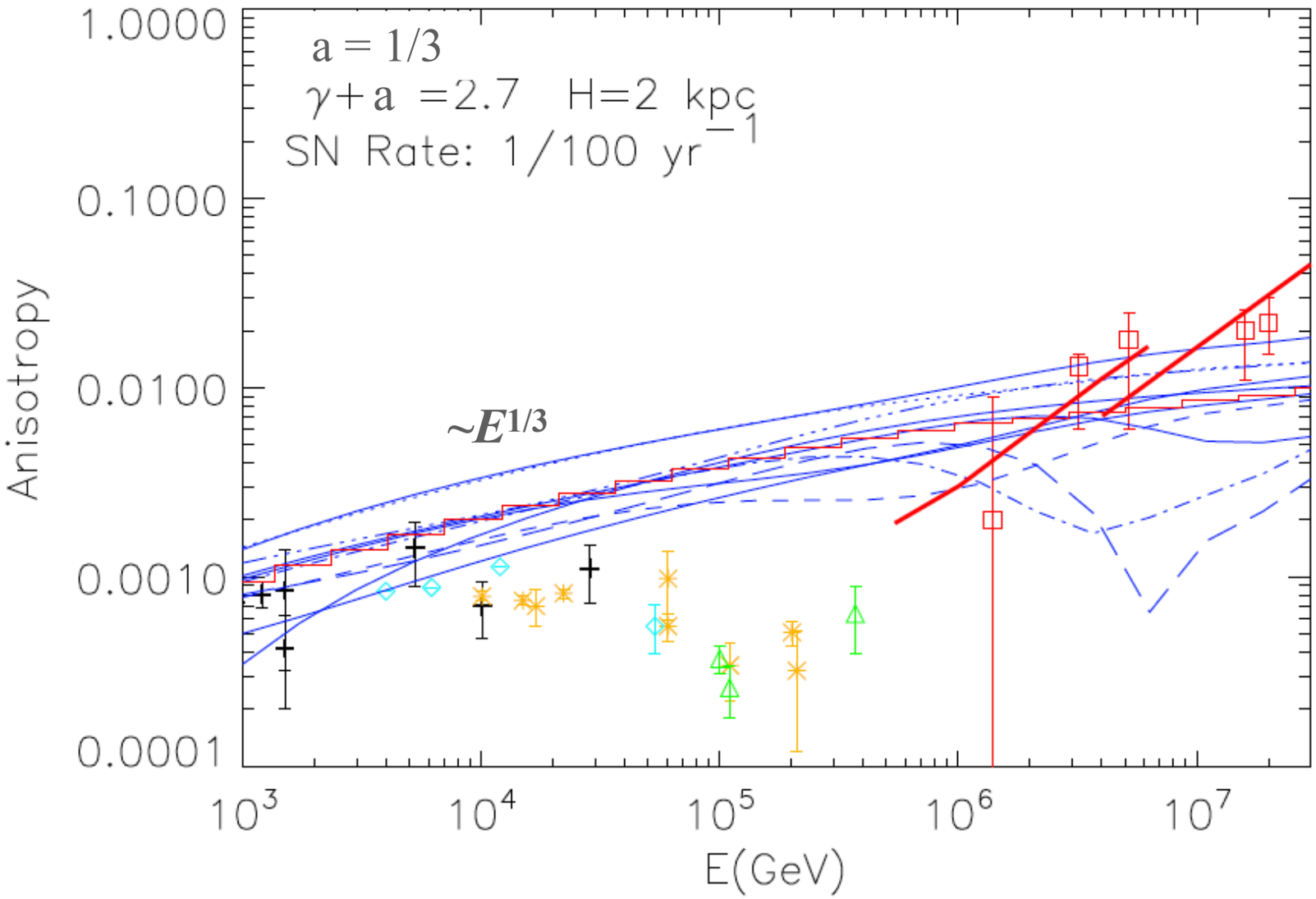
* This gives $\delta \propto D(E) \propto E^a$, where a=0.3-0.6 (from observation + CR propagation model).

Anisotropy increases with energy

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Anisotropy under the pure diffusion mode with "uniform" source distribution

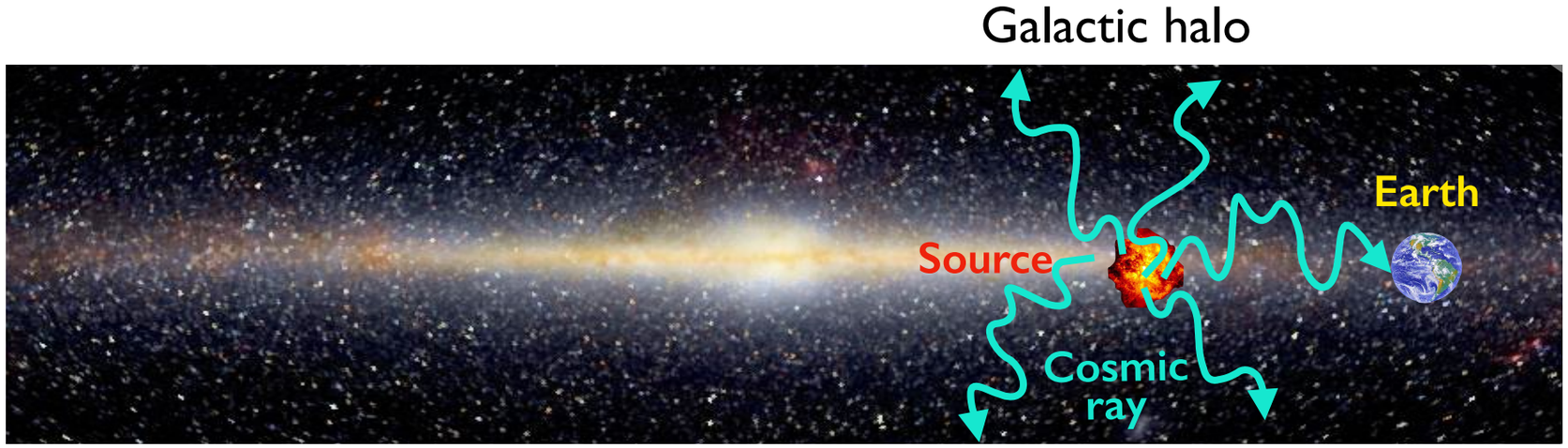


Blasi & Amato, JCAP01(2012)011

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR primary spectrum from a discrete nearby point source

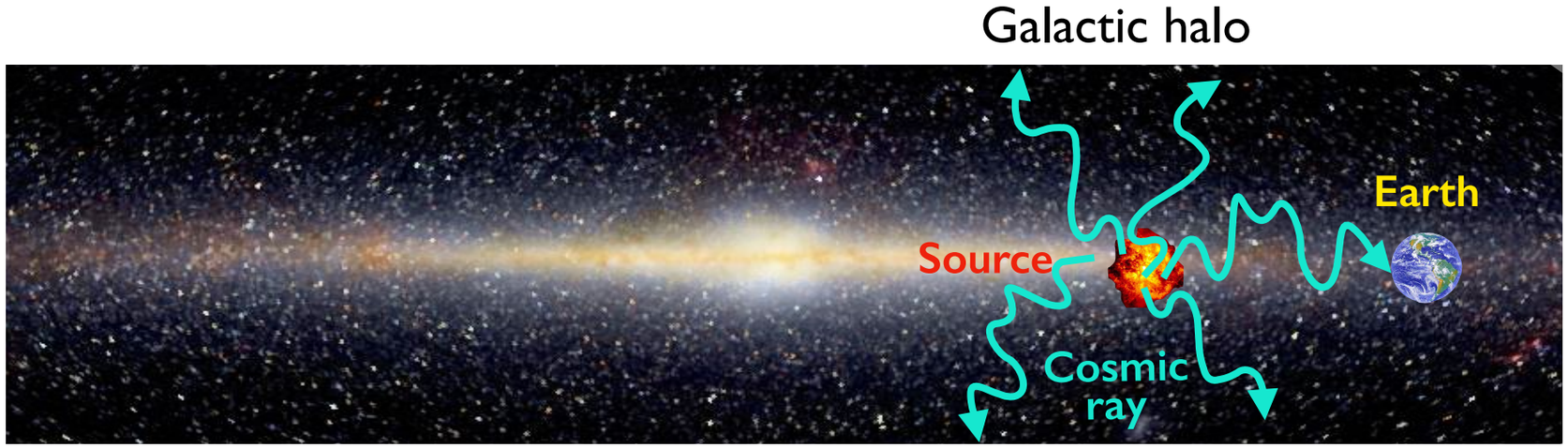


- Nearby sources: Their discrete nature in space and time has to be considered.
- Propagation of cosmic rays from nearby sources has to be treated in a different way.
- Propagation region can be considered to have infinite dimensions:

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR primary spectrum from a discrete nearby point source



- Nearby sources: Their discrete nature in space and time has to be considered.
- Propagation of cosmic rays from nearby sources has to be treated in a different way.
- Propagation region can be considered to have infinite dimensions:
  The reason is CRs from nearby sources do not effectively see the presence of any Galactic boundaries (both in the radial as well as vertical directions) due to their much smaller propagation times to the Earth compared to their escape time-scales from the Galaxy boundaries (Thoudam 2007, MNRAS, 380, L1).

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR primary spectrum from a discrete nearby point source

In the diffusion model, neglecting convection, energy losses and particle reacceleration processes, the propagation of CR primaries from a nearby source can be represented by the equation

$$\nabla \cdot (D_{\rm p} \nabla N_{\rm p}) - 2hn v_{\rm p} \sigma_{\rm p} \delta(z) N_{\rm p} + \mathbb{Q}_{\rm p} = \frac{\partial N_{\rm p}}{\partial t},$$

where the subscript 'p' denotes the primary nuclei,

$N_{\rm p}(\boldsymbol{r}, E, t)$ is the differential number density at a distance $\boldsymbol{r}$ at time t,

E is the kinetic energy per nucleon of the nuclei,

$D_{\rm p}(E)$ is the diffusion coefficient

n is the target density

$\mathbb{Q}_{\rm p}(\boldsymbol{r}, E, t) = Q_{\rm p}(E)\delta(\boldsymbol{r})\delta(t - t_0)$ is the particle production rate from the source.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR primary spectrum from a discrete nearby point source

In the diffusion model, neglecting convection, energy losses and particle reacceleration processes, the propagation of CR primaries from a nearby source can be represented by the equation

$$\nabla \cdot (D_{\rm p} \nabla N_{\rm p}) - 2hn v_{\rm p} \sigma_{\rm p} \delta(z) N_{\rm p} + \mathbb{Q}_{\rm p} = \frac{\partial N_{\rm p}}{\partial t},$$

where the subscript 'p' denotes the primary nuclei,

$N_{\rm p}(\boldsymbol{r}, E, t)$ is the differential number density at a distance $\boldsymbol{r}$ at time t,

E is the kinetic energy per nucleon of the nuclei,

$D_{\rm p}(E)$ is the diffusion coefficient

n is the target density

$\mathbb{Q}_{\rm p}(\boldsymbol{r}, E, t) = Q_{\rm p}(E)\delta(\boldsymbol{r})\delta(t - t_0)$ is the particle production rate from the source.

NOTE: We are also interested in the study of the secondary nuclei which are produced from the fragmentation of primary cosmic rays upon interaction with the interstellar medium. Since the energy per nucleon is an almost conserved quantity in the fragmentation process, we will be dealing with the energy per nucleon in all our relations rather than the total kinetic energy of the nuclei.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR primary spectrum from a discrete nearby point source

$$\nabla \cdot (D_{\rm p} \nabla N_{\rm p}) - 2hn v_{\rm p} \sigma_{\rm p} \delta(z) N_{\rm p} + \mathbb{Q}_{\rm p} = \frac{\partial N_{\rm p}}{\partial t},$$

Using Green function technique and performing proper Fourier and Laplace transforms, the exact unbounded solution of the above equation for a source releasing cosmic rays at time t_0 is given by:

$$N_{\rm p}(\boldsymbol{r}, E, t) = \frac{Q_{\rm p}(E) \exp\left[-\frac{r^2}{4D_{\rm p}(t-t_0)}\right]}{8\pi D_{\rm p}^{3/2}(t-t_0)} \left\{ \frac{{\rm e}^{-q^2/[4(t-t_0)]}}{\sqrt{\pi(t-t_0)}} - b e^{bq+b^2(t-t_0)} {\rm erfc}\left(b\sqrt{t-t_0} + \frac{q}{2\sqrt{t-t_0}} \right) \right\},$$

You may check this solution as an exercise

where $q = |z|/\sqrt{D_{\rm p}}$, $b = 2hnc\sigma_{\rm p}/2\sqrt{D_{\rm p}}$ and $\boldsymbol{r}$ is the distance to the source.

$Q_{\rm p}(E) = A_{\rm p} q_{\rm p}(T)$ with $q_{\rm p}(T)$ given by $q_{\rm p}(T) = k\left(T^2 + 2Tm_{\rm p}\right)^{-(\Gamma+1)/2}(T + m_{\rm p})$,

where $T = A_{\rm p}E$ represents the total kinetic energy of the nuclei,

$A_{\rm p}$ is the mass number, $m_{\rm p}$ is the mass energy,

Γ is the spectral index and k is a normalization constant.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335 (Appendix A)

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Application to nearby supernova remnants

Table 1. Parameters of known SNRs located within a distance of 1.5 kpc from the Earth (see the references given in Thoudam 2007a).

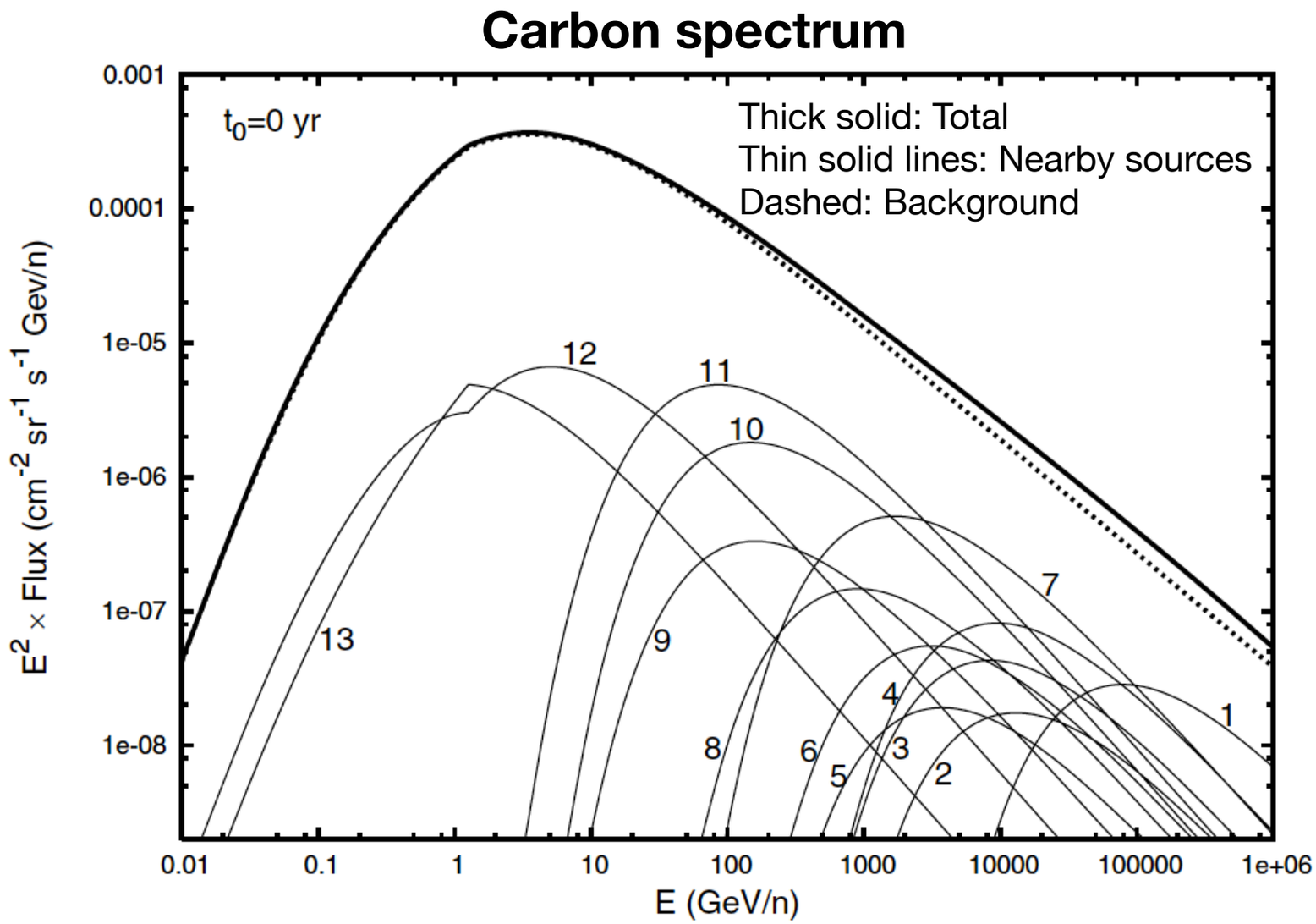
SNR	Distance (kpc)	Age (yr)
G65.3+5.7	1.0	14 000
G73.9+0.9	1.3	10 000
Cygnus Loop	0.4	14 000
HB21	0.8	19 000
G114.3+0.3	0.7	41 000
CTA1	1.4	24 500
HB9	1.0	7700
S147	0.8	4600
Vela	0.3	11 000
G299.2−2.9	0.5	5000
SN185	0.95	1800
Monogem	0.3	86 000
Geminga	0.15	340 000

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Application to nearby supernova remnants



(1) SN185
(2) G73.9+0.9
(3) HB9
(4) S147
(5) CTA1
(6) G65.3+5.7
(7) G299.2–2.9
(8) HB21
(9) G114.3+0.3
(10) Cygnus Loop
(11) Vela
(12) Monogem
(13) Geminga

*Nearby sources can produce significant contribution mostly at high energies.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR secondary spectrum due to cosmic-ray primary produced from a discrete nearby point source

***Source of secondary CRs is the interaction of primary CRs within the volume around the source as, for instance, indicated by the white circle.**

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335 (Appendix A)

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR secondary spectrum due to cosmic-ray primary produced from a discrete nearby point source

The transport of CR secondaries in the Galaxy also follows an equation similar to that of the primaries except for the source term:

$$\nabla \cdot (D_{\rm s} \nabla N_{\rm s}) - 2hnv_{\rm s}\sigma_{\rm s}\delta(z)N_{\rm s} + \mathbb{Q}_{\rm s} = \frac{\partial N_{\rm s}}{\partial t},$$

where the subscript 's' represents the secondary nuclei

For secondaries originating from a single type of primary, we can write the source term as

$$\mathbb{Q}_{\rm s}(\boldsymbol{r}, E, t) = 2hnc\delta(z) \int_E^\infty \frac{\rm d}{{\rm d}E'} \sigma_{\rm ps}(E, E') N_{\rm p}(\boldsymbol{r}, E', t)\, {\rm d}E'.$$

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335 (Appendix A)

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR secondary spectrum due to cosmic-ray primary produced from a discrete nearby point source

The transport of CR secondaries in the Galaxy also follows an equation similar to that of the primaries except for the source term:

$$\nabla \cdot (D_{\rm s} \nabla N_{\rm s}) - 2hn v_{\rm s} \sigma_{\rm s} \delta(z) N_{\rm s} + \mathbb{Q}_{\rm s} = \frac{\partial N_{\rm s}}{\partial t},$$

where the subscript 's' represents the secondary nuclei

For secondaries originating from a single type of primary, we can write the source term as

$$\mathbb{Q}_{\rm s}(\boldsymbol{r}, E, t) = 2hnc\delta(z) \int_E^\infty \frac{\rm d}{{\rm d}E'} \sigma_{\rm ps}(E, E') N_{\rm p}(\boldsymbol{r}, E', t)\, {\rm d}E'.$$

We can approximate the differential production cross-section as $\frac{\rm d}{{\rm d}E'} \sigma_{\rm ps}(E, E') = \sigma_{\rm ps} \delta(E' - E),$

where $\sigma_{\rm ps}$ denotes the total fragmentation cross-section of p to s

E' is the primary energy/nucleon and *E* is the primary energy/nucleon

This simplifies the source term to $\mathbb{Q}_{\rm s}(\boldsymbol{r}, E, t) = 2hnc\sigma_{\rm ps} \delta(z) N_{\rm p}(\boldsymbol{r}, E, t).$

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335 (Appendix A)

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR secondary spectrum due to cosmic-ray primary produced from a discrete nearby point source

The solution for the secondaries at the Earth (considered to be located at the origin) can be obtained as:

$$N_{\rm s}(E,t) = 2hnc\sigma_{\rm ps}\int_0^t {\rm d}t_0 \int {\rm d}V \exp\left[-\frac{(\boldsymbol{r}-\boldsymbol{r}')^2}{4D_{\rm s}(t-t_0)}\right]\left[\frac{1}{8\pi^{3/2}[D_{\rm s}(t-t_0)]^{3/2}} - \frac{2hnc\sigma_{\rm s}}{16D_{\rm s}^2(t-t_0)}\right]$$

$$\times \exp\left[\frac{4h^2n^2c^2\sigma_{\rm s}^2(t-t_0)}{4D_{\rm s}}\right] \times {\rm erfc}\left[\frac{2hnc\sigma_{\rm s}\sqrt{t-t_0}}{2\sqrt{D_{\rm s}}}\right] N_{\rm p}(\boldsymbol{r}', E, t)\delta(z'),$$

where ***r***' denotes the position of the primaries with respect to the point source which in cylindrical coordinates yield an integral over the volume element as

$$\int {\rm d}V = \int_0^\infty r'{\rm d}r' \int_0^{2\pi} {\rm d}\phi' \int_{-\infty}^{\infty} {\rm d}z'.$$

The solution above gives the CR secondary density due to primary CRs produced from a source located at a distance **r**.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335 (Appendix A)

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

CR secondary spectrum due to cosmic-ray primary produced from a discrete nearby point source

The solution for the secondaries at the Earth (considered to be located at the origin) can be obtained as:

$$N_{\rm s}(E,t) = 2hnc\sigma_{\rm ps}\int_0^t {\rm d}t_0 \int {\rm d}V \exp\left[-\frac{(\boldsymbol{r}-\boldsymbol{r}')^2}{4D_{\rm s}(t-t_0)}\right]\left[\frac{1}{8\pi^{3/2}[D_{\rm s}(t-t_0)]^{3/2}} - \frac{2hnc\sigma_{\rm s}}{16D_{\rm s}^2(t-t_0)}\right]$$

$$\times \exp\left[\frac{4h^2n^2c^2\sigma_{\rm s}^2(t-t_0)}{4D_{\rm s}}\right] \times {\rm erfc}\left[\frac{2hnc\sigma_{\rm s}\sqrt{t-t_0}}{2\sqrt{D_{\rm s}}}\right] N_{\rm p}(\boldsymbol{r}', E, t)\delta(z'),$$

If a particular secondary species is produced as the result of fragmentations of more than one type of heavier primaries, the total number of secondaries produced is obtained by simply adding the contributions from the different primary species as

$$N_{\rm s}^{\rm Tot} = \sum_j N_{\rm s}^j, \quad \text{where } j \text{ denotes the primary species.}$$

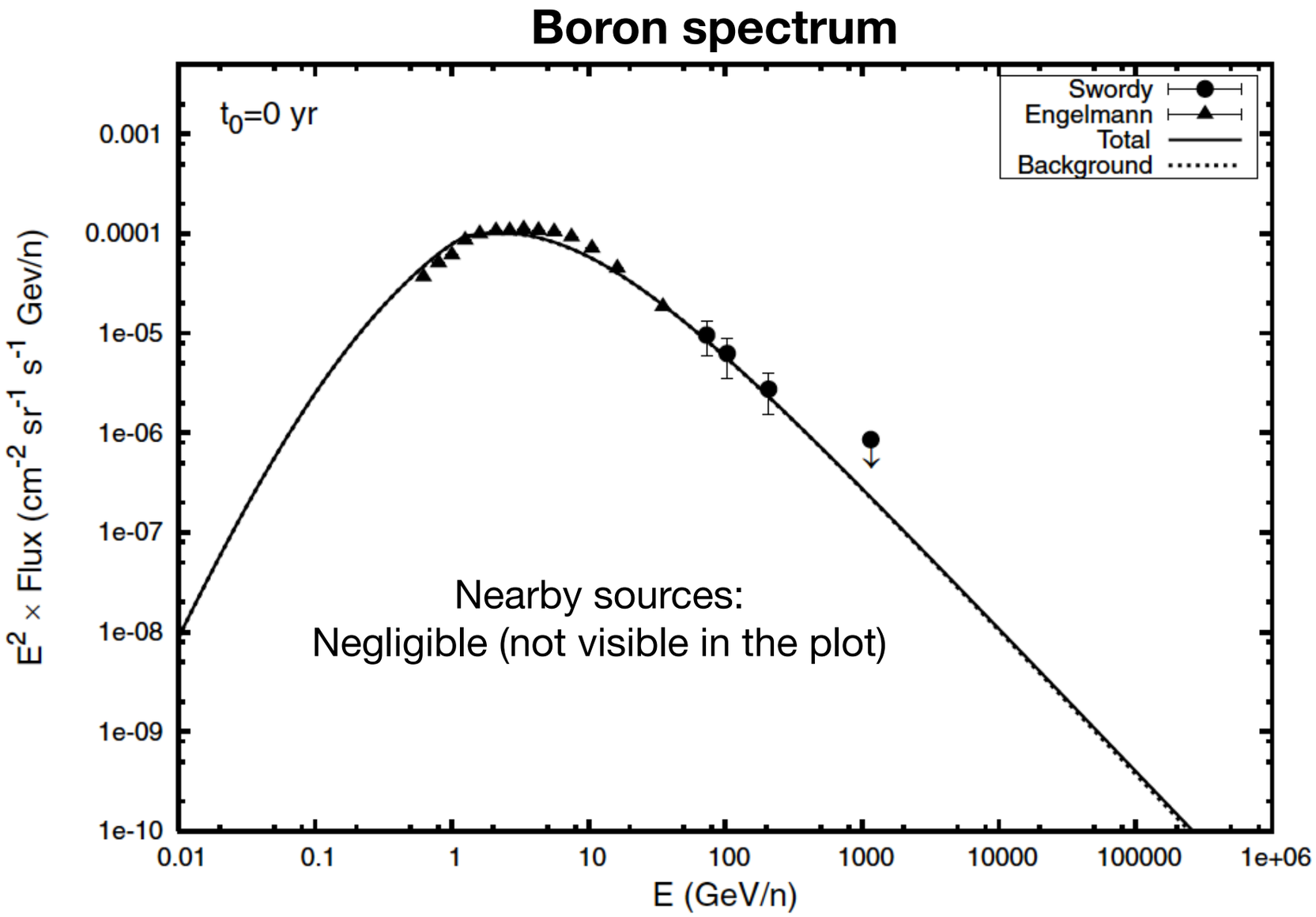
Example: For boron secondaries, carbon and oxygen are the primary species.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335 (Appendix A)

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Application to nearby supernova remnants



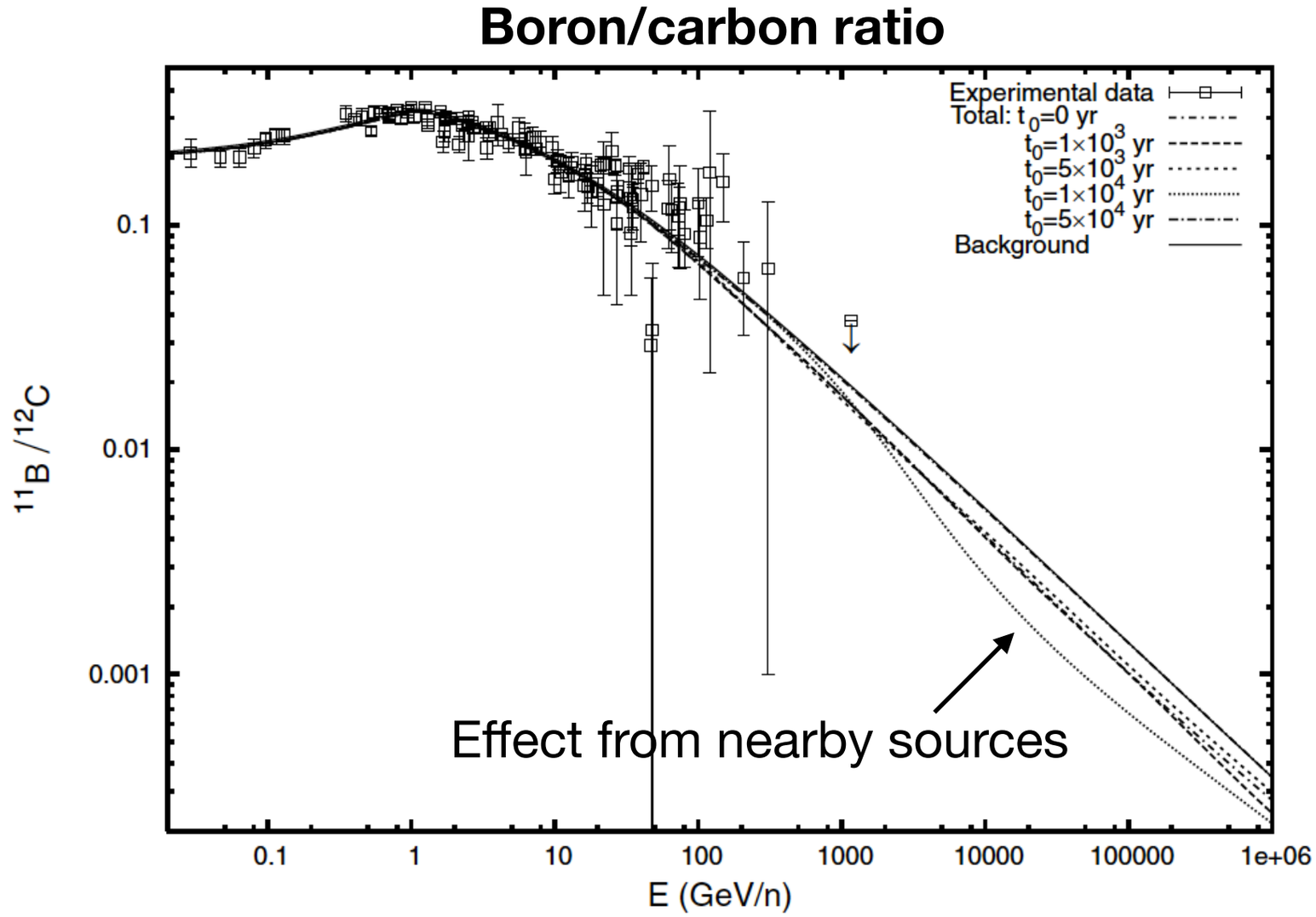
*For secondaries, contribution from nearby sources is negligible.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Application to nearby supernova remnants



*Significant contribution of nearby sources on the primary spectrum can affect the interpretation of the boron-to-carbon ratio.

* That means the ratio may not be reliable enough to determine the diffusion coefficient at high energies.

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

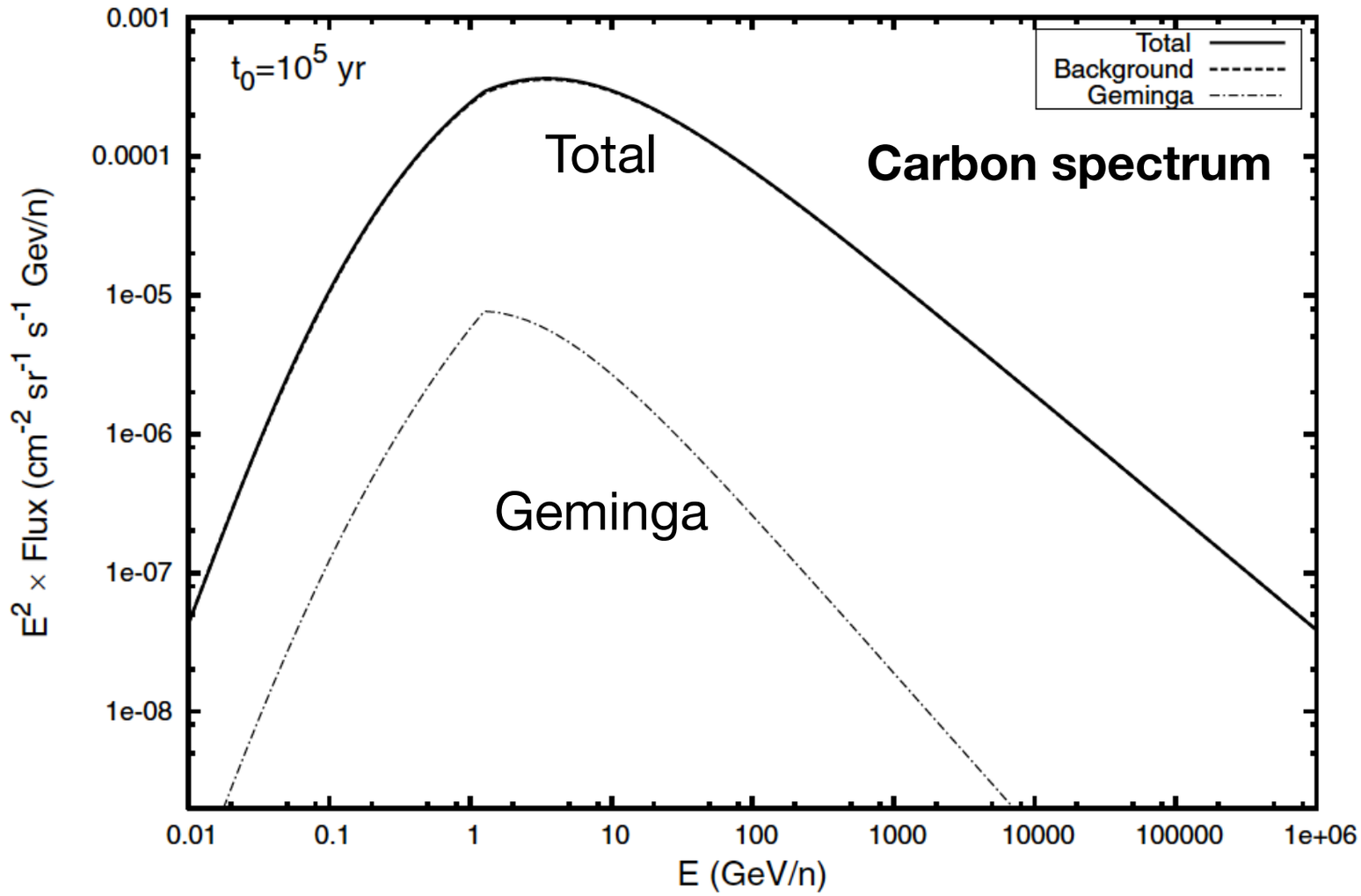
Propagation model I: Pure diffusion

Application to nearby supernova remnants

*But, for a CR release time of $t_0=10^5$ yr which is predicted by diffusive shock acceleration theory, none of the nearby supernova remnants have released their particles yet except for Geminga.

*Then, contribution from nearby sources on the primary cosmic-ray spectrum becomes negligible, and hence, the boron-to-carbon ratio may be reliable to determine the propagation properties of cosmic rays.

*But, this conclusion is true only for the small Galactic region traversed by cosmic rays before reaching the Earth. See next slide.



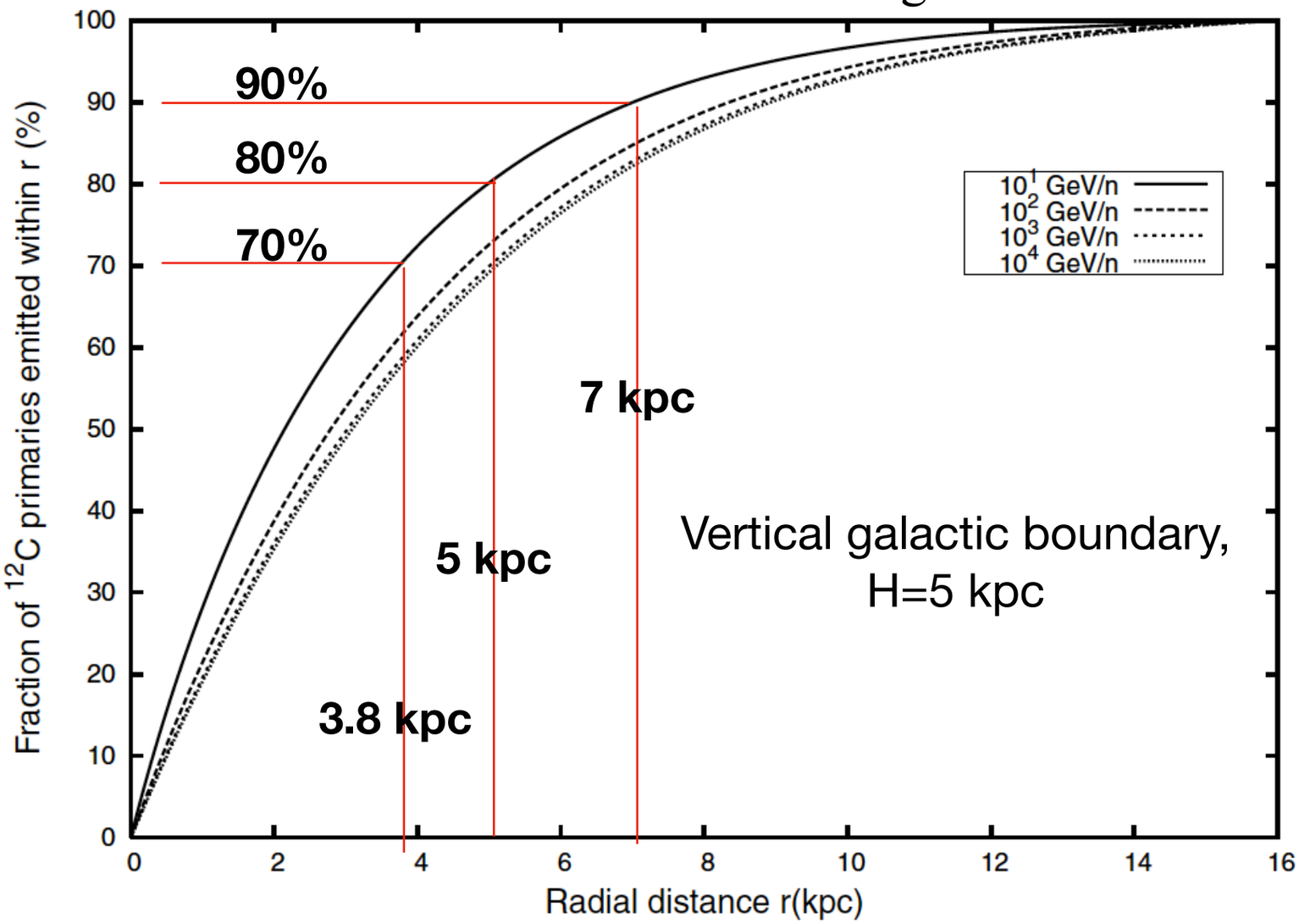
Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

Application to nearby supernova remnants

Fraction of cosmic rays emitted within the radial distance r from the Earth for various energies



This shows that CR nuclei reaching the Earth actually carry information only for a limited Galactic volume!

Thoudam, S., 2008, Mon. Not. R. Astron. Soc. 388, 335

Diffusive propagation of cosmic rays in the Galaxy: In detail

Propagation model I: Pure diffusion

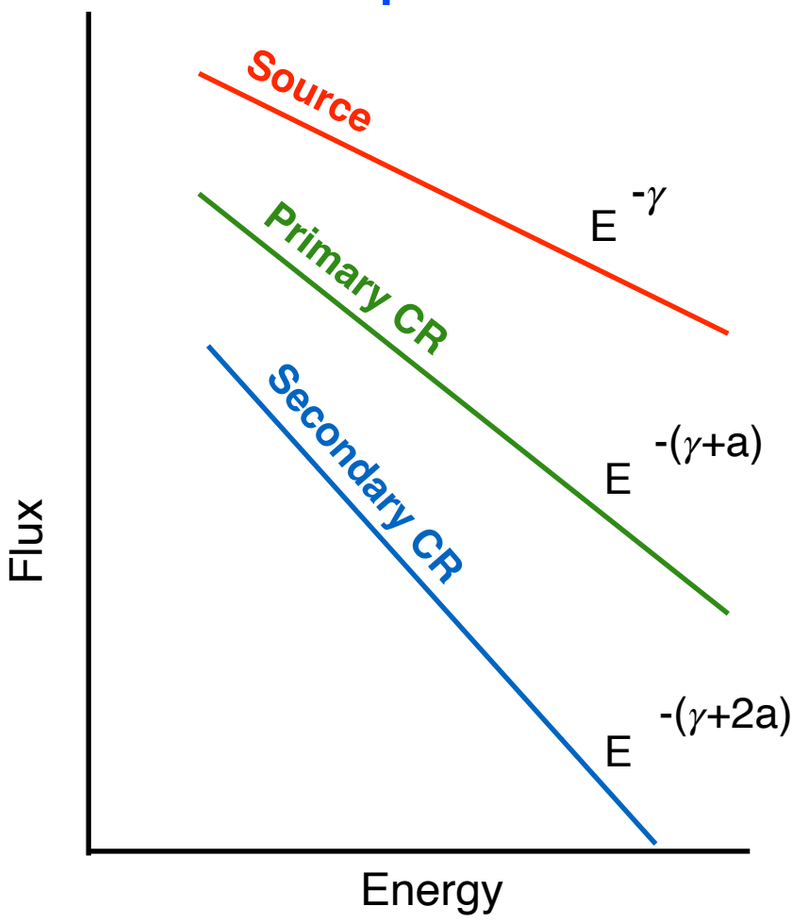
Take home message:

CR nuclei

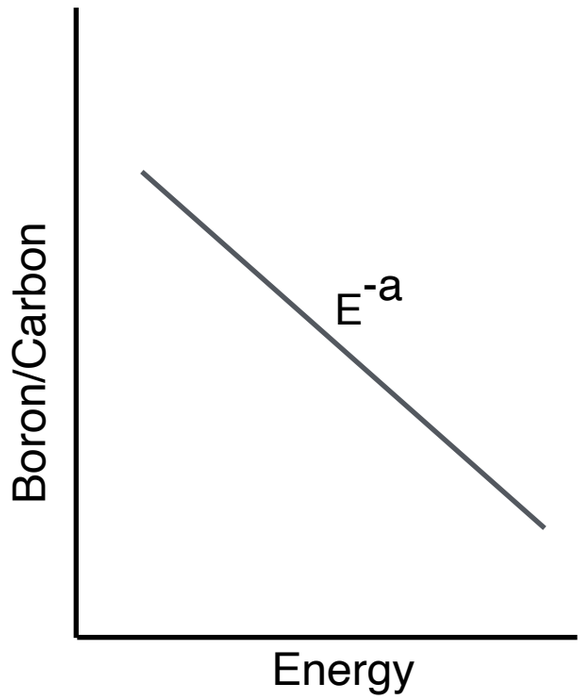
For a primary source spectrum $Q_p(E) \propto E^{-\gamma}$ and $D_p(E) \propto E^{a}$,

- Primary spectrum in the Galaxy: $N_p(E) \propto E^{-(\gamma+a)}$
- Secondary spectrum in the Galaxy: $N_p(E) \propto E^{-(\gamma+2a)}$

Spectrum



Secondary/Primary ratio



Next, we will learn about the propagation of high-energy electrons in the Galaxy